Life insurance mathematics II

Lecture notes for
Topics in Life–insurance (Liv2)

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Draft, February 25, 2026

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Notation

- $\boldsymbol{A} = \{a_{ij}\}$: Matrices are denoted capital letters, and their corresponding elements by the corresponding unbold, lower case letter.
- $\boldsymbol{x}$: Vectors are denoted by lower case bold letters, and are their elements by the corresponding unbold letters, x_j. All vectors with roman letters are column vectors, while all vectors with greek letters are row vectors.
- $\bullet$: Schur or Hadamard matrix product, which is an entry–wise matrix product: $\boldsymbol{A} \bullet \boldsymbol{B} = \{a_{ij} b_{ij}\}$.
- $\boldsymbol{\Delta}(\boldsymbol{x})$: Diagonal matrix which has the vector $\boldsymbol{x}$ as diagonal.
- $\otimes$: Kronecker product between arbitrary matrices, $\boldsymbol{A} \otimes \boldsymbol{B} = \{a_{ij} \boldsymbol{B}\}$.
- $\oplus$: Kronecker sum for square matrices, $\boldsymbol{A} \oplus \boldsymbol{B} = \boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}$.
- $\boldsymbol{A}'$: Matrix or vector transpose.
- $\boldsymbol{e}$: the vector of ones, $\boldsymbol{e} = (1, 1, ..., 1)'$, of adequate dimension.
- $\boldsymbol{A}(x) = \{a_{ij}(x)\}$: A matrix function, i.e. a matrix of functions $a_{ij}(x)$.
- $\prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x$: Product integral of a matrix function.
- $p_{ij}(s,t)$: transition probabilities, p. 22.
- $\boldsymbol{P}(s,t)$: Transition (probability) matrix, $\boldsymbol{P}(s,t) = \{p_{ij}(s,t)\}$.
- $\mathbb{P}_i(\cdot)$: Conditional distribution given $X_0 = i$, i.e. $\mathbb{P}_i(A) = \mathbb{P}(A|X_0 = i)$.
- $b_i(t)$: payment rates.
- $b_{ij}(t)$: Lump sum payments upon transition.
- $\boldsymbol{B}(t)$: Matrix of lump sum payments.
- $\boldsymbol{b}(t)$: Vector containing the payment rates.
- $\boldsymbol{\Lambda}(t)$: Intensity matrix (function) of a time–inhomogeneous Markov process.
- $\boldsymbol{\Lambda}^1(t)$: Transition rates which cause lump sum payments.
- $\boldsymbol{\Lambda}^0(t)$: Transition rates causing no lump sum payments.
- $\boldsymbol{R}(t)$: Reward matrix, $\boldsymbol{R}(t) = \boldsymbol{\Lambda}(t) \bullet \boldsymbol{B}(t) + \boldsymbol{\Delta}(\boldsymbol{b}(t))$.
- $\boldsymbol{C}^{(k)}(t)$: Matrices containing expected values of powers of payments.
- $m_{ij}^{(k)}(s,t)$: Partial moments.
- $\boldsymbol{m}^{(k)}(s,t)$: Partial moment matrix.
- $\boldsymbol{m}_r^{(k)}(s,t)$: Matrix of reduced moments.
- $v_{ij}(s,t)$: Discounted partial moments.
- $\boldsymbol{V}(s,t)$: Matrix of discounted partial moments.
- $\boldsymbol{V}(t)$: Matrix of (partial, state–wise prospective) reserves.
- $\mathrm{RS}_{ij}(t)$: Risk sums.
- δ_{ij}: Kronecker delta, equals 1 if $i = j$ and zero otherwise.
- $\mathbb{P}(X \in s)$: Short notation for the density function of X evaluated in s.

CHAPTER 1

Product integration

Many functions and objects in mathematics are symbols for something that cannot be computed explicitly. Take, for example, the exponential function, $f(x) = \exp(\lambda x)$, $\lambda \in \mathbb{R}$, which can be defined as the unique solution to the differential equation,

$$f'(x) = \lambda f(x),\ f(0) = 1.$$

The exponential function appears in almost any branch of mathematics, and it has a lot of nice properties. Therefore it deserves its own symbol, usually $\exp(x)$ or e^x, which can then be used as building blocks in further derivations and expressions in mathematics.

More generally, we could consider the function f with solution

$$f'(x) = \lambda(x) f(x),\ f(0) = 1. \tag{1}$$

This is a linear differential equation with a variable coefficient. The solution is given by

$$f(x) = \exp\left(\int_0^x \lambda(u)\,\mathrm{d}u\right).$$

Hence it can be expressed by the former exponential function and an integral of the variable coefficients.

In the following we will extend the latter to linear systems of ordinary differential equations (ODE) with variable coefficients,

$$\boldsymbol{Y}'(t) = \boldsymbol{Y}(t)\boldsymbol{A}(t), \boldsymbol{Y}(0) = \boldsymbol{I}, \tag{2}$$

where $\boldsymbol{Y}(t)$ is a $p \times p$ matrix function, $\boldsymbol{A}(t)$ is a $p \times p$ matrix coefficient and $\boldsymbol{I}$ denotes the identity matrix (of adequate order, here $p \times p$). This is just a matrix version of (1).

The unique solution, which has a number of nice properties, can be used extensively in the theory of time–inhomogeneous Markov processes for finding transition probabilities (Komogorov) and in applications towards life–insurance for computing reserves (Thiele) and higher order moments of future payments (Hattendorff). Hence this solution derserves a symbol, called the product integral.

Most texts on multistate Markov models in life insurance deal with the system of differential equations directly instead of collecting them into matrix form and without using their generic solution (the product integral). The use of the product integral, however, allows for a much simplified notion and more transparent derivations.

1. Definition, existence and uniqueness

Definition 1.1: Let $\boldsymbol{A} = \{a_{ij}\}$ be a matrix. Then we define

$$|\boldsymbol{A}| = \{|a_{ij}|\}.$$

If $\boldsymbol{A}^1 = \{a^1_{ij}\}$ and $\boldsymbol{A}^2 = \{a^2_{ij}\}$ are two matrices, then we write

$$\boldsymbol{A}^1 \leq \boldsymbol{A}^2 \iff \forall i, j: \; a^1_{ij} \leq a^2_{ij}.$$

We notice that $|\boldsymbol{D}|$ in the preceeding definition is *not* a matrix norm, but only a convenient notation for applying the absolute value elementwise of a matrix.

We remind the reader about the exponential of a (square) matrix, $\boldsymbol{A}$, which can e.g. be defined in terms of the Taylor expansion

Definition 1.2: The exponential of a square matrix, $\boldsymbol{A}$ is defined by

$$\exp(\boldsymbol{A}) = \sum_{n=0}^{\infty} \frac{\boldsymbol{A}^n}{n!}.$$

This definition is based on simply to plug–in a square matrix into a Taylor expansion for the exponential distribution. This can be done safely (i.e. is well–defined) as the following lemma states.

Lemma 1.3. *For any matrix $\boldsymbol{A}$, the matrix exponential exists and is finite.*

Proof. Let $\boldsymbol{A}$ be an $p \times p$ matrix and let $|\cdot|_p$ denote the Eucledian norm in $\mathbb{R}^p$. Then we use the induced matrix norm

$$\|\boldsymbol{A}\| = \sup_{\boldsymbol{x} \in \mathbb{R}^p} \frac{|\boldsymbol{A}\boldsymbol{x}|_p}{|\boldsymbol{x}|_p} = \sup_{\boldsymbol{x} \in \mathbb{R}^p : |\boldsymbol{x}|_p = 1} |\boldsymbol{A}\boldsymbol{x}|_p.$$

Since

$$\begin{aligned}
\frac{|\boldsymbol{A}\boldsymbol{B}\boldsymbol{x}|_p}{|\boldsymbol{x}|_p} = \frac{|\boldsymbol{A}\boldsymbol{B}\boldsymbol{x}|_p}{|\boldsymbol{B}\boldsymbol{x}|_p} \frac{|\boldsymbol{B}\boldsymbol{x}|_p}{|\boldsymbol{x}|_p} &\leq \left(\sup_{\boldsymbol{x} \in \mathbb{R}^p} \frac{|\boldsymbol{A}\boldsymbol{B}\boldsymbol{x}|}{|\boldsymbol{B}\boldsymbol{x}|} \right) \left(\sup_{\boldsymbol{x} \in \mathbb{R}^p} \frac{|\boldsymbol{B}\boldsymbol{x}|_p}{|\boldsymbol{x}|_p} \right) \\
&\leq \left(\sup_{\boldsymbol{y} \in \mathbb{R}^p} \frac{|\boldsymbol{A}\boldsymbol{y}|_p}{|\boldsymbol{y}|_p} \right) \left(\sup_{\boldsymbol{x} \in \mathbb{R}^p} \frac{|\boldsymbol{B}\boldsymbol{x}|_p}{|\boldsymbol{x}|_p} \right) \\
&= \|\boldsymbol{A}\| \|\boldsymbol{B}\|,
\end{aligned}$$

we have have that

$$\|\boldsymbol{A}\boldsymbol{B}\| \leq \|\boldsymbol{A}\| \|\boldsymbol{B}\|$$

for any matrices $p \times p$ matrices $\boldsymbol{A}$ and $\boldsymbol{B}$.

In particular,

$$\left\| \boldsymbol{A}^n \right\| \leq \|\boldsymbol{A}\|^n.$$

Let $N \in \mathbb{N}$. Then by the triangle inequality (which works for any norm and hence also the matrix norm)

$$\left\| \sum_{n=0}^{N} \frac{\boldsymbol{A}^n}{n!} \right\| \leq \sum_{n=0}^{N} \frac{\|\boldsymbol{A}\|^n}{n!} = \exp(\|\boldsymbol{A}\|) < \infty.$$

This hold for all N, and hence the matrix series converges absolutely. □

We now turn to existence and uniqueness of solutions to (2).

Theorem 1.4 (Uniqueness)**.** *Consider the inhomogeneous system of linear differential equations*

$$\boldsymbol{Y}'(t) = \boldsymbol{Y}(t)\boldsymbol{A}(t), \quad \boldsymbol{Y}(s) = \boldsymbol{C} \tag{3}$$

where $\boldsymbol{Y}(t)$, $\boldsymbol{A}(t)$ and $\boldsymbol{C}$ are $n \times n$ matrices where $\boldsymbol{A}(t)$ is continuous in a certain region $[x_0, x] \ni s, t$. Then (3) has at most one solution.

Proof. Let $\boldsymbol{Y}_1(t)$ and $\boldsymbol{Y}_2(t)$ denote two solutions, and let

$$\boldsymbol{Y}(t) = \boldsymbol{Y}_1(t) - \boldsymbol{Y}_2(t).$$

Then $\boldsymbol{Y}(t)$ is a solution to

$$\boldsymbol{Y}'(t) = \boldsymbol{Y}(t)\boldsymbol{A}(t), \quad \boldsymbol{Y}(s) = \boldsymbol{0},$$

where $\boldsymbol{0}$ denotes the null matrix. Let

$$\boldsymbol{K} = \sup_{x_0 \leq t \leq x} |\boldsymbol{A}(t)| \text{ and } \boldsymbol{G} = \sup_{x_0 \leq t \leq x} |\boldsymbol{Y}(t)|,$$

where the suprema are applied entrywise. Since both $\boldsymbol{A}$ and $\boldsymbol{Y}$ are continuous ($\boldsymbol{Y}$ is even differentiable), the suprema exist. Since $\boldsymbol{Y}(s) = \boldsymbol{0}$,

$$\boldsymbol{Y}(t) = \int_s^t \boldsymbol{Y}(s)\boldsymbol{A}(s)ds$$

so

$$|\boldsymbol{Y}(t)| \leq \int_s^t |\boldsymbol{Y}(u)|\ |\boldsymbol{A}(u)|\,\mathrm{d}u \leq |t-s|\boldsymbol{G}\boldsymbol{K} \leq |x-x_0|\boldsymbol{G}\boldsymbol{K},$$

Then,

$$\begin{aligned} |\boldsymbol{Y}(t)| &\leq \int_s^t |\boldsymbol{Y}(u)|\ |\boldsymbol{A}(u)|\,\mathrm{d}u \\ &\leq \int_s^t (u-s)\boldsymbol{G}\boldsymbol{K}|\boldsymbol{A}(u)|\,\mathrm{d}u \\ &= \frac{1}{2}(t-s)^2\boldsymbol{G}\boldsymbol{K}^2 \\ &\leq \frac{1}{2}(x-x_0)^2\boldsymbol{G}\boldsymbol{K}^2. \end{aligned}$$

Iterating

$$|\boldsymbol{Y}(t)| \leq \frac{1}{k!}(t-s)^k\boldsymbol{G}\boldsymbol{K}^k \leq \frac{1}{k!}(x-x_0)^k\boldsymbol{G}\boldsymbol{K}^k \tag{4}$$

for any $k \in \mathbb{N}$.

For any square matrix we recall that

$$\exp(\boldsymbol{K}(x-x_0)) = \sum_{k=0}^{\infty} \frac{(x-x_0)^k}{k!}\boldsymbol{K}^k,$$

which always exists, which then implies that

$$\frac{(x-x_0)^k}{k!}\boldsymbol{G}\boldsymbol{K}^k \to \boldsymbol{0}$$

as $k \to \infty$. Together with (4), this implies that

$$|\boldsymbol{Y}(t)| \equiv 0.$$

□

Next we turn to existence. Going back to the original $\boldsymbol{Y}(t)$, satisfying

$$\boldsymbol{Y}'(t) = \boldsymbol{Y}(t)\boldsymbol{A}(t), \quad \boldsymbol{Y}(s) = \boldsymbol{C},$$

we get by integration

$$\boldsymbol{Y}(t) = \boldsymbol{C} + \int_s^t \boldsymbol{Y}(u)\boldsymbol{A}(u)\,\mathrm{d}u, \tag{5}$$

which is an equivalent integral equation.

Theorem 1.5 (Existence)**.** *Let*

$$\boldsymbol{Y}_0(t) = \boldsymbol{C}, \quad \boldsymbol{Y}_n(t) = \int_s^t \boldsymbol{Y}_{n-1}(u)\boldsymbol{A}(u)\,\mathrm{d}u.$$

Then the matrix function

$$\boldsymbol{Y}(t) = \sum_{k=0}^{\infty} \boldsymbol{Y}_k(t) \tag{6}$$

converges uniformly and absolutely on finite intervals, and solves the differential equation

$$\boldsymbol{Y}'(t) = \boldsymbol{Y}(t)\boldsymbol{A}(t), \quad \boldsymbol{Y}(s) = \boldsymbol{C}.$$

Proof. First we prove absolute and uniform convergence of the series. Again we take $t \in [x_0, x]$ for some fixed interval. Again we let $\boldsymbol{K} = \sup_{x_0 \le t \le x} |\boldsymbol{A}(t)|$, so that $|\boldsymbol{A}(u)| \le \boldsymbol{K}$ for all $u \in [x_0, x]$. Since $|\boldsymbol{Y}_0(t)| = |\boldsymbol{C}|$, we get that

$$|\boldsymbol{Y}_1(t)| \;=\; \left|\int_s^t \boldsymbol{Y}_0(u)\boldsymbol{A}(u)\,\mathrm{d}u\right| \le \int_s^t |\boldsymbol{Y}_0(u)||\boldsymbol{A}(u)|\,\mathrm{d}u \le |\boldsymbol{C}|\boldsymbol{K}(t-s).$$

In general,

$$|\boldsymbol{Y}_n(t)| \le \frac{1}{n!}(t-s)^n|\boldsymbol{C}|\boldsymbol{K}^n.$$

Since $s, t \in [x_0, x]$, then $t - s \le x - x_0$ so that

$$|\boldsymbol{Y}_n(t)| \le \frac{1}{n!}(x-x_0)^n|\boldsymbol{C}|\boldsymbol{K}^n < \infty$$

as well. But then

$$\sum_{n=0}^{\infty} |\boldsymbol{Y}_n(t)| \leq |\boldsymbol{C}| \sum_{n=0}^{\infty} \frac{(x-x_0)^n \boldsymbol{K}^n}{n!} = |\boldsymbol{C}| \exp(\boldsymbol{K}(x-x_0)) < \infty.$$

Hence the series

$$\boldsymbol{Y}(t) = \sum_{n=0}^{\infty} \boldsymbol{Y}_n(t)$$

converges absolutely and uniformly on $[x_0, x]$. Finally we conclude that

$$\begin{aligned}
\boldsymbol{Y}(t) &= \boldsymbol{Y}_0(t) + \sum_{k=1}^{\infty} \boldsymbol{Y}_k(t) \\
&= \boldsymbol{C} + \sum_{k=1}^{\infty} \int_s^t \boldsymbol{Y}_{k-1}(u) \boldsymbol{A}(u)\, \mathrm{d}u \\
&= \boldsymbol{C} + \int_s^t \sum_{k=1}^{\infty} \boldsymbol{Y}_{k-1}(u) \boldsymbol{A}(u)\, \mathrm{d}u \quad \text{(unif. conv.)} \\
&= \boldsymbol{C} + \int_s^t \boldsymbol{Y}(u) \boldsymbol{A}(u)\, \mathrm{d}u.
\end{aligned}$$

□

Definition 1.6: For a matrix function $\boldsymbol{A}(x)$ we define the product integral

$$\prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x) dx)$$

as the unique solution $\boldsymbol{Y}(t)$ to

$$\boldsymbol{Y}'(t) = \boldsymbol{Y}(t)\boldsymbol{A}(t), \quad \boldsymbol{Y}(s) = \boldsymbol{I} \tag{7}$$

or, equivalently, to

$$\boldsymbol{Y}(t) = \boldsymbol{I} + \int_s^t \boldsymbol{Y}(u)\boldsymbol{A}(u)\, \mathrm{d}u. \tag{8}$$

From the construction (6), we immediately get the alternative representation of the product integral.

Corollary 1.7 (Peano represenation)**.**

$$\begin{aligned}
\prod_s^t (\boldsymbol{I} + \boldsymbol{A}(u) du) &= \sum_{n=0}^{\infty} \int_s^t \int_s^{x_1} \cdots \int_s^{x_{n-1}} \boldsymbol{A}(x_n)\boldsymbol{A}(x_{n-1}) \cdots \boldsymbol{A}(x_1) dx_n\, dx_{n-1} \cdots dx_1 \\
&= \boldsymbol{I} + \int_s^t \boldsymbol{A}(x_1) dx_1 + \sum_{n=2}^{\infty} \int_s^t \int_s^{x_1} \cdots \int_s^{x_{n-1}} \boldsymbol{A}(x_n)\boldsymbol{A}(x_{n-1}) \cdots \boldsymbol{A}(x_1) dx_n\, dx_{n-1} \cdots dx_1
\end{aligned}$$

Corollary 1.8. *The solution to the differential equation*

$$\boldsymbol{Y}'(t) = \boldsymbol{Y}(t)\boldsymbol{A}(t), \quad \boldsymbol{Y}(s) = \boldsymbol{C}$$

or, equivalently, to

$$\boldsymbol{Y}(t) = \boldsymbol{C} + \int_s^t \boldsymbol{Y}(u)\boldsymbol{A}(u)\,\mathrm{d}u,$$

is given by

$$\boldsymbol{Y}(t) = \boldsymbol{C}\prod_s^t(\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x).$$

Proof. By (8),

$$\prod_s^t(\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) = \boldsymbol{I} + \int_s^t \prod_s^u(\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{A}(u)\,\mathrm{d}u$$

from which

$$\begin{aligned}\boldsymbol{Y}(t) &= \boldsymbol{C}\prod_s^t(\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\\ &= \boldsymbol{C} + \int_s^t \boldsymbol{C}\prod_s^u(\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{A}(u)\,\mathrm{d}u\\ &= \boldsymbol{C} + \int_s^t \boldsymbol{Y}(u)\boldsymbol{A}(u)\,\mathrm{d}u.\end{aligned}$$

By uniquess, the result follows. □

2. Properties of the product integral

The product integral is a fundamental solution in the sense that if

$$\boldsymbol{Y}'(t) = \boldsymbol{Y}(t)\boldsymbol{A}(t), \quad \boldsymbol{Y}(s) = \boldsymbol{C}$$

then

$$\boldsymbol{Y}(t) = \boldsymbol{C}\prod_s^t(\boldsymbol{I} + \boldsymbol{A}(x)dx) = \boldsymbol{Y}(s)\prod_s^t(\boldsymbol{I} + \boldsymbol{A}(x)dx).$$

Now take any s, t, u. Then

$$\begin{aligned}\boldsymbol{Y}(t) &= \boldsymbol{Y}(u)\prod_u^t(\boldsymbol{I} + \boldsymbol{A}(x)dx)\\ &= \boldsymbol{Y}(s)\prod_s^u(\boldsymbol{I} + \boldsymbol{A}(x)dx)\prod_u^t(\boldsymbol{I} + \boldsymbol{A}(x)dx)\end{aligned}$$

Now if the matrix function $t \to \boldsymbol{Y}(t)$ is such that $\boldsymbol{Y}(s) = \boldsymbol{I}$, then $\boldsymbol{Y}(t)$ on the left hand side simply equals the product integral from s to t. Hence we conclude the following important result.

Theorem 1.9. *For any s,u,t we have that*

$$\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)dx)=\prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x)dx)\prod_u^t(\boldsymbol{I}+\boldsymbol{A}(x)dx). \tag{9}$$

Proof. □

Since s,t,u are arbitrary, taking $s=t$ we then obtain the following.

Corollary 1.10. *The product integral is an invertible matrix with*

$$\left[\prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\right]^{-1}=\prod_u^s(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)$$

Lemma 1.11 (Commutative property)**.** *If $\boldsymbol{A}(u)\boldsymbol{C}=\boldsymbol{C}\boldsymbol{A}(u)$ for all $u\in[s,t]$, then*

$$\boldsymbol{C}\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(u)\,\mathrm{d}u)=\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(u)\,\mathrm{d}u)\boldsymbol{C}.$$

Proof. Consider

$$\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)=\boldsymbol{I}+\int_s^t\prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{A}(u)\,\mathrm{d}u.$$

Multiply from the right by $\boldsymbol{C}$ to get

$$\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{C}=\boldsymbol{C}+\int_s^t\prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{A}(u)\boldsymbol{C}\,\mathrm{d}u.$$

Then use then commutativity

$$\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{C}=\boldsymbol{C}+\int_s^t\prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{C}\boldsymbol{A}(u)\,\mathrm{d}u.$$

But then by Corollary 1.8, $\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{C}$ must be equal to $\boldsymbol{C}\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)$.

□

Theorem 1.12. *If $\boldsymbol{A}(u)\boldsymbol{B}(v)=\boldsymbol{B}(v)\boldsymbol{A}(u)$ for all $u,v\in[s,t]$, then*

$$\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)dx)\prod_s^t(\boldsymbol{I}+\boldsymbol{B}(x)dx)=\prod_s^t(\boldsymbol{I}+(\boldsymbol{A}(x)+\boldsymbol{B}(x))dx).$$

Proof. Define

$$\boldsymbol{F}(s,t)=\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)dx)\prod_s^t(\boldsymbol{I}+\boldsymbol{B}(x)dx).$$

By Lemma 1.11,

$$\begin{aligned}
\frac{\partial}{\partial t}\boldsymbol{F}(s,t) &= \prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)dx)\boldsymbol{A}(t)\prod_s^t(\boldsymbol{I}+\boldsymbol{B}(x)dx) \\
&\quad +\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)dx)\prod_s^t(\boldsymbol{I}+\boldsymbol{B}(x)dx)\boldsymbol{B}(t) \\
&= \prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)dx)\prod_s^t(\boldsymbol{I}+\boldsymbol{B}(x)dx)\boldsymbol{A}(t) \\
&\quad +\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)dx)\prod_s^t(\boldsymbol{I}+\boldsymbol{B}(x)dx)\boldsymbol{B}(t) \\
&= \boldsymbol{F}(s,t)(\boldsymbol{A}(t)+\boldsymbol{B}(t))
\end{aligned}$$

Since $\boldsymbol{F}(s,s)=\boldsymbol{I}$, the result follows. □

The matrix exponential is a special case of the product integral in the following sense.

Theorem 1.13. *If $\boldsymbol{A}(x)$ all commute, i.e. $\boldsymbol{A}(u)\boldsymbol{A}(v)=\boldsymbol{A}(v)\boldsymbol{A}(u)$ for all $u,v\in[s,t]$, then*

$$\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)dx)=\exp\left(\int_s^t\boldsymbol{A}(u)du\right).$$

In particular, if $\boldsymbol{A}(x)=\boldsymbol{A}\lambda(x)$ for some constant matrix $\boldsymbol{A}$ and function $\lambda(x)$, then

$$\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)dx)=\exp\left(\boldsymbol{A}\int_s^t\lambda(u)du\right)$$

Proof. Our candidate equals

$$\exp\left(\int_s^t\boldsymbol{A}(u)du\right)=\sum_{k=0}^{\infty}\frac{\left(\int_s^t\boldsymbol{A}(u)du\right)^k}{k!}.$$

Differentiation of a product of k matrix functions yields,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{A}_1(t)\boldsymbol{A}_2(t)\cdots\boldsymbol{A}_k(t)) &= \boldsymbol{A}_1'(t)\boldsymbol{A}_2(t)\cdots\boldsymbol{A}_k(t) \\
&\quad +\boldsymbol{A}_1(t)\boldsymbol{A}_2'(t)\cdots\boldsymbol{A}_k(t) \\
&\quad \vdots \\
&\quad +\boldsymbol{A}_1(t)\boldsymbol{A}_2(t)\cdots\boldsymbol{A}_k'(t).
\end{aligned}$$

Now let

$$\boldsymbol{A}_i(t)=\int_s^t\boldsymbol{A}(u)\,\mathrm{d}u$$

for all i. Then $\boldsymbol{A}_i'(t) = \boldsymbol{A}(t)$ for all i, and since $\boldsymbol{A}(u)\boldsymbol{A}(v) = \boldsymbol{A}(v)\boldsymbol{A}(u)$ for all $u, v \in [s,t]$, we get that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_s^t \boldsymbol{A}(u)du\right)^k = k\left(\int_s^t \boldsymbol{A}(u)du\right)^{k-1}\boldsymbol{A}(t).$$

But then

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\exp\left(\int_s^t \boldsymbol{A}(u)du\right) &= \sum_{k=0}^{\infty}\frac{1}{k!}\frac{d}{dt}\left(\int_s^t \boldsymbol{A}(u)du\right)^k \\ &= \sum_{k=1}^{\infty}\frac{1}{k!}k\left(\int_s^t \boldsymbol{A}(u)du\right)^{k-1}\boldsymbol{A}(t) \\ &= \exp\left(\int_s^t \boldsymbol{A}(u)du\right)\boldsymbol{A}(t).\end{aligned}$$

Together with

$$\exp\left(\int_s^t \boldsymbol{A}(u)du\right) = \boldsymbol{I}$$

for $s = t$, the theorem has then been proved. □

Corollary 1.14. *If $\boldsymbol{A}(x) = \boldsymbol{A}$ constant, then*

$$\prod_s^t(\boldsymbol{I} + \boldsymbol{A}dx) = e^{\boldsymbol{A}(t-s)}.$$

Proof. Follows from the above with $\lambda(x) \equiv 1$. □

Corollary 1.15. *Let $r(x)$ be a real and bounded function. Then*

$$\exp\left(-\int_s^t r(x)dx\right)\prod_s^t(\boldsymbol{I} + \boldsymbol{A}(x)dx) = \prod_s^t(\boldsymbol{I} + (\boldsymbol{A}(x) - r(x)\boldsymbol{I})dx).$$

Proof. Since the matrix exponential of a diagonal matrix equals the diagonal matrix containing the exponentials of the diagonal elements (check), it is clear that

$$\begin{aligned}\exp\left(-\int_s^t r(x)dx\right)\boldsymbol{I} &= \exp\left(-\int_s^t r(x)\boldsymbol{I}dx\right) \\ &\overset{\text{Th.1.13}}{=} \prod_s^t(\boldsymbol{I} - r(x)\boldsymbol{I}\,\mathrm{d}x).\end{aligned}$$

But then

$$\exp\left(-\int_s^t r(x)dx\right)\prod_s^t(\boldsymbol{I} + \boldsymbol{A}(x)dx) = \exp\left(-\int_s^t r(x)dx\right)\cdot\boldsymbol{I}\cdot\prod_s^t(\boldsymbol{I} + \boldsymbol{A}(x)dx)$$

$$
\begin{aligned}
&= \prod_s^t (\boldsymbol{I} - r(x)\boldsymbol{I}\,\mathrm{d}x) \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)dx) \\
&\overset{\text{Th.1.12}}{=} \prod_s^t (\boldsymbol{I} + (\boldsymbol{A}(x) - r(x)\boldsymbol{I})dx)
\end{aligned}
$$

since the matrices $r(x)\boldsymbol{I}$ commute with any matrix. □

Theorem 1.16. *The product integral satisfies*

$$\frac{\partial}{\partial s}\prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)dx) = -\boldsymbol{A}(s)\prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)dx).$$

On the other hand, the solution to the system of differential equations

$$\frac{\partial}{\partial s}\boldsymbol{X}(s) = -\boldsymbol{A}(s)\boldsymbol{X}(s)$$

with initial condition $\boldsymbol{X}(t) = \boldsymbol{I}$ *is exactly the function*

$$s \to \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)dx).$$

Proof. For any differentiable matrix function $\boldsymbol{M}(x)$ we have that

$$\frac{d}{dx}\boldsymbol{M}^{-1}(x) = -\boldsymbol{M}^{-1}(x)\boldsymbol{M}'(x)\boldsymbol{M}^{-1}(x).$$

This follows from

$$\boldsymbol{0} = \frac{d}{dx}\boldsymbol{I} = \frac{d}{dx}\left(\boldsymbol{M}(x)\boldsymbol{M}^{-1}(x)\right) = \boldsymbol{M}'(x)\boldsymbol{M}^{-1}(x) + \boldsymbol{M}(x)\frac{d}{dx}\boldsymbol{M}^{-1}(x)$$

and rearranging. Then

$$
\begin{aligned}
\frac{d}{ds}\prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) &= \frac{d}{ds}\left[\prod_t^s (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\right]^{-1} \\
&= -\left[\prod_t^s (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\right]^{-1}\left[\frac{d}{ds}\prod_t^s (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\right]\left[\prod_t^s (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\right]^{-1} \\
&= -\left[\prod_t^s (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\right]^{-1}\prod_t^s (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{A}(s)\left[\prod_t^s (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\right]^{-1} \\
&= -\boldsymbol{A}(s)\left[\prod_t^s (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\right]^{-1} \\
&= -\boldsymbol{A}(s)\prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)
\end{aligned}
$$

□

The next result is the corner stone for computing reserves and higher order moments in multi–state life insurance models.

Theorem 1.17 (Van Loan)**.** *Let $\boldsymbol{A}(x)$, $\boldsymbol{B}(x)$ and $\boldsymbol{C}(x)$ be continuous matrix functions. Then we have that*

$$\prod_s^t \left(\boldsymbol{I} + \begin{pmatrix} \boldsymbol{A}(x) & \boldsymbol{B}(x) \\ \boldsymbol{0} & \boldsymbol{C}(x) \end{pmatrix} \mathrm{d}x \right) = \begin{pmatrix} \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) & \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u) \prod_u^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u \\ \boldsymbol{0} & \prod_s^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x) \end{pmatrix}$$

Proof. By Leibniz' integration rule,

$$\begin{aligned} &\frac{\partial}{\partial t} \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u) \prod_u^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u \\ &= \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u) \frac{\partial}{\partial t} \prod_u^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u + \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(u))\boldsymbol{B}(t) \prod_t^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x) \\ &= \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u) \prod_u^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u\ \boldsymbol{C}(t) + \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(u))\boldsymbol{B}(t). \end{aligned}$$

Let

$$\boldsymbol{B}(s,t) = \begin{pmatrix} \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) & \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u) \prod_u^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u \\ \boldsymbol{0} & \prod_s^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x) \end{pmatrix}$$

denote the proposal. Then

$$\begin{aligned} \frac{\partial}{\partial t}\boldsymbol{B}(s,t) &= \begin{pmatrix} \frac{\partial}{\partial t} \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) & \frac{\partial}{\partial t} \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u) \prod_u^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u \\ \boldsymbol{0} & \frac{\partial}{\partial t} \prod_s^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x) \end{pmatrix} \\ &= \begin{pmatrix} \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) & \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u) \prod_u^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u \\ \boldsymbol{0} & \prod_s^t (\boldsymbol{I} + \boldsymbol{C}(x)\,\mathrm{d}x) \end{pmatrix} \begin{pmatrix} \boldsymbol{A}(t) & \boldsymbol{B}(t) \\ \boldsymbol{0} & \boldsymbol{C}(t) \end{pmatrix} \\ &= \boldsymbol{B}(s,t) \begin{pmatrix} \boldsymbol{A}(t) & \boldsymbol{B}(t) \\ \boldsymbol{0} & \boldsymbol{C}(t) \end{pmatrix}. \end{aligned}$$

Also, $\boldsymbol{B}(s,t) = \boldsymbol{I}$ for $s = t$. Therefore,

$$\boldsymbol{B}(s,t) = \prod_s^t \left(\boldsymbol{I} + \begin{pmatrix} \boldsymbol{A}(u) & \boldsymbol{B}(u) \\ \boldsymbol{0} & \boldsymbol{C}(u) \end{pmatrix} \mathrm{d}u \right).$$

□

Remark 1.18: It is enough to assume continuity of $\boldsymbol{B}(x)$ and $\boldsymbol{C}(x)$ only in the proof above. Also, if we carry out the proof of Lemma 1.17 differentiating w.r.t. s instead of t, for which we may assume continuity of $\boldsymbol{A}(x)$ and $\boldsymbol{B}(x)$ only instead.

Remark 1.19: Lemma 1.17 also holds for functions with a finite number of discontinuities, which can be seen be decomposing the product integrals into product over intervals where the function is continuous and using (9). Indeed, assume that $\boldsymbol{C}(x)$ is continuous in $[s,t]$ apart from $t_0 \in (s,t)$. Then

$$\prod_s^{t_0}\left(\boldsymbol{I}+\begin{pmatrix}\boldsymbol{A}(x) & \boldsymbol{B}(x)\\ \boldsymbol{0} & \boldsymbol{C}(x)\end{pmatrix}\mathrm{d}x\right) = \begin{pmatrix}\prod_s^{t_0}(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x) & \int_s^{t_0}\prod_s^{u}(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u)\prod_u^{t_0}(\boldsymbol{I}+\boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u\\ \boldsymbol{0} & \prod_s^{t_0}(\boldsymbol{I}+\boldsymbol{C}(x)\,\mathrm{d}x)\end{pmatrix}$$

$$\prod_{t_0}^{t}\left(\boldsymbol{I}+\begin{pmatrix}\boldsymbol{A}(x) & \boldsymbol{B}(x)\\ \boldsymbol{0} & \boldsymbol{C}(x)\end{pmatrix}\mathrm{d}x\right) = \begin{pmatrix}\prod_{t_0}^{t}(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x) & \int_{t_0}^{t}\prod_{t_0}^{u}(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u)\prod_u^{t}(\boldsymbol{I}+\boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u\\ \boldsymbol{0} & \prod_{t_0}^{t}(\boldsymbol{I}+\boldsymbol{C}(x)\,\mathrm{d}x)\end{pmatrix}$$

Then using (9),

$$\prod_s^{t}\left(\boldsymbol{I}+\begin{pmatrix}\boldsymbol{A}(x) & \boldsymbol{B}(x)\\ \boldsymbol{0} & \boldsymbol{C}(x)\end{pmatrix}\mathrm{d}x\right) = \prod_s^{t_0}\left(\boldsymbol{I}+\begin{pmatrix}\boldsymbol{A}(x) & \boldsymbol{B}(x)\\ \boldsymbol{0} & \boldsymbol{C}(x)\end{pmatrix}\mathrm{d}x\right)\prod_{t_0}^{t}\left(\boldsymbol{I}+\begin{pmatrix}\boldsymbol{A}(x) & \boldsymbol{B}(x)\\ \boldsymbol{0} & \boldsymbol{C}(x)\end{pmatrix}\mathrm{d}x\right)$$

and ditto on the right hand side expressions, we get for example that the upper right block amounts to

$$\begin{aligned}&\prod_s^{t_0}(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\int_{t_0}^{t}\prod_{t_0}^{u}(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u)\prod_u^{t}(\boldsymbol{I}+\boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u\\ &\quad+\int_s^{t_0}\prod_s^{u}(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u)\prod_u^{t_0}(\boldsymbol{I}+\boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u\prod_{t_0}^{t}(\boldsymbol{I}+\boldsymbol{C}(x)\,\mathrm{d}x\\ &\quad=\int_s^{t}\prod_s^{u}(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{B}(u)\prod_u^{t}(\boldsymbol{I}+\boldsymbol{C}(x)\,\mathrm{d}x)\,\mathrm{d}u.\end{aligned}$$

We next show Leibniz' rule for product integrals. First a lemma.

Lemma 1.20. *If the integrand matrix function $\boldsymbol{A}(x,z)$ depends on an external variable $z \in [c,d]$, such that*

$$\frac{\partial}{\partial z}\boldsymbol{A}(x,z)$$

is bounded over $[s,t]\times[c,d]$, then

$$\frac{\partial}{\partial z}\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x) = \int_s^t \prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\frac{\partial}{\partial z}\boldsymbol{A}(u,z)\prod_u^t(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\mathrm{d}u.$$

Proof. Let

$$\boldsymbol{F}(s,t,z) = \prod_s^t\left(\boldsymbol{I}+\boldsymbol{A}(u,z)\,\mathrm{d}u\right).$$

Then $\boldsymbol{F}(s,t,z)$ satisfies

$$\frac{\partial}{\partial t}\boldsymbol{F}(s,t,z) = \boldsymbol{F}(s,t,z)\boldsymbol{A}(t,z), \quad \boldsymbol{F}(s,s,z) = \boldsymbol{I}.$$

Thus $\frac{\partial}{\partial z}\boldsymbol{F}(s,t,z)$ satisfies

$$\begin{aligned}\frac{\partial}{\partial t}\frac{\partial}{\partial z}\boldsymbol{F}(s,t,z) &= \frac{\partial}{\partial z}\left[\boldsymbol{F}(s,t,z)\boldsymbol{A}(t,z)\right]\\ &= \frac{\partial}{\partial z}\boldsymbol{F}(s,t,z)\boldsymbol{A}(t,z) + \boldsymbol{F}(s,t,z)\frac{\partial}{\partial z}\boldsymbol{A}(t,z)\\ &= \frac{\partial}{\partial z}\boldsymbol{F}(s,t,z)\boldsymbol{A}(t,z) + \prod_s^t\left(\boldsymbol{I}+\boldsymbol{A}(u,z)\,\mathrm{d}u\right)\frac{\partial}{\partial z}\boldsymbol{A}(t,z),\end{aligned}$$

with $\frac{\partial}{\partial z}\boldsymbol{F}(s,s,z) = \boldsymbol{0}$, the latter following from $z \to \boldsymbol{F}(s,s,z) = \boldsymbol{I}$ being constant in z.

To show that

$$\frac{\partial}{\partial z}\boldsymbol{F}(s,t,z) = \int_s^t \prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\frac{\partial}{\partial z}\boldsymbol{A}(u,z)\prod_u^t(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\mathrm{d}u$$

we only need to show that the r.h.s. proposal satisfies the above differential equation. But this follows directly from Leibniz' rule for ordinary integrals since

$$\begin{aligned}&\frac{\partial}{\partial t}\left[\int_s^t \prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\frac{\partial}{\partial z}\boldsymbol{A}(u,z)\prod_u^t(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\mathrm{d}u\right]\\ &\quad= \int_s^t \prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\frac{\partial}{\partial z}\boldsymbol{A}(u,z)\frac{\partial}{\partial t}\prod_u^t(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\mathrm{d}u\\ &\qquad+ \prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\frac{\partial}{\partial z}\boldsymbol{A}(u,z)\prod_u^t(\boldsymbol{I}+\boldsymbol{A}(x,z)\Big|_{u=t}\\ &\quad= \left[\int_s^t \prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\frac{\partial}{\partial z}\boldsymbol{A}(u,z)\prod_u^t(\boldsymbol{I}+\boldsymbol{A}(x,z)\mathrm{d}x)\mathrm{d}u\right]\boldsymbol{A}(t,z)\end{aligned}$$

$$+\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(u,z)\,\mathrm{d}u)\frac{\partial}{\partial z}\boldsymbol{A}(t,z).$$

Hence the proposal in the square bracket indeed satisfies the correct differential equation and initial condition. □

Theorem 1.21 (Leibniz' rule)**.** *Assume that* $\boldsymbol{A}(x,z)$ *is as in Lemma 1.20. Then*

$$\frac{\partial}{\partial s}\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x,s)\,\mathrm{d}x)=-\boldsymbol{A}(s,s)\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x,s)\,\mathrm{d}x)$$
$$+\int_s^t\prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x,s)\mathrm{d}x)\frac{\partial}{\partial s}\boldsymbol{A}(u,s)\prod_u^t(\boldsymbol{I}+\boldsymbol{A}(x,s)\mathrm{d}x)\mathrm{d}u$$

Proof. Define

$$\boldsymbol{F}(a,s)=\prod_a^t(\boldsymbol{I}+\boldsymbol{A}(x,s)\,\mathrm{d}x)$$
$$\boldsymbol{g}(s)=(g_1(s),g_2(s))=(s,s).$$

Then

$$\boldsymbol{H}(s)=\boldsymbol{F}(\boldsymbol{g}(s))=\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x,s)\,\mathrm{d}x).$$

Let $h^{ij}(s)$ and $f^{ij}(a,s)$ denote the ijth element of $\boldsymbol{H}(s)$ and $\boldsymbol{F}(a,s)$, respectively. Then

$$h^{ij}(s)=f^{ij}(\boldsymbol{g}(s)).$$

Taking derivative,

$$\frac{\mathrm{d}}{\mathrm{d}s}h^{ij}(s)=f_a^{ij}(\boldsymbol{g}(s))g_1'(s)+f_s^{ij}(\boldsymbol{g}(s))g_2'(s)$$
$$=f_a^{ij}(s,s)+f_s^{ij}(s,s),$$

so that

$$\frac{\mathrm{d}}{\mathrm{d}s}\boldsymbol{H}(s)=\boldsymbol{F}_a(s,s)+\boldsymbol{F}_s(s,s).$$

Thus

$$\frac{\mathrm{d}}{\mathrm{d}s}\boldsymbol{H}(s)=-\boldsymbol{A}(s,s)\prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x,s)\,\mathrm{d}x)$$
$$+\int_s^t\prod_s^u(\boldsymbol{I}+\boldsymbol{A}(x,s)\mathrm{d}x)\frac{\partial}{\partial s}\boldsymbol{A}(u,s)\prod_u^t(\boldsymbol{I}+\boldsymbol{A}(x,s)\mathrm{d}x)\mathrm{d}u$$

□

3. Numerical computation of product integrals

Computing product integrals numerically does usually not cause in problems. We may identify the product integral of a matrix function $\boldsymbol{A}(x)$,

$$\boldsymbol{Y}(t) = \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x),$$

as the solution to (3).

If we let $x_0 = s < x_1 < x_2 < \cdots < x_N = t$ with

$$x_{j+1} = x_j + \Delta, \quad \Delta = \frac{t-s}{N}$$

then we could approximate

$$\boldsymbol{Y}'(x) \approx \frac{1}{\Delta}\left(\boldsymbol{Y}(x+\Delta) - \boldsymbol{Y}(x)\right)$$

or

$$\boldsymbol{Y}(x+\Delta) \approx \boldsymbol{Y}(x) + \Delta \boldsymbol{Y}'(x).$$

But

$$\boldsymbol{Y}'(x) = \boldsymbol{Y}(x)\boldsymbol{A}(x)$$

so

$$\boldsymbol{Y}(x+\Delta) \approx \boldsymbol{Y}(x) + \boldsymbol{Y}(x)\boldsymbol{A}(x)\Delta = \boldsymbol{Y}(x)(\boldsymbol{I} + \boldsymbol{A}(x)\Delta).$$

This formula also explains the idea behind the notation of the product integral.

By iterating the above formula, starting with $\boldsymbol{Y}(s) = \boldsymbol{Y}(x_0) = \boldsymbol{I}$, we can then obtain an approximation for $\boldsymbol{Y}(t)$. This way of solving differential equations numerically is referred to as Eulers method. The Euler method is not very precise, and requires a very fine grid.

In turn, the following algorithm written in R, is a so–called fourth order Runge–Kutta method, and computes the solution to the system of differential equations much more precisely than Eulers method:

```
prodint<-function(A,s,t,n){
  x0 <- s
  y0 <- diag(nrow(A(s)))
  h<-(t-s)/n
  for(i in 1:n) {
    s1 <- h * y0%*%A(x0)
    s2 <- h * (y0 +s1/2)%*%A(x0 + h/2)
    s3 <- h * (y0 +s2/2)%*%A(x0 + h/2)
    s4 <- h * (y0 + s3)%*%A(x0 + h)
    y0 <- y0 + s1/6 + s2/3 + s3/3+ s4/6
    x0 <- x0 + h
  }
  return(y0)
}
```

The input is a matrix $\boldsymbol{A}$, integration limits s,t and number of partitions n. Output is an approximation to the product integral of $\boldsymbol{A}(x)$ from s to t.

CHAPTER 2

Markov jump processes

1. Markov jump processes

In this chapter we consider Markov processes in continuous time that take values in a discrete (finite or at most countable) state space. By nature, such processes are piecewise constant, and transitions occur via jumps. They are often referred to as Markov jump processes or continuous-time Markov chains.

The value of the process at a jump time must be defined. It can either be the value just before the jump or the value just after. The most common convention is to assign it the value after the jump. This assumption makes Markov jump processes continuous from the right (and with limits from the left), i.e., they are so-called càdlàg processes (Continue à droite, limite à gauche).

Definition 2.1 (Markov property)**:** A continuous-time stochastic process $\{X_t\}_{t\geq 0}$ taking values in a countable set E is called a Markov jump process with state space E if for all $t > s > t_n > \cdots > t_1 > 0$ and $j, i, i_n, \ldots, i_0 \in E$, we have that

$$\mathbb{P}(X_t = j \mid X_s = i, X_{t_n} = i_n, \ldots, X_{t_1} = i_1, X_0 = i_0) = \mathbb{P}(X_t = j \mid X_s = i). \tag{10}$$

In the above definition we may think of s as the present, t a time in the future and $t_0, ..., t_n$ as times in the history. Hence the Markov property states that the distribution of the future given the present and the past only depends on the present. Or quivalently, the future and the past are conditionally independent given the present. The latter follows from

$$\begin{aligned}
\mathbb{P}(X_t = j, X_{t_n} = i_n, ..., X_{t_0} = i_0 | X_s = i) &= \frac{\mathbb{P}(X_t = j, X_s = i, X_{t_n} = i_n, ..., X_{t_0} = i_0)}{\mathbb{P}(X_s = i)} \\
&= \frac{\mathbb{P}(X_t = j | X_s = i, X_{t_n} = i_n, ..., X_{t_0} = i_0)\mathbb{P}(X_s = i, X_{t_n} = i_n, ..., X_{t_0} = i_0)}{\mathbb{P}(X_s = i)} \\
&\overset{(10)}{=} \mathbb{P}(X_t = j | X_s = i)\frac{\mathbb{P}(X_s = i, X_{t_n} = i_n, ..., X_{t_0} = i_0)}{\mathbb{P}(X_s = i)} \\
&= \mathbb{P}(X_t = j | X_s = i)\mathbb{P}(X_{t_n} = i_n, ..., X_{t_0} = i_0 | X_s = i).
\end{aligned}$$

An example of a *sample path* (or realisation), $t \to X(t, \omega)$ for some $\omega \in \Omega$ in the sample space, is given in Figure 1 as the blue lines. Also the càdlàg is clearly indicated by empty and solid circles at the times of transition.

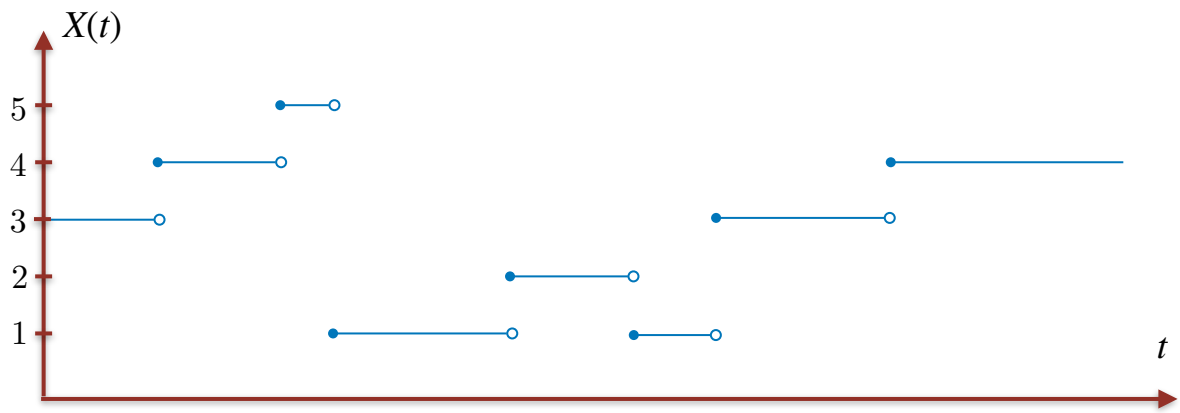

Figure 1. The blue lines show a sample path of a Markov jump process $X(t)$ with 5 states. The value of the process at the exact moment of a jump is indicated by a solid circle (hence right–continuous).

For $s \leq t$ and $i, j \in E$, we refer to

$$p_{ij}(s,t) = \mathbb{P}(X_t = j \mid X_s = i) \tag{11}$$

as the *transition probabilty* of the process for going from state i to state j during $[s,t]$. If $i = j$ the process may transition from i to i both with or without any jumps occuring during $[s,t]$. T

The process is called time-homogeneous if the transition probabilities $p_{ij}(s,t)$ only depend on $t-s$, in which case we write $p_{ij}(h) = \mathbb{P}(X_{t+h} = j \mid X_t = i)$, referred to as the h–step transition probabilities.

The transition probabilities are arranged in a transition matrix,

$$\boldsymbol{P}(s,t) = \{p_{ij}(s,t)\}_{i,j \in E}, \quad s \leq t,$$

or, in the time–homogeneous case,

$$\boldsymbol{P}(h) = \{p_{ij}(h)\}_{i,j \in E}, \quad h \geq 0.$$

The so–called finite–dimensional distributions of a Markov process are fully determined by the its initial distribution and the transition probabilities.

Theorem 2.2 (finite–dimensional distributions)**.** *Let $\{X_t\}_{t \geq 0}$ be a Markov process with initial distribution $\pi_i = \mathbb{P}(X_0 = i)$ and transition probabilities $p_{ij}(s,t)$. Then for any set of time points $t_1 < t_2 < \cdots < t_n$ and states $i_0, i_1, ..., i_n$, we have that*

$$\mathbb{P}(X_0 = i_0, X_{t_1} = i_1, ..., X_{t_n} = i_n) = \pi_{i_0} p_{i_0 i_1}(0,t_1) p_{i_1 i_2}(t_1,t_2) \cdots p_{i_{n-1} i_n}(t_{n-1,t_n}).$$

Proof. Follows by repeated use of the Markov property. □

Definition 2.3: When conditioning on X_0 it is common to use the notation

$$\mathbb{P}_i(A) = \mathbb{P}(A | X_0 = i).$$

Hence Theorem 2.2 can be rephrased as

$$\mathbb{P}_i(x_{t_1} = i_1, ..., X_{t_n} = i_n) = p_{i,i_1}(0,t_1)p_{i_1 i_2}(t_1,t_2)\cdots p_{i_{n-1}i_n}(t_{n-1,t_n}),$$

at least when $\pi_i > 0$. If we condition on $X_{t_k} = i_k$, then by the Markov property

$$\begin{aligned}\mathbb{P}_i(x_{t_1} = i_1, ..., X_{t_n} = i_n | X_{t_k} = i_k) &= \mathbb{P}_i(x_{t_{k+1}} = i_{k+1}, ..., X_{t_n} = i_n | X_{t_k} = i_k) \\ &= p_{i_k i_{k+1}}(t_k, t_{k+1})\cdots p_{i_{n-1}i_n}(t_{n-1,t_n}).\end{aligned}$$

But the last expression is just finite–dimensional distribution of a Markov process starting at t_k in state i_k, i.e. of the shifted process $X(t-t_k)$ conditionally on it starts at i_k. Hence, at any time s, conditionally on $X(s) = i$, say, the development of $X(t)$ from that point onwards is simply a new Markov process, $Y(s) = X(t-s)$, starting at i and with transition probabilities $\tilde{p}_{jk}(u,v) = p_{jk}(u+s, v+s)$. This is illustrated in Figure 2.

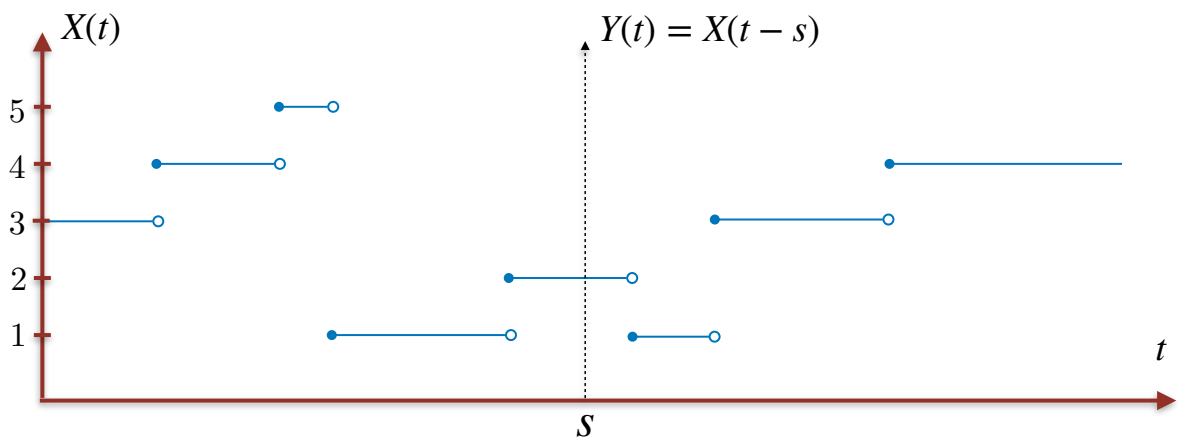

Figure 2. Conditioning on the state of $X(s)$, the continuation of the process beyond s is the same as starting a new Markov process at time s with shifted transition probabilities.

While transition probabilities can be thought of as the probabilties of a future state given a current state over an arbitrary period of time, we next look into what is the probability of a jump in the process at a given moment of time, i.e. the probability of an instantaneous transition.

Assumption 2.4: For any $s \geq 0$, we assume the existence of the limits

$$\boldsymbol{\Lambda}(s) = \{\lambda_{ij}(s)\}_{i,j\in E} = \lim_{h\downarrow 0} \frac{\boldsymbol{P}(s,s+h) - \boldsymbol{I}}{h}, \tag{12}$$

and that all $s \to \lambda_{ij}(s)$ are continuous. The matrices $\boldsymbol{\Lambda}(s)$ are called intensity matrices. In the special case of time–homogenous processes, $\boldsymbol{\Lambda}(s) = \boldsymbol{\Lambda}$ is a constant matrix.

An intensity matrix has an infinitesimal interpretation as

$$\boldsymbol{P}(s,s+h) = \boldsymbol{I} + h\boldsymbol{\Lambda}(s) + o(h),$$

so that

$$p_{ij}(s,s+h) \quad = \quad \delta_{ij}+\lambda_{ij}(s)h+o(h) \tag{13}$$

for all $i,j \in E$ and where $\delta_{ij} = 1\{i=j\}$ is the indicator for whether $i=j$.

The elements $\lambda_{ij}(s)$ of the matrix $\mathbf{\Lambda}(s)$ then satisfies

$$\lambda_{ij}(s) \geq 0,\ i \neq j, \quad \lambda_{ii}(s) \leq 0 \ \text{ and } \ \lambda_{ii}(s) = -\sum_{j \neq i} \lambda_{ij}(s), \tag{14}$$

which follows from

$$\mathbf{\Lambda}(s)\boldsymbol{e} = \lim_{h \downarrow 0} \frac{\boldsymbol{P}(s,s+h)\boldsymbol{e}-\boldsymbol{e}}{h} = \mathbf{0},$$

or, equivalently, summing over all j on both sides in (13).

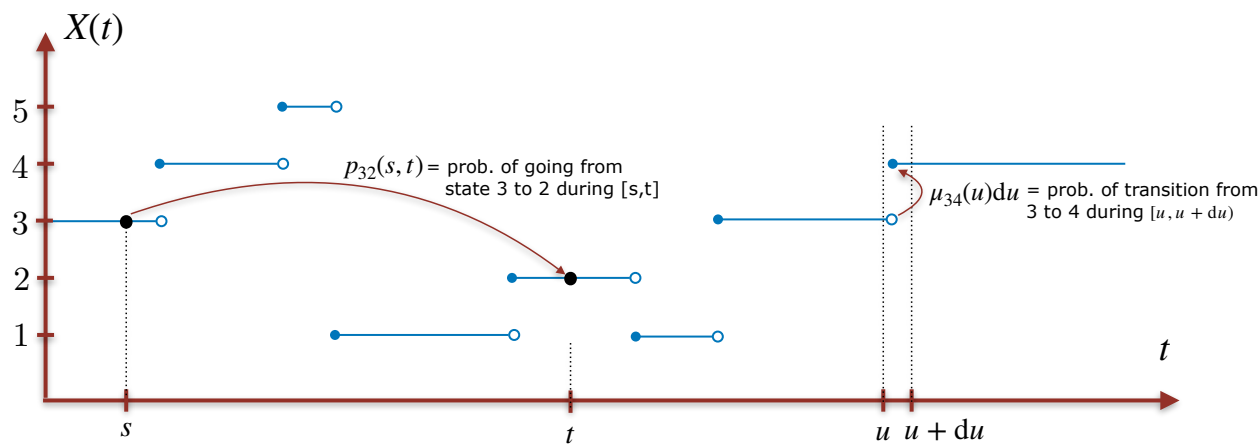

Figure 3. The blue lines show a trajectory of a time–inhomogeneous Markov jump process $X(t)$ with 5 states. The value of the process at the exact moment of a jump is indicated by a solid circle (hence right–continuous). Highlighted are examples of transitions over longer times as opposed to instantaneous transitions.

Let $\lambda_i(s) = -\lambda_{ii}(s)$. Then from (13), we conclude that

$$\lambda_{ij}(t)\,\mathrm{d}t = \text{probability of a jump from state } i \text{ to } j \text{ during } [t,t+\mathrm{d}t)$$

for $i \neq j$, and

$$1+\lambda_{ii}(t)\,\mathrm{d}t = 1-\lambda_i(t)\,\mathrm{d}t = \text{probability of no jumps during } [t,t+\mathrm{d}t).$$

Hence, the probability of a jump out of state i during $[t,t+dt)$ is

$$\lambda_i(t)dt = -\lambda_{ii}(t)dt.$$

Conditionally on there is a jump during $[t,t+dt)$, the probability $q_{ij}(t)$ that the process will jump to state j is therefore

$$q_{ij}(t) = \lim_{h \downarrow 0} \frac{\lambda_{ij}(t)\,\mathrm{d}t}{\lambda_i(t)\,\mathrm{d}t}$$

$$
\begin{aligned}
&= \lim_{h\downarrow 0} \frac{\lambda_{ij}(t)+o(h)/h}{\lambda_i(t)h+o(h)} \\
&= \lim_{h\downarrow 0} \frac{\lambda_{ij}(t)+o(h)/h}{\lambda_i(t)h+o(h)} \\
&= \frac{\lambda_{ij}(t)}{\lambda_i(t)}. \qquad (15)
\end{aligned}
$$

The number $\lambda_i(t)$, referred to as a rate or intensity, is in fact a hazard rate.

The sample path properties for the Markov jump process are then as follows. Starting at state i, the time until the first jump out of it, T_1 say, is then the same as the waiting time for the first arrival in a Poisson process with rates $\lambda_i(t)$. Its density is given by (**??**),

$$f_{T_1}(s) = \mathrm{e}^{-\int_0^s \lambda_i(x)\,\mathrm{d}x}\lambda_i(s).$$

The process then jumps to a state, j, with probability $q_{ij}(s)$. Once in state j at time s, the process then starts over again a Markov jump process with intensities $\lambda_{ij}(s+\cdot)$. This follows from the strong Markov property, which is dealt in the following section, since times of jump are stopping times. This is illustrated in Figure 4.

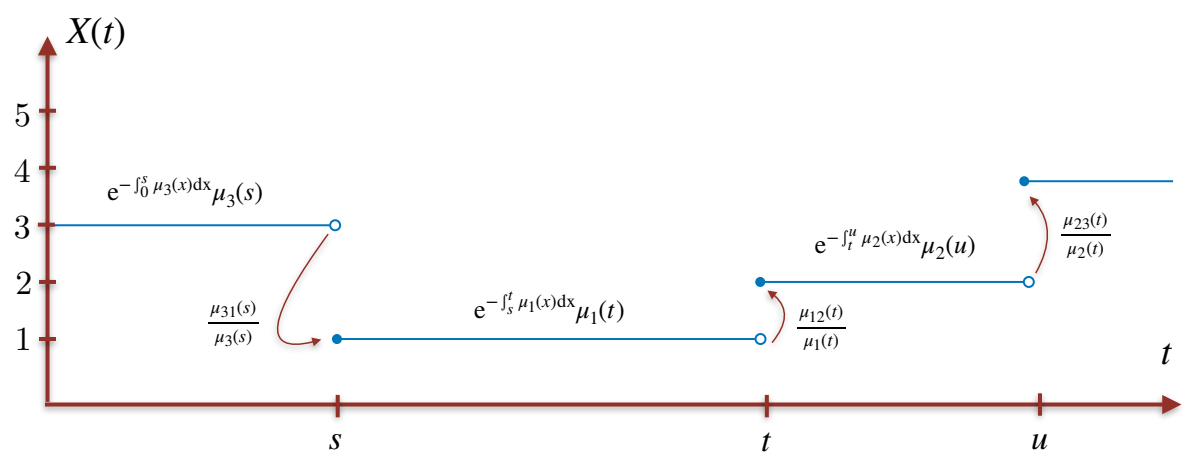

Figure 4. An illustration of sample path properties of $X(t)$. The formulas above the blue lines (sojourns) correspond to the densities of the sojourns times. Once a sojourn time has been drawn, a jump must occur. The formulas next to the red curved arrows indicate the probability of a transition to indicated new state.

Now that we have a complete characterisation of the sample paths, we now turn to the celebrated result of Kolmogorov, which will provide us with an important tool for computing the transition probabilities.

Theorem 2.5 (Kolmogorov' differential equations)**.** *Relations bewteen $\boldsymbol{\Lambda}(s)$, $\boldsymbol{\Lambda}(t)$ and $\boldsymbol{P}(s,t)$ are given by kolmogorov' backwards and forward differential equations, respectively:*

$$\frac{\partial}{\partial s}\boldsymbol{P}(s,t) = -\boldsymbol{\Lambda}(s)\boldsymbol{P}(s,t) \quad \textit{and} \quad \frac{\partial}{\partial t}\boldsymbol{P}(s,t) = \boldsymbol{P}(s,t)\boldsymbol{\Lambda}(t),$$

the solution of which is given by

$$\boldsymbol{P}(s,t) = \prod_s^t (\boldsymbol{I} + \boldsymbol{\Lambda}(x)\,\mathrm{d}x).$$

In the time–homogeneous case we have that

$$\frac{d}{dt}\boldsymbol{P}(t) = \boldsymbol{\Lambda}\boldsymbol{P}(t) = \boldsymbol{P}(t)\boldsymbol{\Lambda}$$

with solution

$$\boldsymbol{P}(t) = \exp(\boldsymbol{\Lambda}t).$$

Remark 2.6: The terms backward and forward are related to how the differential equations are solved. For the backward equation,

$$\frac{\partial}{\partial s}\boldsymbol{P}(s,t) = -\boldsymbol{\Lambda}(s)\boldsymbol{P}(s,t),$$

starting from t, one solves for $s < t$, i.e. backwards in time, whereas for the the forward equation,

$$\frac{\partial}{\partial t}\boldsymbol{P}(s,t) = \boldsymbol{P}(s,t)\boldsymbol{\Lambda}(t),$$

starting from s, one solves ahead in time.

Proof. Conditioning on whether there is a jump or not in $[s,t]$,

$$\begin{aligned} p_{ij}(s,t) &= \delta_{ij}\underbrace{\mathrm{e}^{-\int_s^t \lambda_i(u)\,\mathrm{d}u}}_{\substack{\text{prob. of no}\\ \text{jump}}} + \int_s^t \underbrace{\mathrm{e}^{-\int_s^u \lambda_i(u)\,\mathrm{d}u}\lambda_i(u)}_{\substack{\text{density of a}\\ \text{jump at } u}} \sum_{k\neq j} \underbrace{q_{ik}(u)}_{\substack{\text{prob. of jump}\\ \text{from } i \text{ to } k}} \quad p_{kj}(u,t)\,\mathrm{d}u \\ &= \delta_{ij}\mathrm{e}^{-\int_s^t \lambda_i(u)\,\mathrm{d}u} + \sum_{k\neq j}\int_s^t \mathrm{e}^{-\int_s^u \lambda_i(u)\,\mathrm{d}u}\lambda_{ik}(u)p_{kj}(u,t)\,\mathrm{d}u. \end{aligned}$$

The integrand is continuous, its integral is differentiable, so we conclude that $p_{ij}(s,t)$ is differentiable in both s and t. We can now obtain the results by differentiation w.r.t. s or t, using Leibniz' rule.

However, since we now know about the differentiability, we may proceed with a direct probabilistic argument as follows. For backward equation, we condition on what happens in $(s, s+\mathrm{d}s]$:

$$p_{ij}(s,t) = \underbrace{(1-\lambda_i(s)\,\mathrm{d}s)}_{\substack{\text{prob. of no}\\ \text{jumps}}} p_{ij}(s+\mathrm{d}s,t) + \sum_{k\neq i} \underbrace{\lambda_{ik}(s)\,\mathrm{d}s}_{\substack{\text{prob. of a}\\ \text{jump from } i \text{ to } k}} \quad p_{kj}(s+\mathrm{d}s,t).$$

Rearrange to

$$\frac{p_{ij}(s,t)-p_{ij}(s+\mathrm{d}s,t)}{\mathrm{d}s}=-\lambda_i(s)p_{ij}(s+\mathrm{d}s,t)+\sum_{k\neq i}\lambda_{ik}(s)p_{kj}(s+\mathrm{d}s,t).$$

Since $\lambda_{ii}(s)=-\lambda_i(s)$, this amounts to

$$-\frac{\partial}{\partial s}p_{ij}(s,t)=\sum_k \lambda_{ik}(s)p_{kj}(s,t).$$

On matrix form, this yields the backward equation.

Regarding the forward equation, we condition on what happens at time t and consider

$$\begin{aligned}p_{ij}(s,t+\mathrm{d}t)&=p_{ij}(s,t)(1-\lambda_j(t)\,\mathrm{d}t)+\sum_{k\neq j}p_{ik}(s,t)\lambda_{kj}(t)\,\mathrm{d}t\\&=p_{ij}(s,t)+\sum_k p_{ik}(s,t)\lambda_{kj}(t)\,\mathrm{d}t.\end{aligned}$$

Rearranging as before yields the result. □

From Theorem 1.9 we also obtain Chapman–Kolmogorov's equations

$$\boldsymbol{P}(s,t)=\boldsymbol{P}(s,u)\boldsymbol{P}(u,t).$$

In the time–homogeneous case this amounts to

$$\boldsymbol{P}(s+t)=\boldsymbol{P}(s)\boldsymbol{P}(t).$$

Example 2.7: Consider the matrix

$$\boldsymbol{\Lambda}(u)=\boldsymbol{\Lambda}=\begin{pmatrix}-5&3&2\\3&-4&1\\0&0&0\end{pmatrix}.$$

First we notice, that the row sums all sum to zero, and that the off–diagonal elements are non–negative. Hence it is an intensity matrix. Let $\{X_t\}_{t\geq 0}$ denote a Markov jump process with $\boldsymbol{\Lambda}$ as intensity matrix. We let row i correspond to the ith state of the process, $i=1,2,3$.

Let us for example think of i being the state of an insured, so that state 1 as active, 2 as invalid and 3 as dead.

First we consider some information we can deduce from the intensity matrix. We notice that state 3 is *absorbing*, since once entered the process will remain in this state.

Next we find the probabilities (15). They will not depend on u either, and will be

$$q_{12}=\frac{3}{5},q_{13}=\frac{2}{5},q_{21}=\frac{3}{4},q_{23}=\frac{1}{4},$$

and the rest being zero. This means, for example, that given $X_t=2$, the probability that the process will jump to state 3 next time it jumps will be 1/4, and with 3/4 of probability it will jump to state 1.

Next we consider the transition probabilities. The transition probability matrix is given by $\exp(\boldsymbol{\Lambda} x)$. To find this, we compute its eigenvalues, which amounts to

$$\lambda_1 = -7.541381, \ \lambda_2 = -1.458619, \text{ and } \ \lambda_3 = 0,$$

with corresponding eigenvectors

$$\boldsymbol{v}_1 = \begin{pmatrix} -0.7630200 \\ 0.6463749 \\ 0 \end{pmatrix}, \ \boldsymbol{v}_2 = \begin{pmatrix} -0.6463749 \\ -0.7630200 \\ 0 \end{pmatrix} \text{ and } \boldsymbol{v}_3 = \begin{pmatrix} 0.5773503 \\ 0.5773503 \\ 0.5773503 \end{pmatrix}.$$

Let

$$\boldsymbol{M} = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\} = \begin{pmatrix} -0.7630200 & -0.6463749 & 0.5773503 \\ 0.6463749 & -0.7630200 & 0.5773503 \\ 0 & 0 & 5773503 \end{pmatrix}.$$

Then

$$\exp(\boldsymbol{\Lambda} x) = \boldsymbol{M} \exp(\boldsymbol{D} x) \boldsymbol{M}^{-1}.$$

For $x = 1$, this gives

$$\exp(\boldsymbol{\Lambda}) = \begin{pmatrix} 0.0974715 & 0.1144348 & 0.7880937 \\ 0.1144348 & 0.1356164 & 0.7499487 \\ 0.0000000 & 0.0000000 & 1.0000000 \end{pmatrix}.$$

From this we can also deduce all the transition probabilities. For example, given that $X_0 = 1$ the probability that $X_1 = 1$ is 0.0974715. As we see, there is a possibility that process leaves state 1 and comes back at a later point. The probability, given $X_0 = 1$, that the process never leaves state 1 up to time 1, is in turn

$$\exp(-5 \cdot 1) = 0.00674,$$

which is much smaller than 0.0974715. □

2. The strong Markov property

Definition 2.8 (Stopping time)**:** Let $\{X_t\}_{t \geq 0}$ be a Markov jump process and let $\mathcal{F}_t = \sigma(X_s : s \leq t)$. A nonnegative random variable τ is called a stopping time for $\{X_t\}_{t \geq 0}$ if $\{\tau \leq t\} \in \mathcal{F}_t$ for all t.
The σ-algebra of the process up to a stopping τ, $\mathcal{F}_\tau$, is defined by the measurable sets A for which

$$\forall t \geq 0 : \quad A \cap \{\tau \leq t\} \in \mathcal{F}_t.$$

Theorem 2.9 (Strong Markov property)**.** *Every Markov jump process $\{X_t\}_{t\geq 0}$ satisfies the strong Markov property, i.e., for all $0 \leq h_1 \leq h_2 \leq \cdots \leq h_n$, we have that*

$$\mathbb{P}(X_{\tau+h_1} = i_1, \ldots, X_{\tau+h_n} = i_n \mid \mathcal{F}_\tau) = \mathbb{P}(X_{\tau+h_1} = i_1, \ldots, X_{\tau+h_n} = i_n \mid X_\tau)$$

on $\{\tau < \infty\}$. For time–homogeneous processes this further reduces to

$$\mathbb{P}(X_{\tau+h_1} = i_1, \ldots, X_{\tau+h_n} = i_n \mid \mathcal{F}_\tau) = \mathbb{P}_{X_\tau}(X_{h_1} = i_1, \ldots, X_{h_n} = i_n)$$

also on $\{\tau < \infty\}$.

Proof. If τ is deterministic, then the statement of the theorem is the usual Markov property, which holds by assumption. Now assume that τ is a stopping time that can take at most a countable number of values (including possibly $+\infty$),

$$\tau \in \{t_1, t_2, \ldots\} \cup \{+\infty\}.$$

The assertion of the theorem is by the definition of conditional expectation equivalent to the statement that for every $B \in \mathcal{F}_\tau$, we have

$$\int_{B\cap\{\tau<\infty\}} \mathbb{P}_{X_\tau}(X_{\tau+h_1}{=}i_1, \ldots, X_{\tau+h_n}{=}i_n)\,\mathrm{d}\mathbb{P} = \int_{B\cap\{\tau<\infty\}} 1\{X_{\tau+h_1} = i_1, \ldots, X_{\tau+h_n}{=}i_n\}\,\mathrm{d}\mathbb{P}.$$

Since $\{\tau < \infty\} = \bigcup_{i=1}^\infty \{\tau = t_i\}$ (excluding $+\infty$), we then have that

$$\begin{aligned}
&\int_{B\cap\{\tau<\infty\}} \mathbb{P}_{X_\tau}(X_{\tau+h_1} = i_1, \ldots, X_{\tau+h_n} = i_n)\,\mathrm{d}\mathbb{P} \\
&= \sum_{i=1}^\infty \int_{B\cap\{\tau=t_i\}} \mathbb{P}_{X_\tau}(X_{\tau+h_1} = i_1, \ldots, X_{\tau+h_n} = i_n)\,\mathrm{d}\mathbb{P} \\
&= \sum_{i=1}^\infty \int_{B\cap\{\tau=t_i\}} \mathbb{P}_{X_{t_i}}(X_{t_i+h_1} = i_1, \ldots, X_{t_i+h_n} = i_n)\,\mathrm{d}\mathbb{P} \\
&\overset{(*)}{=} \sum_{i=1}^\infty \int_{B\cap\{\tau=t_i\}} 1\{X_{t_i+h_1} = i_1, \ldots, X_{t_i+h_n} = i_n\}\,\mathrm{d}\mathbb{P} \\
&= \int_{B\cap\{\tau<\infty\}} 1\{X_{\tau+h_1} = i_1, \ldots, X_{\tau+h_n} = i_n\}\,\mathrm{d}\mathbb{P}.
\end{aligned}$$

In $(*)$ we have used the ordinary Markov property and the definition of $\mathcal{F}_\tau$ being the σ–algebra satisfying $B \cap \{\tau = t_i\} \in \mathcal{F}_{t_i}$.

In the the general case is. Here we approximate the stopping time τ by

$$\tau_m = \frac{[2^m\tau + 1]}{2^m} \downarrow \tau$$

as $m \to \infty$. Since τ_m takes only countably many values, we have that the strong Markov property holds for τ_m, i.e.,

$$\mathbb{P}(X_{\tau_m+h_1} = i_1, \ldots, X_{\tau_m+h_n} = i_n \mid \mathcal{F}_{\tau_m}) = \mathbb{P}_{X_{\tau_m}}(X_{\tau_m+h_1} = i_1, \ldots, X_{\tau_m+h_n} = i_n)$$

on $\tau_m < \infty$, or equivalently, for all $B \in \mathcal{F}_{\tau_m}$ we have that

$$\mathbb{E}(1\{X_{\tau_m+h_1} = i_1, \ldots, X_{\tau_m+h_n} = i_n\}1_B) = \mathbb{E}(1_B\mathbb{P}_{X_{\tau_m}}(X_{\tau_m+h_1} = i_1, \ldots, X_{\tau_m+h_n} = i_n)).$$

Now let $B \in \mathcal{F}_\tau$, which is hence also contained in every $\mathcal{F}_{\tau_m}$ as well. By right continuity of Markov jump processes, $X_{\tau_m} \to X_\tau$, and dominated convergence,

$$\mathbb{E}(1\{X_{\tau_m+h_1} = i_1, \ldots, X_{\tau_m+h_n} = i_n\}1_B) \to \mathbb{E}(1\{X_{\tau+h_1} = i_1, \ldots, X_{\tau+h_n} = i_n\}1_B)$$

as $m \to \infty$. This settles the left hand side. Concerning the right hand side,

$$\mathbb{P}(\cdot \mid X_{\tau_m} = j) = \frac{\mathbb{P}(\cdot \cap \{X_{\tau_m} = j\})}{\mathbb{P}(X_{\tau_m} = j)} \to \frac{\mathbb{P}(\cdot \cap \{X_\tau = j\})}{\mathbb{P}(X_\tau = j)} = \mathbb{P}(\cdot \mid X_\tau = j),$$

we have that $\mathbb{P}_{X_{\tau_m}}(\cdot) \to \mathbb{P}_{X_\tau}(\cdot)$, and so by dominated convergence,

$$\mathbb{E}(1_B \mathbb{P}_{X_{\tau_m}}(X_{h_1} = i_1, \ldots, X_{h_n} = i_n)) \to \mathbb{E}(1_B \mathbb{P}_{X_\tau}(X_{h_1} = i_1, \ldots, X_{h_n} = i_n))$$

as $m \to \infty$. This concludes the proof of the first part. The details of the time–homogeneous case is left to the reader. □

Often τ is defined in such way that X_τ is deterministic. For example, if τ represents the first time that $\{X_t\}_{t\geq 0}$ is in state i, then $X_\tau = i$, and consequently $\mathbb{P}_{X_\tau}(\cdot) = \mathbb{P}_i(\cdot)$. In Figure 5, for example, we consider τ to be the first time the process enters state 2. Then the strong Markov property states that the future of the process only depends on the state at time τ (which is obviously state 2). In other words, a new process starting at τ in state 2, $Y(t) = X(t-\tau)$, is again a Markov process with shifted transition probabilties.

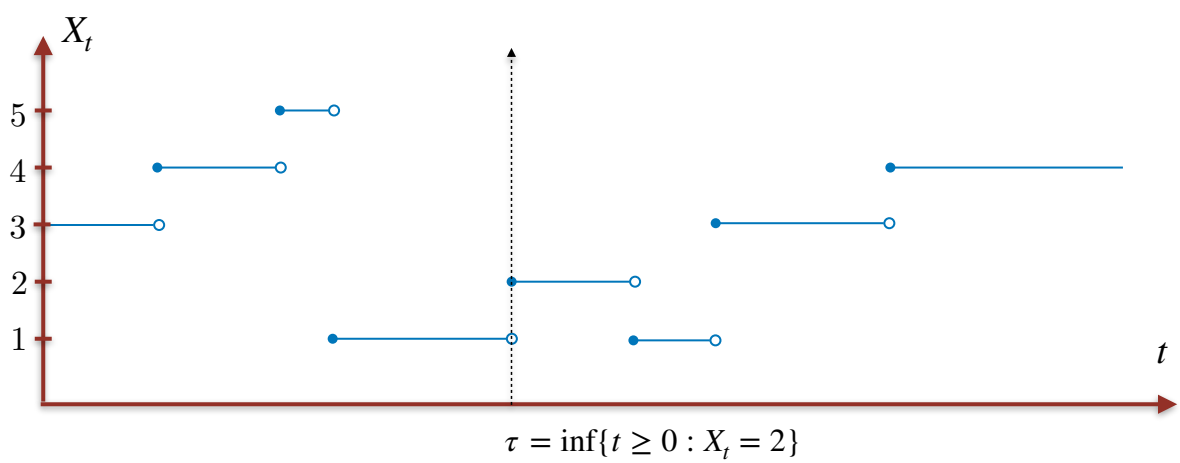

Figure 5. An illustration of sample path properties of $X(t)$. The formulas above the blue lines (sojourns) correspond to the densities corresponding sojourns. From these the sojourn times can be drawn (simulated). Once a sojourn time has been drawn, a jump must occur. The formulas next to the red curved arrows indicate the probability of a transition to indicated new state.

CHAPTER 3

A model for stochastic interest rates

1. Recap on hazard rates

Let X be a random variable with density f, its c.d.f. F and survival function $\bar{F} = 1 - F$. Since

$$f(x) = F'(x) \approx \frac{F(x + \mathrm{d}x) - F(x)}{\mathrm{d}x}$$

then we have the following well known interpretation of the density,

$$f(x)\,\mathrm{d}x = \mathbb{P}(X \in (x, x + \mathrm{d}x]).$$

This is the unconditional probability that $X \in (x, x + \mathrm{d}x]$. The conditional probability of this evnent given $X > x$, is given by

$$\begin{aligned}
\mathbb{P}(X \in (x, x + \mathrm{d}x] | X > x) &= \frac{\mathbb{P}(X \in (x, x + \mathrm{d}x])}{\mathbb{P}(X > x)} \\
&= \frac{f(x)\,\mathrm{d}x}{\bar{F}(x)} \\
&= -\frac{\bar{F}'(x)}{\bar{F}(x)}\,\mathrm{d}x \\
&= -\frac{\mathrm{d}}{\mathrm{d}x} \log \bar{F}(x)\,\mathrm{d}x.
\end{aligned}$$

Defining

$$h(x) = -\frac{\mathrm{d}}{\mathrm{d}x} \log \bar{F}(x)$$

we get

$$\mathbb{P}(X > x) = \bar{F}(x) = \exp\left(-\int_0^x h(y)\,\mathrm{d}y\right).$$

The interpreation of $h(x)$ is that $h(x)\,\mathrm{d}x$ is the probability of instantaneous realisation of X just after x given $X > x$. For example, if X denotes the life–time of a person, then $h(x)\,\mathrm{d}x$ is the probability that a person aged x will die right at that moment, i.e. $h(x)$ is the intensity or force of mortality at that moment. It is also referred to as the hazard rate, and $h(x)$ the hazard function. As we see, there is a one–to–one correspondence between hazard rates and survival functions (and hence also c.d.f. and densities).

Now consider stochastic mortality rates, or which is the same, stochastic hazard rates. Then $\{h(x)\}_{x\geq 0}$ can be thought of as a stochastic process. Let $\mathcal{F}_x = \sigma(h(u) : u \leq x)$ be the σ–algebra representing the history of the process up to time x. Then

$$\mathbb{P}(X > x|\mathcal{F}_x) = \exp\left(-\int_0^x h(y)\,\mathrm{d}y\right) \tag{16}$$

since conditionally on $\mathcal{F}_x$, the process $h(y)$ is to be considered as deterministic up to time x. But then, taking expectations, we get

$$\mathbb{P}(X > x) = \mathbb{E}\left(\exp\left(-\int_0^x h(y)\,\mathrm{d}y\right)\right). \tag{17}$$

2. Phase–type distributions

Markov models in life–insurance are often illustrated by means of flow diagrams like in Figures 1 and 2.

In these models, the ultimate state of death is an absorbing state. The time until death from a certain state, which is the time it takes for the Markov process to reach the absorbing state, is hence random and dependent on the current state. The distribution of this random time is said to have a phase–type distribution.

Let $\{X(t)\}_{t\geq 0}$ denote the Markov process with a finite state space E, say, and intensity matrices $\mathbf{\Lambda}(t) = \{\lambda_{ij}(t)\}_{i,j\in E}$.

Example 3.1 (Life-death model)**:** Let (x) be an insured with mortality rates μ_{x+t}. We let $X(t)$ denote a Markov process with state–space $E = \{0, 1\}$ and behaving according to Figure 1. The states correspond to the states "alive" and "dead", respectively. Then the time–dependent intensity matrices are then on the form

$$\mathbf{\Lambda}(s) = \begin{pmatrix} \lambda_{00}(s) & \lambda_{01}(s) \\ \lambda_{10}(s) & \lambda_{11}(s) \end{pmatrix} = \begin{pmatrix} -\mu_{x+t}(s) & \mu_{x+t}(s) \\ 0 & 0 \end{pmatrix}.$$

The time until death starting in state 0 (alive) at time $t = 0$ (age $= x$) is the time until absorption

$$\tau = \inf\{s > 0 : X_s = 1\}.$$

Figure 1. Simplest alive-death model.

□

Example 3.2 (Disability model)**:** We now consider a disability model for (x). Let $X(t)$ denote the Markov process describing the state of (x) at time $x+t$, being either "active", "disabled" or "dead", see Figure 2. Then its state–space can be written as $E = \{0,1,2\}$ and its intensity matrix is seen to be

$$\mathbf{\Lambda} = \begin{pmatrix} -\lambda_0(s) & \lambda_{01}(s) & \lambda_{02}(s) \\ \lambda_{10}(s) & -\lambda_1(s) & \lambda_{12}(s) \\ 0 & 0 & 0 \end{pmatrix}.$$

The rate $\lambda_0(s)$ is the rate of either becoming disabled or die at time s. The rate $\lambda_{01}(s)$ is the rate of becoming disabled and $\lambda_{02}(s)$ the mortality rate for (x) as an active person. Furthermore,

$$\lambda_0(s) = \lambda_{01}(s) + \lambda_{02}(s)$$

is the rate of an active (x) either becoming disabled or dying at time s.

In the second row, $\lambda_{12}(s)$ is the mortality of (x) now being disabled at time s. In the simple life–death model, the mortality rate μ_{x+t} is simply based on (x) being alive, but not on his condition of being either active or disabled as in this model. The rate $\lambda_{10}(s)$ is the recovery rate from disability at time s.

In order to compute the distribution of the time until death,

$$\tau = \inf\{s > 0 : X_s = 2\},$$

we will now have to consider the possibility of (x) going through the states "active" and "disabled" before death. These states are also known as "phases".

In this model, "active" and "disabled exhaust all options for possible conditions of (x), and in order to compute his life–time distribution, it would be wrong to use only λ_{02} or λ_{12} as mortality rates. Here λ_{02} will only be the correct mortality rate for an active (x) dying, while λ_{12} is only correct for a disabled (x) dying. The mortality rate μ_{x+t} for (x) simply being "alive" is then obtain from the hazard rate of τ, i.e.

$$\mu_{x+t} = \frac{f_\tau(t)}{\bar{F}_\tau(t)} = -\frac{\mathrm{d}}{\mathrm{d}t} \log \bar{F}_\tau(t),$$

where $f_\tau(t)$ denotes the density function for τ and $\bar{F}_\tau(t) = \mathbb{P}(\tau > t)$ the survival function for τ.

Knowing f_τ and $\bar{F}_\tau$, we could, in principle, recover the simple alive"–"dead" model from the more refined disability model.

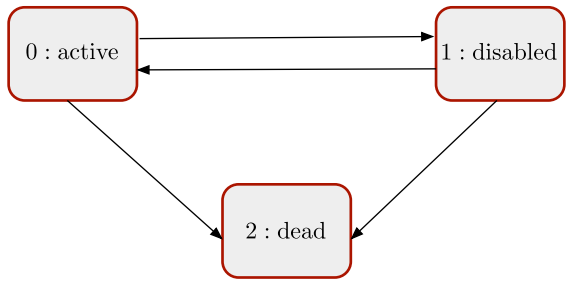

Figure 2. Simple disability model.

□

A Markovian life–insurance model, which includes an absorbing state (death), can then be written as follows in a general set–up.

Let $\{X_t\}_{t\geq 0}$ be a time–inhomogeneous Markov jump process on the finite state–space $E = \{1,2,...,p,p+1\}$, say, where states $1,2,...,p$ are transient (states of the insured) and $p+1$ absorbing (death). Thus the intensity matrix of $\{X_t\}_{t\geq 0}$ is on the form

$$\boldsymbol{\Lambda}(t) = \begin{pmatrix} \boldsymbol{T}(t) & \boldsymbol{t}(t) \\ \boldsymbol{0} & 0 \end{pmatrix}.$$

Here, for any time $t \geq 0$, $\boldsymbol{t}(t) = -\boldsymbol{T}(t)\boldsymbol{e}$, where $\boldsymbol{e}$ denotes the column vector $\boldsymbol{e} = (1,1,...,1)'$. This follows from the row sums of intensity matrices being zero.

Assume that $\mathbb{P}(X_0 = p+1) = 0$ and define $\boldsymbol{\pi} = (\pi_1,...,\pi_p)$. Then $\boldsymbol{\pi}$ is called the initial distribution of the Markov process. In life insurance $\boldsymbol{\pi}$ will often be $(1,0,...,0)$, which means that the process starts in the active state at time zero.

Definition 3.3: Let

$$\tau = \inf\{t \geq 0 : X_t = p+1\}$$

denote the time until absorption of $\{X_t\}$. The distribution of τ is then said to be an inhomogeneous phase–type distribution with representation $(\boldsymbol{\pi}, \boldsymbol{T}(t))$ and we write $\tau \sim \mathrm{IPH}(\boldsymbol{\pi}, \boldsymbol{T}(t))$.

Lemma 3.4. *We have the following decomposition:*

$$\boldsymbol{P}(s,t) = \prod_s^t (\boldsymbol{I} + \boldsymbol{\Lambda}(u)du) = \begin{pmatrix} \prod_s^t (\boldsymbol{I} + \boldsymbol{T}(u)du) & \boldsymbol{e} - \prod_s^t (\boldsymbol{I} + \boldsymbol{T}(u)du)\boldsymbol{e} \\ \boldsymbol{0} & 1 \end{pmatrix}, \tag{18}$$

where $\boldsymbol{e} = (1,1,...,1)'$.

Proof. From Van Loan (Lemma 1.17, p. 15) and using that $\boldsymbol{t}(u) = -\boldsymbol{T}(u)\boldsymbol{e}$, we get that

$$\begin{aligned} \boldsymbol{P}(s,t) &= \prod_s^t (\boldsymbol{I} + \boldsymbol{\Lambda}(u)du) \\ &= \prod_s^t \left(\boldsymbol{I} + \begin{pmatrix} \boldsymbol{T}(u) & \boldsymbol{t}(u) \\ \boldsymbol{0} & 0 \end{pmatrix} du \right) \\ &= \begin{pmatrix} \prod_s^t (\boldsymbol{I} + \boldsymbol{T}(x)\,\mathrm{d}x) & \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{T}(x)\,\mathrm{d}x)\boldsymbol{t}(u)\,\mathrm{d}u \\ \boldsymbol{0} & 1 \end{pmatrix}. \end{aligned}$$

But

$$\begin{aligned} \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{T}(x)\,\mathrm{d}x)\boldsymbol{t}(u)\,\mathrm{d}u &= -\int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{T}(x)\,\mathrm{d}x)\boldsymbol{T}(u)\boldsymbol{e}\,\mathrm{d}u \\ &= -\int_s^t \frac{\partial}{\partial u} \prod_s^u (\boldsymbol{I} + \boldsymbol{T}(x)\,\mathrm{d}x)\,\mathrm{d}u\boldsymbol{e} \\ &= \boldsymbol{e} - \prod_s^t (\boldsymbol{I} + \boldsymbol{T}(x)\,\mathrm{d}x)\boldsymbol{e}. \end{aligned}$$

□

Theorem 3.5. *Assume that* $\tau \sim IPH(\boldsymbol{\pi}, \boldsymbol{T}(t))$. *Then the density* f *and the distribution function* F *of* τ *are given by*

$$\begin{aligned} f(x) &= \boldsymbol{\pi} \left(\prod_0^x (\boldsymbol{I} + \boldsymbol{T}(u)du) \right) \boldsymbol{t}(x) \\ F(x) &= 1 - \boldsymbol{\pi} \left(\prod_0^x (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u) \right) \boldsymbol{e}. \end{aligned}$$

Proof. Since $(\boldsymbol{\pi}, 0)\boldsymbol{P}(0,t)$ is the distribution of X_t, we get from (18) that

$$\mathbb{P}(X_t = j) = \boldsymbol{\pi} \prod_0^t (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u)\boldsymbol{e}_j.$$

The density f for τ is now readily derived as

$$f(x)\,\mathrm{d}x = \mathbb{P}(\tau \in (x, x + \mathrm{d}x])$$

$$
\begin{aligned}
&= \sum_{j=1}^{p} \boldsymbol{\pi} \prod_{0}^{x} (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u) \boldsymbol{e}_j t_j(x)\,\mathrm{d}x \\
&= \boldsymbol{\pi} \prod_{0}^{x} (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u) \boldsymbol{t}(x)\,\mathrm{d}x,
\end{aligned}
$$

so that

$$
f(x) = \boldsymbol{\pi} \prod_{0}^{x} (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u) \boldsymbol{t}(x). \tag{19}
$$

The distribution function F for τ is obtained by

$$
\begin{aligned}
1 - F(s) &= \mathbb{P}(\tau > s) \\
&= \mathbb{P}(X_s \in \{1, 2, ..., p\}) \\
&= \sum_{j=1}^{p} \mathbb{P}(X_s = j) \\
&= \boldsymbol{\pi} \prod_{0}^{x} (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u) \boldsymbol{e}.
\end{aligned}
$$

□

We can think about a phase–type distribution as a distribution with a stochastic hazard rate (i.e. a hazard process) with

$$
h(u) = \lambda_{X(u)}(u)
$$

for some (deterministic) hazard functions $\lambda_i(u)$ and Markov jump process $\{X(u)\}_{u \geq 0}$. Hence we get the following representation for the c.d.f. of a phase–type distribution.

Theorem 3.6. *Assume that $\tau \sim IPH(\boldsymbol{\pi}, \boldsymbol{T}(t))$ and let $\{X(u\}_{u \geq 0}$ denote the underlying Markov jump process. Then the survival function $\bar{F}$ of τ has the following representation*

$$
\bar{F}(x) = \mathbb{E}\left(\exp\left(-\int_0^x t_{X(u)}(u)\,\mathrm{d}u \right) \right),
$$

where $\boldsymbol{t}(u) = (t_1(u), ..., t_p(u))\prime = -\boldsymbol{T}(u)\boldsymbol{e}$ are the exit rates.

We shall extend this result in Theorem 4.4, where an alternative probabilistic proof is given.

We the following incremental property as well.

Theorem 3.7. *If $\tau \sim IPH(\boldsymbol{\pi}, \boldsymbol{T}(t))$ then*

$$\mathbb{P}(\tau > s+t | \tau > s) = \frac{\boldsymbol{\pi} \prod_0^s (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u)}{\boldsymbol{\pi} \prod_0^s (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u)\boldsymbol{e}} \prod_s^{s+t} (1 + \boldsymbol{T}(u)\,\mathrm{d}u)\boldsymbol{e}$$

so that

$$\tau - s | \{\tau > s\} \sim IPH(\boldsymbol{\alpha}, \boldsymbol{S}(\cdot)),$$

where $\boldsymbol{S}(u) = \boldsymbol{T}(s+u)$ and

$$\boldsymbol{\alpha} = \frac{\boldsymbol{\pi} \prod_0^s (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u)}{\boldsymbol{\pi} \prod_0^s (\boldsymbol{I} + \boldsymbol{T}(u)\,\mathrm{d}u)\boldsymbol{e}}.$$

Proof. Make a drawing! □

Corollary 3.8. *If $\tau \sim IPH(\boldsymbol{\pi}, \boldsymbol{T}(t))$ and $\boldsymbol{T}(t_1)$ and $\boldsymbol{T}(t_2)$ commute for all $t_1, t_2 \geq 0$, then the density f and the distribution function F of τ are given by*

$$\begin{aligned} f(x) &= \boldsymbol{\pi} \exp\left(\int_0^x \boldsymbol{T}(u)\,\mathrm{d}u\right) \boldsymbol{t}(x) \\ F(x) &= 1 - \boldsymbol{\pi} \exp\left(\int_0^x \boldsymbol{T}(u)\,\mathrm{d}u\right) \boldsymbol{e}. \end{aligned}$$

Proof. If all $\boldsymbol{T}(u)$ commute with each other, then by Theorem 1.13, p. 12, the product integral becomes a matrix exponential, i.e.

$$\prod_s^t (\boldsymbol{I} + \boldsymbol{T}(u)du) = \exp\left(\int_s^t \boldsymbol{T}(u)du\right).$$

□

An important special case where all matrices $\boldsymbol{T}(x)$ commute is when

$$\boldsymbol{T}(x) = \lambda(x)\boldsymbol{T} \tag{20}$$

for some non–negative real function $\lambda(x)$ and constant intensity matrix $\boldsymbol{T}$. Here $\lambda(x)$ scales the evolution of time in the same way for all states involved.

Definition 3.9: Let $\mathrm{IPH}(\boldsymbol{\pi}, \boldsymbol{T}(x))$ be a phase–type representation. If $\boldsymbol{T}(x)$ satisfies (20), then we write

$$\mathrm{IPH}(\boldsymbol{\pi}, \boldsymbol{T}(x)) = \mathrm{IPH}(\boldsymbol{\pi}, \boldsymbol{T}, \lambda(x)).$$

If $\lambda(x) \equiv 1$, then we simplly write $\mathrm{PH}(\boldsymbol{\pi}, \boldsymbol{T})$, which corresponds to the classical time–homogeneous phase–type distributions.

If $\tau \sim \mathrm{IPH}(\boldsymbol{\pi}, \boldsymbol{T}, \lambda(x))$, then τ can be obtained from a simple transformation of a $X \sim \mathrm{PH}(\boldsymbol{\pi}, \boldsymbol{T})$.

Theorem 3.10. $\tau \sim \mathrm{IPH}(\boldsymbol{\pi}, \boldsymbol{T}, \lambda)$ *if and only if*

$$\int_0^\tau \lambda(u)\,\mathrm{d}u \sim \mathrm{PH}(\boldsymbol{\pi}, \boldsymbol{T}).$$

Proof. Let $X \sim \mathrm{PH}(\boldsymbol{\pi}, \boldsymbol{T})$. Since $\lambda(u) > 0$ we notice that

$$\tau > y \quad \Longleftrightarrow \quad \int_0^\tau \lambda(u)\,\mathrm{d}u > \int_0^y \lambda(u)\,\mathrm{d}u.$$

Hence

$$\begin{aligned} \mathbb{P}\left(\int_0^\tau \lambda(u)\,\mathrm{d}u > \int_0^y \lambda(u)\,\mathrm{d}u\right) &= \mathbb{P}(\tau > y) \\ &= \boldsymbol{\pi} \exp\left(\boldsymbol{T} \int_0^y \lambda(u)\,\mathrm{d}u\right) \boldsymbol{e}, \end{aligned}$$

where the latter equals $\mathbb{P}\left(X > \int_0^y \lambda(u)\,\mathrm{d}u\right)$ for $X \sim \mathrm{PH}(\boldsymbol{\pi}, \boldsymbol{T})$. □

Finally, we mention the fact that the class of distributions $\mathrm{PH}(\boldsymbol{\pi}, \boldsymbol{T})$ is dense in the class of distributions on the positive reals. This means, that for any given distribution F on $(0, \infty)$, we can find a sequence $(\boldsymbol{\pi}_n, \boldsymbol{T}_n)$, $n \geq 1$ such that

$$\tau_n \stackrel{\sim}{\to} X,$$

(weak convergence) where $X \sim F$ and $\tau_n \sim \mathrm{PH}(\boldsymbol{\pi}_n, \boldsymbol{T}_n)$. This, of course, also implies that the different classes of IPH are also dense since they contain the class PH as a special case. Another proof of this follows from the continuous mapping theorem and Theorem 3.10.

3. Markovian interest rates

The classical Markovian life–insurance models provide fine examples of time–inhomogeneous phase–type distribution, describing the distribution of the time until death from different states. In these examples, the transition rates are known with specific biometric interpretations.

We now turn to a more surprising application of phase–type distributions towards stochastic interest rates, in which the transition rates do not possess a physical interpretation, but are simply present for the sake of fitting a model adequately.

The model of interest rates is as follows. Let $\{X(t)\}_{t \geq 0}$ be a Markov jump process defined on a finite state–space $E = \{1, 2, ..., p\}$ with intensity matrix $\boldsymbol{M}(t) = \{\mu_{ij}(t)\}_{i,j \in E}$.

We then define the spot (interest) rate $r(u)$ at time u by

$$r(u) = r_{X(u)}(u), \tag{21}$$

where $r_i(u)$, $i = 1, ..., p$, are deterministic functions. Let $\boldsymbol{r}(t) = \left(r_1(t), \ldots, r_p(t)\right)'$ denote the column vector of interest rate functions.

The spot rates give rise to a zero–coupon bond with expiration $T > 0$, defined in terms of its bond prices as

$$B(t,T) = \mathbb{E}^{\mathbb{Q}}\left(\mathrm{e}^{-\int_t^T r_{X(u)}\,\mathrm{d}u} \middle| \mathcal{F}(t) \right), \quad 0 \le t \le T, \tag{22}$$

where $\mathcal{F}(t) = \sigma(X(u) : 0 \le u \le t)$ is the σ–algebra generated by $\{X(u)\}_{u\ge 0}$ and $\mathbb{Q}$ is a risk–neutral measure (this is not important at the moment and we will return to this point later). It means that the transitions move according to $\mathbb{Q}$.

If all possible interest rates are non–negative, then $\boldsymbol{T}(u) = \boldsymbol{M}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u))$ are sub–intensity matrices. If $\boldsymbol{\pi} = (\pi_1, ..., \pi_p)$ with $\pi_i = \mathbb{P}(X(0) = i)$ denotes the initial distribution, then the pair $(\boldsymbol{\pi}, \boldsymbol{T}(u))$ defines a representation for an IPH. The corresponding exit rate vectors are

$$\boldsymbol{t}(u) = -\boldsymbol{T}(u)\boldsymbol{e} = -(\boldsymbol{M}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u)))\boldsymbol{e} = \boldsymbol{\Delta}(\boldsymbol{r}(u))\boldsymbol{e} = \boldsymbol{r}(u).$$

But then from Theorem 3.6 we get that if

$$\begin{aligned}
\mathbb{P}(\tau > T) &= \mathbb{E}^{\mathbb{Q}}\left(\mathrm{e}^{-\int_0^T t_{X(u)}(u)\,\mathrm{d}u} \right) \\
&= \mathbb{E}^{\mathbb{Q}}\left(\mathrm{e}^{-\int_0^T r_{X(u)}(u)\,\mathrm{d}u} \right) \\
&= B(0,T),
\end{aligned}$$

with $\tau \sim \mathrm{PH}(\boldsymbol{\pi}, \boldsymbol{T}(u))$, and where the last equality follows from (22) with $t = 0$.

More generally:

Theorem 3.11. *Let*

$$\boldsymbol{D}(t,T) = \prod_t^T (\boldsymbol{I} + (\boldsymbol{M}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u)))\,\mathrm{d}u). \tag{23}$$

Then price of the zero–coupon bond (22) satisfies

$$B(t,T) = \mathbb{E}^{\mathbb{Q}}\left(\exp\left(-\int_t^T r_{X(u)}(u)\,\mathrm{d}u \right) \middle| X(t) \right) = \boldsymbol{e}'_{X(t)} \boldsymbol{D}(t,T)\boldsymbol{e}. \tag{24}$$

If $r_k(t) \ge 0$ for all k, then $T \to B(t,T)$ is the survival function of an IPH distributed random variable, $\tau(t)$, with initial distribution $\boldsymbol{e}'_{X(t)}$ and intensity matrices $\boldsymbol{M}(x+t) - \boldsymbol{\Delta}(\boldsymbol{r}(x+t))$, $x \ge 0$.

Proof. This is simply a new Markov jump process initiating at time t in state $X(t)$, and the right–hand side is simply an expression for the survival function of the time until absorption, as started from t. □

A slight modification may allow for negative interest. Assume that all $r_i(x)$ are bounded from below by a ρ. We may e.g. take ρ as

$$\rho = \max\left(0, -\min_{i\in E}\inf_{x\geq 0} r_i(x)\right).$$

We may take $\rho = 0$ if all interest rates are non–negative. Otherwise, $-\rho$ provides a lower bound for all of them. Then we have the following result.

Theorem 3.12. *The price of the zero–coupon bond (22) satisfies*

$$B(t,T) = \mathbb{E}^{\mathbb{Q}}\left(\exp\left(-\int_t^T r_{X(u)}(u)\,\mathrm{d}u\right)\middle| X(t)\right) = \boldsymbol{e}'_{X(t)}\boldsymbol{D}(t,T)\boldsymbol{e}. \tag{25}$$

Conditional on $X(t) = i$,

$$T \to \mathrm{e}^{-\rho(T-t)}B(t,T)$$

is the survival function for an IPH distributed random variable, $\tau(t)$, with initial distribution $\boldsymbol{e}'_i$ and intensity matrices $\boldsymbol{M}(x+t) - \boldsymbol{\Delta}(\boldsymbol{r}(x+t)) - \rho\boldsymbol{I}$, $x \geq 0$.

Proof. Formula (25) follows as for (24) above. Next, by Corollary 1.15, p. 13, we notice that

$$e^{-\rho(t-s)}\prod_s^t\left(\boldsymbol{I} + [\boldsymbol{M}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u))]\mathrm{d}u\right) = \prod_s^t\left(\boldsymbol{I} + [\boldsymbol{M}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u)) - \rho\boldsymbol{I}]\mathrm{d}u\right).$$

The matrix $\boldsymbol{M}(x) - \boldsymbol{\Delta}(\boldsymbol{r}(x)) - \rho\boldsymbol{I}$ is now a sub–intensity matrix and the result follows. □

Corollary 3.13. *Conditional on $X(t) = i$, the forward rate $f(t,T)$ equals the hazard rate at T for the random variable $\tau(t) \sim \mathrm{PH}(\boldsymbol{e}_i, \boldsymbol{M}(x+t) - \boldsymbol{\Delta}(\boldsymbol{r}(T)) - \rho\boldsymbol{I})$, less ρ. i.e.*

$$f(t,T) = \frac{f_{\tau(t)}(T)}{1 - F_{\tau(t)}(T)} - \rho. \tag{26}$$

The yield curve equals

$$T \to Y(t,T) \quad = \quad -\frac{\log \bar{F}_{\tau(t)}(T)}{T-t} - \rho,$$

Proof. We recall that the forward rate $f(t,T)$ is defined by

$$f(t,T) = -\frac{\partial}{\partial T}\log B(t,T).$$

Using Theorem 3.12, we may write

$$B(t,T) = \mathrm{e}^{\rho(T-t)}\bar{F}_{\tau(t)}(T),$$

where $\bar{F}_{\tau(t)}(T) = 1 - F_{\tau(t)}(T)$ denotes the survival function for $\tau(t) \sim \mathrm{IPH}(\boldsymbol{e}'_{X(t)}, \boldsymbol{M}(x+t) - \boldsymbol{\Delta}(\boldsymbol{r}(x+t)) - \rho \boldsymbol{I})$. Then clearly

$$-\frac{\partial}{\partial T} \log B(t,T) = -\rho + \frac{f_{\tau(t)}(T)}{1 - F_{\tau(t)}(T)},$$

where $f_{\tau(t)}$ denotes the density function for $\tau(t)$.

Similarly, the yield curve then equals

$$\begin{aligned} T \to Y(t,T) &= -\frac{\log(B(t,T))}{T-t} \\ &= -\frac{\log \bar{F}_{\tau(t)}(T)}{T-t} - \rho. \end{aligned}$$

□

Another immediate consequence of Theorem 3.12 is the following.

Corollary 3.14. *Assume that all interest rates are non–negative. Then conditional on $X(t) = i$, the random variable $\tau(t) \sim \mathrm{IPH}(\boldsymbol{e}'_i, \boldsymbol{M}(t+x) - \boldsymbol{\Delta}(\boldsymbol{r}(t+x)))$, $x \geq 0$ then has a c.d.f. given by*

$$F_{\tau(t)}(T) = 1 - B(t,T) = \mathbb{E}^{\mathbb{Q}} \left(\int_t^T r_{X(y)}(y) \mathrm{e}^{-\int_t^y r_{X(u)}(u)\,\mathrm{d}u} \,\mathrm{d}y \middle| X(t) = i \right).$$

Proof. This follows from Theorem 3.12 with $\rho = 0$ and

$$f_{\tau(t)}(y) = -\frac{\partial}{\partial y} B(t,y) = \mathbb{E}^{\mathbb{Q}} \left(r_{X(y)}(y) \mathrm{e}^{-\int_t^y r_{X(u)}(u)\,\mathrm{d}u} \middle| X(t) = i \right).$$

Integrating the expression then yields the result. □

For the case where $t = 0$, the above results are reduced to the following.

Corollary 3.15. *Assume that all interest rates are non–negative. Let $\tau \sim \mathrm{IPH}(\boldsymbol{\pi}, \boldsymbol{M}(x) - \boldsymbol{\Delta}(\boldsymbol{r}(x)))$ and let $\boldsymbol{\pi} = (\pi_1, ..., \pi_p)'$ denote the (initial) distribution of $X(0)$. Then*

$$\mathbb{P}(\tau > T) = \mathbb{E}^{\mathbb{Q}} \left(\exp\left(-\int_0^T r_{X(u)}(u)\,\mathrm{d}u \right) \right), \tag{27}$$

$$F_\tau(T) = \mathbb{E}^{\mathbb{Q}} \left(\int_0^T r_{X(y)}(y) \mathrm{e}^{-\int_0^y r_{X(u)}(u)\,\mathrm{d}u} \,\mathrm{d}y \right), \tag{28}$$

$$f(0,T) = \frac{f_\tau(T)}{1 - F_\tau(T)}. \tag{29}$$

The density $f_\tau(t)$ then has the interpretation of being the expected present value of the current interest rate accumulated in a small time interval around t, and $F_\tau(T)$ is the present value of the total accumulated interest rate during $[0,T]$.

The Markovian interest rates have two nice and useful properties for life–insurance models: they can be fitted to real bond data (or theoretical models), and they fit nicely into the framework of product integral representations for reserves and higher–order moments.

We first consider how to fit to real data or calibrate to other known models.

4. Estimation of phase–type distributions

First, we need to assess the nature of the interest rate data. Stochastic interest rate models are often calibrated to bond prices at different maturities or, equivalently, to yield curves.

We will use the bond prices at different maturities as our basic observable data. Assume that we have observed $B(0,T_i)$ at maturities $T_1,T_2,...,T_N$. Then this corresponds to observing $\mathbb{P}(\tau > T_i),\ i=1,...,N,$ where $\tau \sim \mathrm{IPH}(\boldsymbol{\pi},\boldsymbol{T}(u))$. This is again equivalent to observing a histogram with breaks at T_i, bin widths of $T_{i+1}-T_i$ and corresponding probability mass of $B(0,T_i)-B(0,T_{i+1})$.

The histogram comprises information of a hypothetical sample of realisations of τ, but the individual values of the τ's in the sample are unknown. Since we do not have a sample size either, we shall consider the data to be *weighted data*, where the data points are the midpoints of the bins, i.e.

$$\tau_i = \frac{T_i+T_{i+1}}{2}, i=1,...,N-1,$$

with weights

$$w_i = T_{i+1}-T_i, i=1,...,N.$$

Hence, we need to be able to fit the weighted data $\boldsymbol{y}=(\tau_i,w_i),\ i=1,...,N-1$. We assume that the phase–type distribution is time–homogeneous $\mathrm{PH}(\boldsymbol{\pi},\boldsymbol{T})$.

The likelihood function is then given by

$$L(\boldsymbol{y};\boldsymbol{\pi},\boldsymbol{T}) = \prod_{i=1}^{n}\left(\boldsymbol{\pi}\mathrm{e}^{\boldsymbol{T}\tau_i}\boldsymbol{t}\right)^{w_i}$$

or log–likelihood

$$\ell(\boldsymbol{y};\boldsymbol{\pi},\boldsymbol{T}) = \sum_{i=1}^{n} w_i \log\left(\boldsymbol{\pi}\mathrm{e}^{\boldsymbol{T}\tau_i}\boldsymbol{t}\right).$$

There is no explicit expression for the maximum likelihood estimator. Since the number of parameters is large, a general–purpose maximiser can become unstable and slow.

Alternatively, since τ_i are the times until absorption of an underlying Markov jump process, the problem can also be seen as an incomplete data problem. Here, the complete data are the full trajectories until absorption, while the incomplete data are the times until absorption.

There are several ways to deal with incomplete data estimation, and one of the most popular ones is the so-called Estimation–Maximisation (EM) algorithm, which turns out to be particularly well-suited for our case.

Assume that the parameters to be estimated are $\boldsymbol{\theta}(=(\boldsymbol{\pi},\boldsymbol{T})$ in our case), where $\boldsymbol{y}$ denotes the vector of incomplete data and $\boldsymbol{X}$ the random vector corresponding to the complete (unknown) data. Let ℓ_c denote the complete data likelihood. Then the EM algorithm works as follows.

0. initialize with some "arbitrary" $\boldsymbol{\theta}_0$ and let $n=0$.
1. (E-step) Calculate the function
$$h:\boldsymbol{\theta}\to\mathbb{E}_{\boldsymbol{\theta}_n}\left(\ell_c(\boldsymbol{\theta};\boldsymbol{X})\mid\boldsymbol{Y}=\boldsymbol{y}\right).$$
2. (M-step) Let $\boldsymbol{\theta}_{n+1}:=\operatorname{argmax}_{\boldsymbol{\theta}}h(\boldsymbol{\theta})$.
3. n=n+1; GOTO 1.

In step 0, we initiate the algorithm by choosing any starting value $\boldsymbol{\theta}_0$ for $\boldsymbol{\theta}$. Under these parameters, the conditional expectation of the complete data likelihood given our incomplete data is calculated, and in step 2, we find the parameters $\boldsymbol{\theta}_1$ that maximise the expectation. Replacing $\boldsymbol{\theta}_0$ by $\boldsymbol{\theta}_1$, we may then continue in this way. It can be shown that the likelihood increases in every iteration step, which makes the algorithm convergent. It is not certain if it converges to a global or only local maximum, so different initiations may be needed to verify this point.

Let us analyse how this algorithm looks for phase-type data.

(a) Complete data likelihood ℓ_c.

Data is now full trajectories of the underlying Markov jump processes, which generate the times until absorption $\tau_1,\ldots,\tau_p$. Let us consider one of them, $\boldsymbol{y}=\tau_1$ say. Then $\boldsymbol{X}$ is the full trajectory which gets absorbed exactly at time τ_1. Describe the trajectory by (i_k,u_k), $k=0,\ldots,n$, where i_k are the kth state visited ($k=0$ meaning the initial state), and u_k is the time spent in that state until jumping out of it again.

Denote
$$\boldsymbol{\Lambda}=\{\lambda_{ij}\}_{i,j=1,\ldots,p+1}=\begin{pmatrix}\boldsymbol{T} & \boldsymbol{t}\\ \boldsymbol{0} & 0\end{pmatrix},$$
$\lambda_i=-\lambda_{ii}$ and $q_{ij}=-t_{ij}/t_{ii}$.

Since the sojourn times are conditional indpendent and exponentially distributed, we get the complete data likelihood L_c to be
$$\begin{aligned}L_c(\tau_1;\boldsymbol{\pi},\boldsymbol{T})&=\pi_{i_0}\mathrm{e}^{-\lambda_{i_0}u_0}\lambda_{i_0}q_{i_0i_1}\cdots q_{i_{n-1}i_n}\mathrm{e}^{-\lambda_{i_n}u_n}\lambda_{i_n}q_{i_n,p+1}\\&=\pi_{i_0}\mathrm{e}^{-\lambda_{i_0}u_0}\lambda_{i_0i_1}\cdots\lambda_{i_{n-1}i_n}\mathrm{e}^{-\lambda_{i_n}u_n}\lambda_{i_n,p+1}\end{aligned}$$

$$= \left(\prod_{i=1}^{p} \pi^{B_i}\right)\left(\prod_{i=1}^{p}\prod_{j\neq i} \lambda_{ij}^{N_{ij}}\right)\left(\prod_{i=1}^{p} \mathrm{e}^{-Z_i\lambda_i}\right)\left(\prod_{i=1}^{p} \lambda_{i,p+1}^{N_i}\right)$$

$$= \left(\prod_{i=1}^{p} \pi^{B_i}\right)\left(\prod_{i=1}^{p}\prod_{j\neq i} \lambda_{ij}^{N_{ij}}\mathrm{e}^{-Z_i\lambda_{ij}}\right)\left(\prod_{i=1}^{p} \lambda_{i,p+1}^{N_i}\mathrm{e}^{-Z_i\lambda_{i,p+1}}\right),$$

where $B_i = \{Z(0) = i\}$ the indicator for whether initiation in state i has taken place, N_{ij} the total number of jumps from i to j, $i \neq j$, Z_i the total time spent in state i until absorption, and N_i the indicator for wether absorption takes place through a jump from i.

For i.i.d. data $\boldsymbol{y} = (\tau_1, ..., \tau_n)$, the complete data likelihood has the exact same form, with the only changes of B_i noe being the number of initiations from state i, etc.

The complete data log–likelihood then becomes

$$\begin{aligned}\ell_c(\tau_1;\boldsymbol{\pi},\boldsymbol{T}) &= \sum_{i=1}^{p} B_i\log(\pi_i) + \sum_{i=1}^{p}\sum_{j\neq i}\Big[N_{ij}\log(\lambda_{ij}) - Z_i\lambda_{ij}\Big] + \sum_{i=1}^{p} N_i\Big[\log(\lambda_{i,p+1}) - \lambda_{i,p+1}Z_i\Big]\\ &= \sum_{i=1}^{p} B_i\log(\pi_i) + \sum_{i=1}^{p}\sum_{j\neq i}\Big[N_{ij}\log(t_{ij}) - Z_it_{ij}\Big] + \sum_{i=1}^{p} N_i\left[\log(t_i) - t_iZ_i\right].\end{aligned}$$

It is then easy to see that the maximum of $\ell_c(\tau_1;\boldsymbol{\pi},\boldsymbol{T})$ is attained for

$$\hat{\pi}_i = \frac{B_i}{n}, \quad \hat{t}_{ij} = \frac{N_{ij}}{Z_i}, \quad \text{and} \quad \hat{t}_i = \frac{N_i}{Z_i}.$$

Another important observation is that $\ell_c(\tau_1;\boldsymbol{\pi},\boldsymbol{T})$ is linear in the sufficient statistics B_i, N_i, N_{ij} and Z_i. This is equivalent to the model being a (curved) exponential family.

(b) The EM–algorithm

In this case, E–step reduces to computing the conditional expected values of the sufficient statistics given (incomplete) data, and the M–step will be as the estimates above, replacing the sufficient statistics by their conditional expectations. Indeed, the E–step becomes

$$\begin{aligned}&\mathbb{E}_{\boldsymbol{\theta}_n}\left(\ell_c(\boldsymbol{\theta};\boldsymbol{X}) \mid \boldsymbol{Y} = \boldsymbol{y}\right)\\ &\quad= \sum_{i=1}^{p}\mathbb{E}(B_i|\boldsymbol{Y}=\boldsymbol{y})\log(\pi_i) + \sum_{i=1}^{p}\sum_{j\neq i}\Big[\mathbb{E}(N_{ij}|\boldsymbol{Y}=\boldsymbol{y})\log(t_{ij}) - \mathbb{E}(Z_i|\boldsymbol{Y}=\boldsymbol{y})t_{ij}\Big]\\ &\qquad+ \sum_{i=1}^{p}\mathbb{E}(N_i|\boldsymbol{Y}=\boldsymbol{y})\left[\log(t_i) - t_iZ_i\right]\end{aligned}$$

while the M–step amounts to

$$\hat{\pi}_i = \frac{\mathbb{E}(B_i|\boldsymbol{Y}=\boldsymbol{y})}{n}, \quad \hat{t}_{ij} = \frac{\mathbb{E}(N_{ij}|\boldsymbol{Y}=\boldsymbol{y})}{\mathbb{E}(Z_i|\boldsymbol{Y}=\boldsymbol{y})}, \quad \text{and} \quad \hat{t}_i = \frac{\mathbb{E}(N_i|\boldsymbol{Y}=\boldsymbol{y})}{\mathbb{E}(Z_i|\boldsymbol{Y}=\boldsymbol{y})}.$$

Hence, it is enough to compute the conditional expectation.

First consider $n = 1$ and $y = \tau_1$. Then

$$\begin{aligned}\mathbb{E}(B_i \mid Y = y) &= \mathbb{E}(1\{Z(0) = i\} \mid Y = y)\\ &= \mathbb{P}(Z(0) = i \mid Y = y)\\ &= \frac{\mathbb{P}(Z(0) = i, Y \in \mathrm{d}y)}{\mathbb{P}(Y \in \mathrm{d}y)}\\ &= \frac{\mathbb{P}(Z(0) = i)\mathbb{P}(Y \in \mathrm{d}y \mid X_0 = i)}{\mathbb{P}(Y \in \mathrm{d}y)}\\ &= \frac{\pi_i \boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}y}\boldsymbol{t}}{\boldsymbol{\pi}\mathrm{e}^{\boldsymbol{T}y}\boldsymbol{t}}.\end{aligned}$$

Regarding the occupations in state i,

$$\begin{aligned}\mathbb{E}(Z_i \mid Y = y) &= \mathbb{E}\left(\int_0^y 1\{Z(u) = i\}\,\mathrm{d}u \;\middle|\; Y = y\right)\\ &= \int_0^y \mathbb{P}(Z(u) = i \mid Y = y)\,\mathrm{d}u\\ &= \int_0^y \frac{\mathbb{P}(Y \in dy \mid Z(u) = i)\mathbb{P}(Z(u) = i)}{\mathbb{P}(Y \in \mathrm{d}y)}\,\mathrm{d}u\\ &= \int_0^y \frac{\boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}(y-u)}\boldsymbol{t}\boldsymbol{\pi}\mathrm{e}^{\boldsymbol{T}u}\boldsymbol{e}_i}{\boldsymbol{\pi}\mathrm{e}^{\boldsymbol{T}y}\boldsymbol{t}}\,\mathrm{d}u.\end{aligned}$$

This integral can be computed using Van Loan by

$$\exp\left(\begin{pmatrix}\boldsymbol{T} & \boldsymbol{t}\boldsymbol{\pi}\\ \boldsymbol{0} & \boldsymbol{T}\end{pmatrix} y\right) = \begin{pmatrix}\mathrm{e}^{\boldsymbol{T}y} & \int_0^y \mathrm{e}^{\boldsymbol{T}(y-u)}\boldsymbol{t}\boldsymbol{\pi}\mathrm{e}^{\boldsymbol{T}u}\,\mathrm{d}u\\ \boldsymbol{0} & \mathrm{e}^{\boldsymbol{T}y}\end{pmatrix}, \tag{30}$$

which then provides all ingredients for computing $\mathbb{E}(Z_i \mid Y = y)$.

Converning N_{ij}, we write

$$\begin{aligned}\mathbb{E}(N_{ij} \mid Y = y) &= \frac{\mathbb{E}(N_{ij}1\{Y \in \mathrm{d}y\})}{f_Y(y)}\\ &= \sum_\ell \frac{\mathbb{E}(N_{ij}1\{Z(y-) = \ell, Y \in [y, y+\mathrm{d}y)\})}{f_Y(y)\,\mathrm{d}y}\\ &= \sum_\ell \frac{\mathbb{E}(N_{ij} \mid Z(y-) = \ell, Y \in [y, y+\mathrm{d}y))\mathbb{P}(Y \in [y, y+\mathrm{d}y) \mid Z(y-) = \ell)\mathbb{P}(Z(y-) = \ell)}{f_Y(y)\,\mathrm{d}y}\\ &= \sum_\ell \frac{\mathbb{E}(N_{ij} \mid Z(y-) = \ell)t_\ell\,\mathrm{d}y\mathbb{P}(Z(y-) = \ell)}{f_Y(y)\,\mathrm{d}y}\\ &= \sum_\ell \frac{\mathbb{E}(N_{ij}1\{Z(y-) = \ell\})}{f_Y(y)}t_\ell\\ &= \sum_{k,\ell} \pi_k \frac{\mathbb{E}(N_{ij}1\{Z(y-) = \ell\} \mid Z(0) = k)}{f_Y(y)}t_\ell.\end{aligned}$$

Now the expectation

$$\mathbb{E}(N_{ij}1\{Z(y-) = \ell\} \mid Z(0) = k) = \mathbb{E}(N_{ij}1\{Z(y) = \ell\} \mid Z(0) = k)$$

is nothing but the partial reserve (with zero interest rate) for the case where the only payment is 1 when there is a jump from i to j, and zero otherwise, given $Z(0) = k$ and on the event that $Z(y) = \ell$. Hence, it must be the $k\ell$th element of

$$\int_0^y \mathrm{e}^{\boldsymbol{T}(y-u)} \boldsymbol{e}_i \boldsymbol{e}_j' t_{ij} \mathrm{e}^{\boldsymbol{T}u} \,\mathrm{d}u,$$

i.e.

$$\begin{aligned}\mathbb{E}(N_{ij} 1\{Z(y-) = \ell\} \mid Z(0) = k) &= \int_0^y \boldsymbol{e}_k' \mathrm{e}^{\boldsymbol{T}(y-u)} \boldsymbol{e}_i \boldsymbol{e}_j' t_{ij} \mathrm{e}^{\boldsymbol{T}u} \boldsymbol{e}_\ell \,\mathrm{d}u \\ &= t_{ij} \boldsymbol{e}_j' \int_0^y \mathrm{e}^{\boldsymbol{T}(y-u)} \boldsymbol{e}_\ell \boldsymbol{e}_k' t_{ij} \mathrm{e}^{\boldsymbol{T}u} \,\mathrm{d}u \boldsymbol{e}_i.\end{aligned}$$

Thus

$$\begin{aligned}\mathbb{E}(N_{ij} \mid Y = y) &= t_{ij} \sum_{k,\ell} \pi_k t_\ell \boldsymbol{e}_j' \int_0^y \mathrm{e}^{\boldsymbol{T}(y-u)} \boldsymbol{e}_\ell \boldsymbol{e}_k' t_{ij} \mathrm{e}^{\boldsymbol{T}u} \,\mathrm{d}u \boldsymbol{e}_i \\ &= t_{ij} \boldsymbol{e}_j' \int_0^y \mathrm{e}^{\boldsymbol{T}(y-u)} \boldsymbol{t} \boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}u} \,\mathrm{d}u \boldsymbol{e}_i.\end{aligned}$$

Again, this expression can be obtained from (30).

Finally, for N_i we get

$$\begin{aligned}\mathbb{E}(N_i \mid Y = y) &= \mathbb{P}(Z(y-) = i \mid Y = y) \\ &= \frac{\mathbb{P}(Y \in \mathrm{d}y \mid Z(y-) = i)\mathbb{P}(Z(y-) = i)}{\mathbb{P}(Y \in \mathrm{d}y)} \\ &= \frac{\boldsymbol{\pi} e^{\boldsymbol{T}y} \boldsymbol{e}_i t_i}{\boldsymbol{\pi} e^{\boldsymbol{T}y} \boldsymbol{t}}.\end{aligned}$$

For n datapoints, we simply have to sum over the formulae:

$$\begin{aligned}\mathbb{E}(B_i \mid Y = y) &= \sum_{k=1}^n \frac{\pi_i \boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}} \\ \mathbb{E}(Z_i \mid Y = y) &= \sum_{k=1}^n \int_0^{\tau_k} \frac{\boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}(\tau_k - u)} \boldsymbol{t} \boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}u} \boldsymbol{e}_i}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}} \,\mathrm{d}u \\ \mathbb{E}(N_{ij} \mid Y = y) &= \sum_{k=1}^n t_{ij} \boldsymbol{e}_j' \int_0^{\tau_k} \mathrm{e}^{\boldsymbol{T}(\tau_k - u)} \boldsymbol{t} \boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}u} \,\mathrm{d}u \boldsymbol{e}_i \\ \mathbb{E}(N_i \mid Y = y) &= \sum_{k=1}^n \frac{\boldsymbol{\pi} e^{\boldsymbol{T}\tau_k} \boldsymbol{e}_i t_i}{\boldsymbol{\pi} e^{\boldsymbol{T}\tau_k} \boldsymbol{t}}.\end{aligned}$$

(c) Weighted data.

Now suppose that there are repeated data points: there are M different data points τ_k, which are all repeated $n_k \geq 1$ times such that $n_1 + \cdots n_M = n$. Then we get

$$\mathbb{E}(B_i \mid Y = y) = \sum_{k=1}^M n_k \frac{\pi_i \boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}}$$

$$\mathbb{E}(Z_i \mid Y = y) = \sum_{k=1}^{M} n_k \int_0^{\tau_k} \frac{\boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}(\tau_k - u)} \boldsymbol{t}\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}u} \boldsymbol{e}_i}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}} \,\mathrm{d}u$$

$$\mathbb{E}(N_{ij} \mid Y = y) = \sum_{k=1}^{M} n_k t_{ij} \boldsymbol{e}_j' \int_0^{\tau_k} \mathrm{e}^{\boldsymbol{T}(\tau_k - u)} \boldsymbol{t}\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}u} \,\mathrm{d}u \boldsymbol{e}_i$$

$$\mathbb{E}(N_i \mid Y = y) = \sum_{k=1}^{M} n_k \frac{\boldsymbol{\pi} e^{\boldsymbol{T}\tau_k} \boldsymbol{e}_i t_i}{\boldsymbol{\pi} e^{\boldsymbol{T}\tau_k} \boldsymbol{t}}.$$

Then

$$\hat{\pi}_i = \frac{\mathbb{E}(B_i | \boldsymbol{Y} = \boldsymbol{y})}{n} = \sum_{k=1}^{M} \frac{n_k}{n} \frac{\pi_i \boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}}.$$

Similarly, if we divide through with n in the remaining formulae,

$$\frac{\mathbb{E}(Z_i \mid Y = y)}{n} = \sum_{k=1}^{M} \frac{n_k}{n} \int_0^{\tau_k} \frac{\boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}(\tau_k - u)} \boldsymbol{t}\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}u} \boldsymbol{e}_i}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}} \,\mathrm{d}u$$

$$\frac{\mathbb{E}(N_{ij} \mid Y = y)}{n} = \sum_{k=1}^{M} \frac{n_k}{n} t_{ij} \boldsymbol{e}_j' \int_0^{\tau_k} \mathrm{e}^{\boldsymbol{T}(\tau_k - u)} \boldsymbol{t}\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}u} \,\mathrm{d}u \boldsymbol{e}_i$$

$$\frac{\mathbb{E}(N_i \mid Y = y)}{n} = \sum_{k=1}^{M} \frac{n_k}{n} \frac{\boldsymbol{\pi} e^{\boldsymbol{T}\tau_k} \boldsymbol{e}_i t_i}{\boldsymbol{\pi} e^{\boldsymbol{T}\tau_k} \boldsymbol{t}},$$

then the M–step estimates remain the same.

The fraction $w_i = n_i/n$ is a weight. In a histogram, the data points are in the bins, and the weight is the probability contribution. We may take e.g., the midpoint of a bin as a representative for the data falling into that bin.

If we want to fit to a histogram, it can be a problem if the sum of the weights is (much) less than 1. This can happen if the survival function we consider, or equivalently the bond price, is not observed at very many points. If the bond price for maturity 30 years, for example, equals 0.38, then the survival function of the corresponding phase-type distribution at the highest observed point is 0.38, significantly larger than the other weights. This is because we do not have observations for maturities larger than 30. In this case, it is better to consider the largest maturity to be a right-censored data point.

(d) Censored data.

We consider only right censoring, since we do not need other types. Assuming that τ_n, the largest value, is the only right censored point in the sample, we must modify the expectation of $\mathbb{E}(B_i \mid Y = \tau_n)$ to $\mathbb{E}(B_i \mid Y > \tau_n)$ and similarly for the rest. Thus we get

$$\mathbb{E}(B_i \mid Y = y) = \sum_{k=1}^{n-1} \frac{\pi_i \boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}} + \frac{\pi_i \boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}\tau_n} \boldsymbol{e}}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_n} \boldsymbol{e}}$$

$$\mathbb{E}(Z_i \mid Y = y) = \sum_{k=1}^{n-1} \int_0^{\tau_k} \frac{\boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}(\tau_k - u)} \boldsymbol{t}\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}u} \boldsymbol{e}_i}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{t}} \,\mathrm{d}u + \int_0^{\tau_n} \frac{\boldsymbol{e}_i' \mathrm{e}^{\boldsymbol{T}(\tau_n - u)} \boldsymbol{e}\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}u} \boldsymbol{e}_i}{\boldsymbol{\pi} \mathrm{e}^{\boldsymbol{T}\tau_k} \boldsymbol{e}} \,\mathrm{d}u$$

$$\mathbb{E}(N_{ij} \mid Y = y) = \sum_{k=1}^{n-1}\sum_{k=1}^{n-1} t_{ij}\boldsymbol{e}_j' \int_0^{\tau_k} \mathrm{e}^{\boldsymbol{T}(\tau_k - u)}\boldsymbol{t}\boldsymbol{\pi}\mathrm{e}^{\boldsymbol{T}u}\,\mathrm{d}u\boldsymbol{e}_i + t_{ij}\boldsymbol{e}_j' \int_0^{\tau_n} \mathrm{e}^{\boldsymbol{T}(\tau_n - u)}\boldsymbol{e}\boldsymbol{\pi}\mathrm{e}^{\boldsymbol{T}u}\,\mathrm{d}u\boldsymbol{e}_i$$

$$\mathbb{E}(N_i \mid Y = y) = \sum_{k=1}^{n-1} \frac{\boldsymbol{\pi} e^{\boldsymbol{T}\tau_k}\boldsymbol{e}_i t_i}{\boldsymbol{\pi} e^{\boldsymbol{T}\tau_k}\boldsymbol{t}}.$$

Hence, all density expressions are simply replaced by survival function expressions instead, and in the last equation, there is no contribution to N_i for the right censored observation since absorption did not take place yet by time τ_n.

(e) Fitting to IPH.

It is possible to fit to an IPH$(\boldsymbol{\pi}, \boldsymbol{T}, \lambda)$ with relatively minor modifications using Theorem 3.10. In each iteration, the data are transformed back to corresponding PH$(\boldsymbol{\pi}, \boldsymbol{T})$ data, in which the appropriate E and M steps are then performed. This results in updated parameters $(\boldsymbol{\pi}, \boldsymbol{T})$. The function $\lambda(x) = \lambda(x;\theta)$ may be parameter dependent as well, and a separate optimisation of λ is performed. In the next iteration, data are no transformed according to the updated $\lambda(x;\theta)$, etc.

It can be shown that this is again an EM–algorithm.

(f) Computation. Fortunately, we can make use of the R package "matrixdist" for fitting to weighted, censored phase–type data.

5. Estimation of PH bond models

Consider bond prices $B(0,T_i)$ available at different maturities $T_1, T_2, ..., T_n$. Then according to Theorem 3.12 we have that

$$B(0,T_i) = \boldsymbol{\pi}\boldsymbol{D}(0,T_i)\boldsymbol{e} = \mathrm{e}^{\rho T_i}\mathbb{P}(\tau > T_i), \quad i = 1,2,...,n,$$

for some $\rho > 0$ and where $\tau \sim$ IPH$(\boldsymbol{\pi}, \boldsymbol{M}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u)) - \rho\boldsymbol{I})$. Then ρ must satisfy that

$$\mathrm{e}^{-\rho T_i}B(0,T_i) \le 1, \quad i = 1,2,...,n.$$

This can be achieved by choosing

$$\rho = \max_{i\in\{1,...,n\}} \left(\frac{\log B(0,T_i)}{T_i}\right).$$

In the life–insurance context in Denmark, by regulation, the bond prices (discounting factors) must be computed from discrete forward rates, $f_d(0,T_i)$, published by the Danish Financial Supervisory Authority. Thus

$$B(0,T_i) = (1 + f_d(0,T_i))^{-T_i}$$

from which

$$\frac{\log B(0,T_i)}{T_i} = -\log(1 + f_d(0,T_i)).$$

Hence

$$\rho = \max_i \left(-\log(1+f_d(0,T_i))\right) = -\min_i \log(1+f_d(0,T_i)). \tag{31}$$

Hence calibrating to data $B(0,T_i)$, $i=1,...,n$ can be done by fitting PH or IPH distributions to $\mathrm{e}^{-\rho T_i}B(0,T_i)$ using an EM algorithm. The possible interest rates can either be selected by the EM algorithm (referred to as unrestricted interest rates) or fixed to values (or functions) of our choice (restricted interest rates).

In the former case, we obtain a maximum likelihood estimate $(\hat{\boldsymbol{\pi}},\hat{\boldsymbol{T}}(x))$ for the parameters. The estimate for $\boldsymbol{M}(x)$ is then readily obtained from

$$\hat{\boldsymbol{M}}(x) = \hat{\boldsymbol{T}}(x) + \boldsymbol{\Delta}(\boldsymbol{t}(x)).$$

To find the induced interest rates, we also have from Theorem 3.12 that

$$\hat{\boldsymbol{T}}(x) = \hat{\boldsymbol{M}}(x) - \boldsymbol{\Delta}(\boldsymbol{r}(x)) - \rho\boldsymbol{I}$$

so we conclude that the estimated exit rates $\boldsymbol{t}(x)$ must satisfy

$$\boldsymbol{t}(x) = \boldsymbol{r}(x) + \rho\boldsymbol{e},$$

where $\boldsymbol{e}$ is the vector of ones. Hence, the induced interest rates are given by

$$\boldsymbol{r}(x) = \boldsymbol{t}(x) - \rho\boldsymbol{e}.$$

Neither the transition rates nor the interest rates are unique, but the resulting discount factor (bond price) is invariant under different representations, which is all that matters regarding reserving in the insurance context.

If, in turn, we decide to choose the possible range of interest rates $r_i(x)$ ourselves, then the EM–algorithm is modified not to update the exit rates.

In this case, the exit rates will be fixed at

$$\boldsymbol{t}(x) = \boldsymbol{r}(x) + \rho\boldsymbol{e}$$

so

$$\hat{\boldsymbol{M}}(x) = \hat{\boldsymbol{T}}(x) + \boldsymbol{\Delta}(\boldsymbol{r}(x)) + \rho\boldsymbol{I}.$$

While the parametrisation of the transition rates may not be unique, the interest rates remain fixed.

Remark 3.16: By fixing the exit rates, the class of phase–type distributions may no longer be dense. If, for example, we fix $\boldsymbol{r}(x) = r\boldsymbol{e}$, then only the exponential distribution with parameter r is reachable.

We now present two examples of fitting to real data and one example to a theoretical model. The estimation is computed using the R–package *matrixdist*.

Example 3.17 (Fitting to observed bond prices with restricted interest rates)**:** Bond prices, $B(0,T)$ as of 31/12/2003 (time zero) with maturities $T = 1,2,...,30$ years are available from the Danish Financial Supervisory Authority and given by 0.9755051, 0.9434934, 0.9059545, 0.8679149, 0.8251354, 0.7857250, 0.7472528, 0.7075066, 0.6679984, 0.6286035, 0.5951316, 0.5625969, 0.5310441, 0.5005108, 0.4710280, 0.4448469, 0.4197550, 0.3958013, 0.3728296, 0.3508858, 0.3319907, 0.3140894, 0.2970098, 0.2808430, 0.2654229, 0.2508400, 0.2369349, 0.2237965, 0.2112725, 0.1994495, respectively.

This corresponds to an empirical survival distribution to which we can then fit phase–type distributions of different dimensions. Regarding the discretisation, we let $0.5+i$, $i = 0,...,29$ denote the data points with probability mass $B(i) - B(i+1)$, where $B(0) = 1$, and a right censored data point at 30 with probability mass $B(30) = 0.1994495$. Since all observed bond prices are less than one, we have $\rho = 0$, corresponding to an environment with non-negative interest rates.

We used $p = 2,3,4,5,10$ and 15 phases, with state–wise interest rates being $r_i^p = i/(10p)$, $i = 1,...,p$ for the different dimensions p. Underlying this choice is the assumption that the interest rates fluctuate between 1% and 10%, and the r_i's are obtained as the points that divide the interval $[0,0.1]$ into p, including the right endpoint. The vectors $\boldsymbol{r}^p = (r_1^p,...,r_p^p)'$ will serve as exit rate vectors of the phase–type distributions to be fitted.

In Figure 3 (left), we have plotted the phase–type fits to the empirical survival curve for dimensions $p = 2,3,4,5$. At dimension 3, we obtain a decent fit and excellent fits for dimensions 4 and 5. The likelihood values for 4 cases are -3.178171, -3.16838,-3.166633, and -3.166182. Further experimentation with dimensions 10 and 15 resulted in likelihoods of -3.165002 and -3.164654, respectively. However, the plots of bond prices and yields are indistinguishable from the plots corresponding to dimension 5, see Figure 4. We can also assess the quality of the fits by plotting the estimated density function vs. the weighted data, shown in Figure 5. Again the plots for dimensions $p = 5,10,15$ are almost indistinguishable. Therefore, we conclude that dimension 4 or 5 will suffice to *approximate* the bond prices.

The estimates of the sub–intensity matrix $\boldsymbol{M} - \boldsymbol{\Delta}(\boldsymbol{r})$ (under a risk neutral measure $\mathbb{Q}$) for dimensions $p = 3,4,5$ are given by

$$\begin{pmatrix} -0.13 & 0.1 & 0 \\ 0 & -0.41 & 0.34 \\ 0.14 & 0 & -0.24 \end{pmatrix}, \begin{pmatrix} -0.25 & 0.22 & 0.01 & 0 \\ 0.14 & -1.11 & 0.75 & 0.18 \\ 0.06 & 0.29 & -0.63 & 0.2 \\ 0.09 & 0.22 & 0.65 & -1.05 \end{pmatrix}, \begin{pmatrix} -0.26 & 0.02 & 0.06 & 0.07 & 0.08 \\ 0.07 & -1.68 & 0.69 & 0.23 & 0.65 \\ 0.19 & 0.32 & -1.99 & 0.93 & 0.48 \\ 0.04 & 0.35 & 0.27 & -1.2 & 0.46 \\ 0.07 & 0.82 & 0.07 & 0.8 & -1.85 \end{pmatrix}.$$

To fit the bond prices, the initial distributions of Markov processes were all on the form $(1,0,...,0)$ of appropriate dimension, i.e., initiation in state 1. □

Example 3.18 (Fitting to 2019 bond prices with unrestricted interest rates)**:** To illustrate the applicability of our methods also in the case of a negative interest rate environment, we can instead fit to bond prices as of 31/12/2019 from the Danish Financial Supervisory

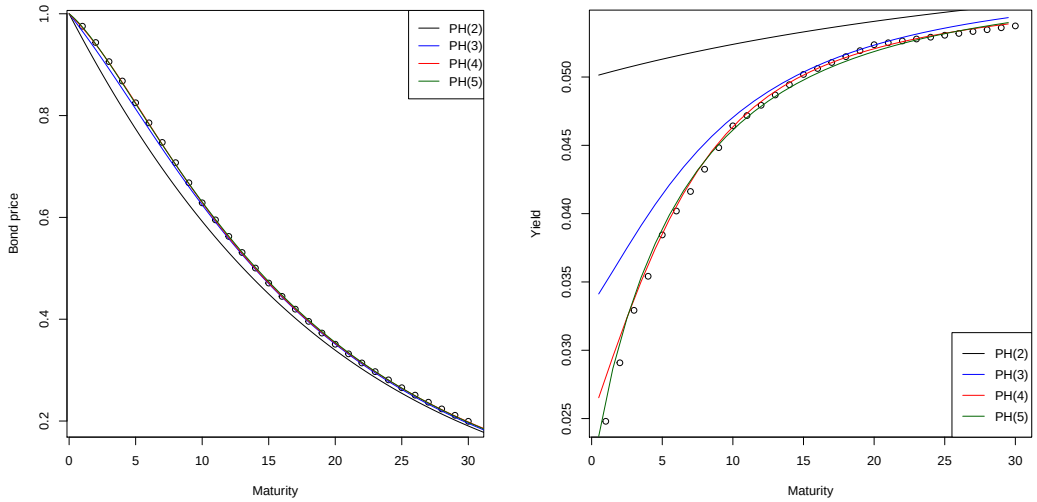

Figure 3. Phase–type fits to Zero-coupon bond prices (left) and corresponding yield curves (right) for dimension $p = 2, 3, 4, 5$.

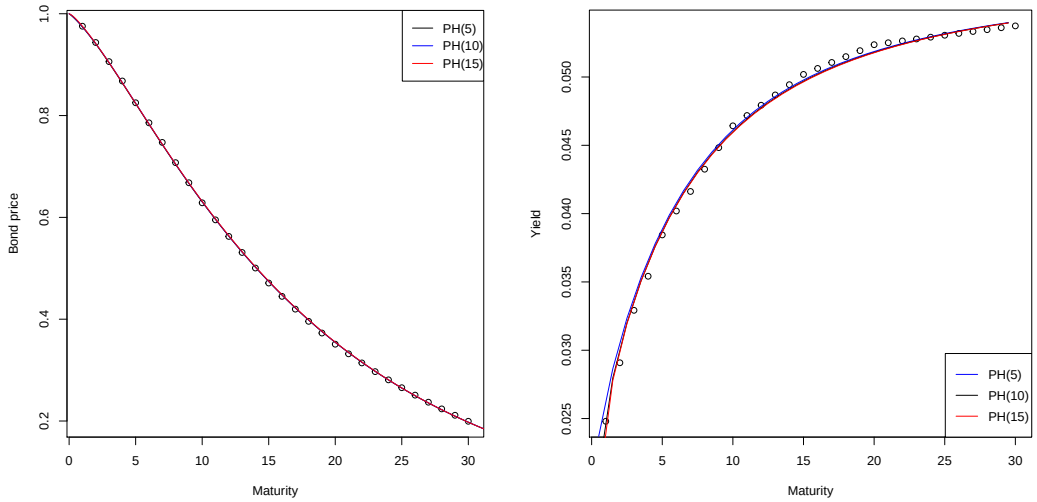

Figure 4. Phase–type fits to Zero-coupon bond prices (left) and corresponding yield curves (right) for dimension $p = 5, 10, 15$.

Authority; this dataset consists of maturities of $T = 1, 2, ..., 120$ years. In this case, we let the EM algorithm choose the necessary positive and negative interest rates.

The first five years have bond prices above one and given by 1.00231736, 1.00403337, 1.00445679, 1.00382807, and 1.00197787, which reflects the (slightly) negative interest rate environment at the time. From (31), we get $\rho = 0.002314677$ as the exponential factor to down-scale prices to below one. In Figure 6, we show the phase–type fits to the bond prices. We have used the subclass of time–homogeneous Coxian distributions, where initiation is

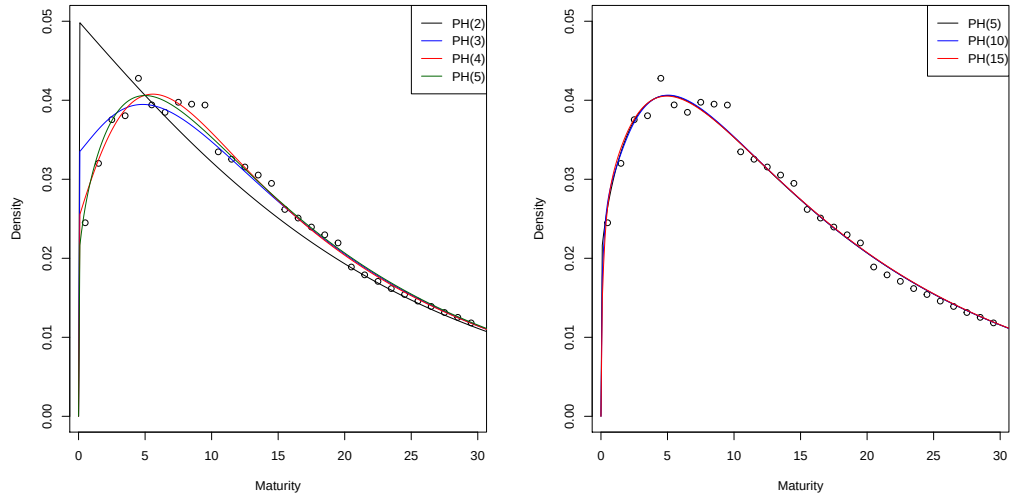

Figure 5. Fitted phase–type densities vs. weighted data for $p = 2,3,4,5$ (left) and $p = 5,10,15$ (right).

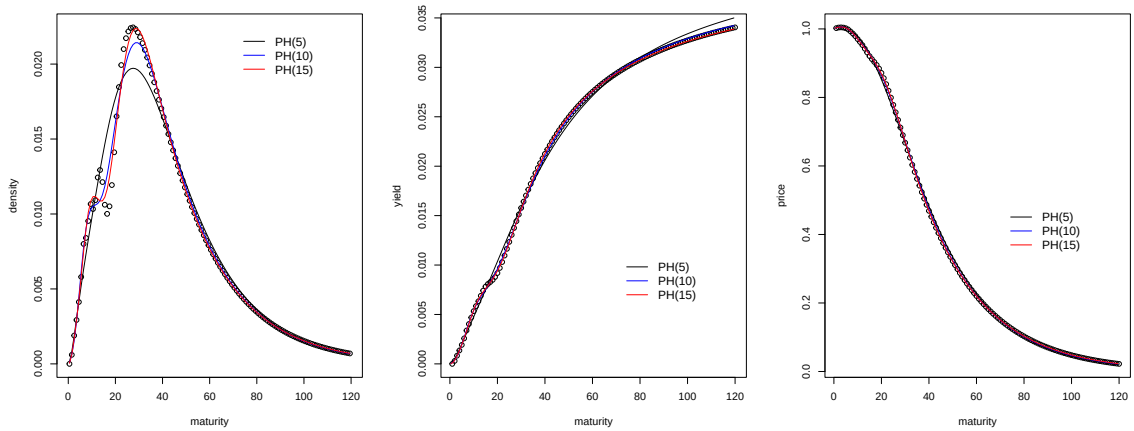

Figure 6. Fitted phase-type densities (left), corresponding yield curves (middle) and bond prices (right) for dimensions $p = 5,10,15$ based on bond price data as of 31/12/2019.

always in state 1, and the only possible transitions are from a state, i say, to the following, $i+1$, or to exit to the absorbing state.

If the primary purpose is using the fits as a discounting factor in a life–insurance model, then probably all fits could be used (right plot). If the yield curve fitting is the concern, then only dimensions 10 and 15 seem to catch the appropriate curvature. Regarding the probability density of the phase–type, the 15-dimensional fit is the best.

To exemplify, we consider the 10–dimensional fit. The fitted intensity matrix, $\hat{\boldsymbol{M}}$, for $\{X(u)\}_{u\geq 0}$, is given by

$$\begin{pmatrix} -0.5212 & 0.5212 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & -0.5212 & 0.5212 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & -0.5185 & 0.5185 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & -0.5161 & 0.5161 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.5152 & 0.5152 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.4664 & 0.4664 & 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.3099 & 0.3099 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.3099 & 0.3099 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.3099 & 0.3099 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \end{pmatrix}$$

The matrix contains six different parameter values. The matrix structure is carried over from the phase–type fit to the (discounted) bond prices. The blocks with the same parameters correspond to Erlang blocks, i.e. convolution of exponential distributions with the same parameter.

The induced (estimated) interest rates (in %) are, respectively,

$$-\rho, -\rho, 0.03468739, 0.28218594, -\rho, 4.64627655, -\rho, -\rho, -\rho, 3.86252219.$$

These should also be counted as parameters. □

Example 3.19 (Fitting to a Vasiček model)**:** Consider the Vasiček model

$$dr(u) = k(\theta - r(u))\,\mathrm{d}u + \sigma dW^{\mathbb{Q}}(u),$$

where $W^{\mathbb{Q}}(u)$ denotes the Wiener process under the equivalent martingale measure $\mathbb{Q}$ where $k = 0.5$, $\theta = 0.1$, $\sigma = 0.2$ and $r(0) = 0.01$. We approximate this model a state–wise deterministic interest model, $r_{X(u)}(u)$ based on an underlying Markov process with 5 and 10 states, using $\boldsymbol{r}(u) = (r_1, ..., r_5) = (0.02, 0.04, 0.06, 0.08, 0.1)$ and $\boldsymbol{r}(u) = (r_1, ..., r_{10}) = (0.01, ..., 0.1)$, respectively. We take the Markov processes as time–homogeneous with 5 and 10 states, respectively, since the Vasiček model is stationary. We obtained the following phase–type representations for $p = 5$

$$\boldsymbol{\pi} = \begin{pmatrix} 0.76 \\ 0.24 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{M} - \boldsymbol{\Delta}(\boldsymbol{r}) = \begin{pmatrix} -0.02 & 0 & 0 & 0 & 0 \\ 0 & -1.72 & 0 & 1.68 & 0 \\ 0 & 0 & -2.19 & 0 & 2.13 \\ 0 & 0 & 1.09 & -1.17 & 0 \\ 0.35 & 0 & 0 & 0 & -0.45 \end{pmatrix}.$$

and, for $p = 10$, $\boldsymbol{\pi} = (0.63, 0.37, 0, 0, 0, 0, 0, 0, 0, 0)'$ and

$$\boldsymbol{M} - \boldsymbol{\Delta}(\boldsymbol{r}) = \begin{pmatrix} -0.06 & 0.01 & 0.01 & 0.01 & 0 & 0 & 0.01 & 0 & 0 & 0 \\ 0 & -5.89 & 5.87 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -7.21 & 0 & 0 & 0 & 0 & 0 & 7.18 & 0 \\ 0 & 0.06 & 0.02 & -2.02 & 0.14 & 0.05 & 0.1 & 1.51 & 0 & 0.09 \\ 0 & 0.12 & 0.01 & 0.92 & -2.58 & 0.01 & 0.08 & 1.3 & 0 & 0.08 \\ 0 & 0.01 & 0 & 3.28 & 0.86 & -4.32 & 0.11 & 0 & 0 & 0 \\ 0.01 & 0.22 & 0.07 & 2.18 & 2.25 & 0.44 & -5.9 & 0.54 & 0.04 & 0.09 \\ 0 & 0 & 0 & 0 & 0.01 & 0 & 0 & -2.61 & 0 & 2.51 \\ 0 & 0 & 0 & 0 & 0.02 & 2.75 & 0.13 & 0 & -2.99 & 0 \\ 0.42 & 0 & 0 & 0 & 0 & 0.01 & 0 & 0.08 & 0.01 & -0.63 \end{pmatrix}$$

respectively. The corresponding intensity matrices, $\hat{\boldsymbol{M}}$, then provide us with risk–neutral transition rates between the possible interest rates, $r = 0.02, 0.04, 0.06, 0.1$ or $r = 0.01, ..., 0.1$.

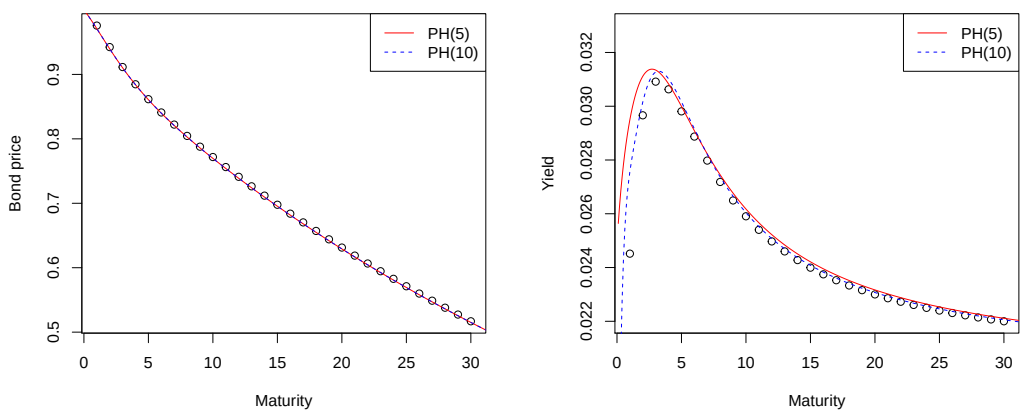

Figure 7. Phase–type density approximations to corresponding Vasiček values. The red curve corresponds to an approximation with a 5–dimensional state–space, whereas the blue curve corresponds to a 10–dimensional state–space.

Figure 7 illustrates the approximations. While the bond prices provide excellent fits, the yield curves are more revealing in exposing discrepancies.

The assessment regarding the quality of the fits depends on their use. When computing reserves or higher-order moments in life insurance with stochastic interest rates being independent of the biometric process, the interest rates are merely used as a discounting factor, in which case even a 5–state approximation to the Vasiček model with a state-wise deterministic interest rate model will provide a good fit. □

CHAPTER 4

Multi–state Markov models with Markovian interest rates

1. Markovian interest rates

Consider a Markov process $\{X_t\}_{t\geq 0}$ taking values in a state–space E, which defines the biometric states of an insured.

In this chapter, in turn, we will assign a possibly different interest rate to each state. Let

$$\boldsymbol{r}(t) = (r_i(t))_{i\in E}$$

denote the vector of interest rates in the different states.

The state–space E may either be the exact same for both the interest rate process and biometric process, or entirely different state–spaces if, e.g. the Markovian interest rate model of Chapter 3 is to be employed.

In that case, there is a Markov jump process $X_r = \{X_r(t)\}_{t\geq 0}$ with state–space $E_r = \{1,2,...,p\}$ and intensity matrix $\boldsymbol{\Lambda}_r(t) = \{\lambda^r_{ij}(t)\}_{t\geq 0}$, say, such that the corresponding bond prices $B(t,T)$ are given as in Theorem 3.12.

Additionally, we let $X_b = \{X_b(t)\}_{t\geq 0}$ denote the Markov jump process governing the transition between the biometric states with the state–space $E_b = \{1,2,...,q\}$ and intensity matrix $\boldsymbol{\Lambda}_b(t) = \{\lambda^b_{ij}(t)\}_{t\geq 0}$. Hence, the Markov jump process can be written in the form

$$X(t) = (X_b(t), X_r(t)) \tag{32}$$

with state–space $E = E_b \times E_r$.

The product space will be ordered lexicographically, which means that state (i,j) is identified with state $k(i,j) = i(p-1)+j,\ j = 1,...,p,\ i = 1,...,q$ of the state–space $\{1,2,...,pq\}$ and $(i,j) < (\tilde{i},\tilde{j})$ iff $k(i,j) < k(\tilde{i},\tilde{j})$.

The processes X_b and X_r may or may not be independent, and the payment processes likewise may or may not be independent of X_r. In the independent case the processes X_b and X_r are defined on each their state–space, and the common state–space will be the product set of the two. If the processes share states, with the possibility of having simultaneous jumps, then we obtain dependency of the processes. Such a case could, e.g. be a rise in the interest rate causing an increased intensity of jumping to surrender or free–policy states.

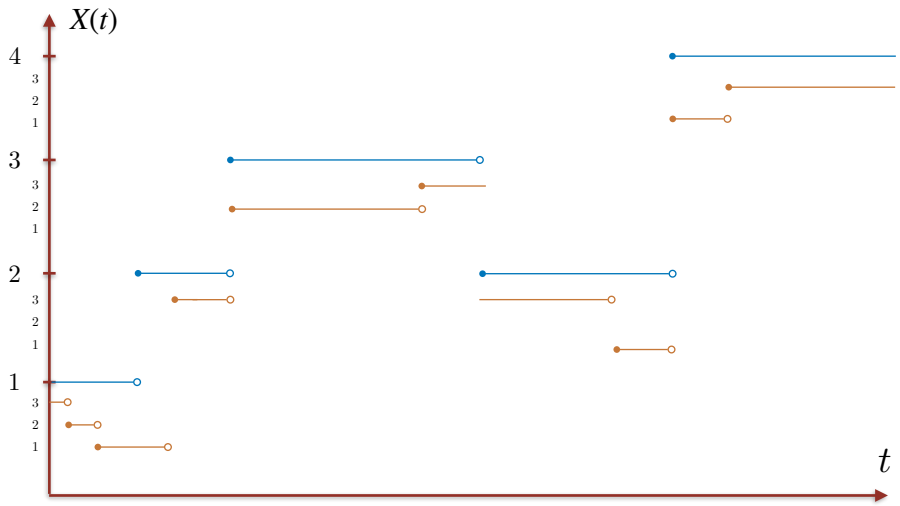

Figure 1. Lexicographical ordering: for each biometric state (blue), several sub-states (orange) define the underlying interest rate level.

2. The general setup

Let $\{X(t)\}_{t\geq 0}$ be a Markov process on the form (32) with state–space $E = E_b \times E_r$.

Definition 4.1: A payment process $\{B(t)\}_{t\geq 0}$ is a random function on the form

$$\mathrm{d}B(t) = \sum_{i\in E} 1\{X(t-) = i\}\left(b_i(t)\,\mathrm{d}t + \sum_{j\in E} b_{ij}(t)\,\mathrm{d}N_{ij}(t)\right), \tag{33}$$

where $b_i(t)$ are continuous payment rates (negative if premiums), and $b_{ij}(t)$ lump sum payments, which occur according to the counting measure $N_{ij}(t)$. The payment process defines the (random) **cash flow**.

The intensity matrix is decomposed into

$$\mathbf{\Lambda}(t) = \mathbf{\Lambda}^0(t) + \mathbf{\Lambda}^1(t), \tag{34}$$

where $\mathbf{\Lambda}^1(t)$ is a non–negative matrix and, consequently, $\mathbf{\Lambda}^0(t)$ a sub–intensity matrix, i.e. row sums are non–positive. The counting process $N_{ij}(t)$ is linked to the transitions of X in

the following way. Upon transition from i to j, $i \neq j$, in X at time t, a lump sum payment of $b_{ij}(t)$ will be triggered with probability

$$\frac{\lambda_{ij}^1(t)}{\lambda_{ij}^0(t)+\lambda_{ij}^1(t)}. \tag{35}$$

If $i=j$, then $N_{ii}(t)$ is an inhomogeneous Poisson process with intensity $\lambda_{ii}^1(t)$, and a lump sum during a sojourn in state i will then be triggered in $[t,t+\mathrm{d}t)$ with probability $\lambda_{ii}^1(t)\,\mathrm{d}t$. We set

$$\lambda_{ii}^0(t) = -\lambda_{ii}^1(t) - \sum_{k\neq i}\lambda_{ik}^1(t) - \sum_{k\neq i}\lambda_{ik}^0(t) = -\sum_{k}\lambda_{ik}^1(t) - \sum_{k\neq i}\lambda_{ik}^0(t).$$

Hence $\lambda_{ii}^1(t)$ are artificially added to the state i, which is compensated for in the diagonal for $\boldsymbol{\Lambda}^0(t)$, such that the intensity matrix $\boldsymbol{\Lambda}(t)$ remains unaffected.

To compute reserves and higher order moments, we let $\boldsymbol{b}(t) = (b_i(t))_{i\in E}$ denote the vector containing the continuous rates, and define matrices

$$\boldsymbol{B}(t) = \left\{b_{ij}(t)\right\}_{i,j\in E}, \tag{36}$$

$$\boldsymbol{R}(t) = \boldsymbol{\Lambda}^1(t)\bullet\boldsymbol{B}(t) + \boldsymbol{\Delta}(\boldsymbol{b}(t)), \tag{37}$$

$$\boldsymbol{C}^{(k)}(t) = \boldsymbol{\Lambda}^1(t)\bullet\boldsymbol{B}^{\bullet k}(t), \quad k\geq 2, \tag{38}$$

where $\boldsymbol{\Delta}(\boldsymbol{b}(t))$ denotes the diagonal matrix with $\boldsymbol{b}(t)$ as diagonal. The operator $\bullet$ denotes Schur (entrywise) matrix product, defined by $\boldsymbol{A}\bullet\boldsymbol{B} = \{a_{ij}b_{ij}\}$ for matrices $\boldsymbol{A} = \{a_{ij}\}$ and $\boldsymbol{B} = \{b_{ij}\}$.

Hence $\boldsymbol{B}(t)$ is the matrix containing the lump payments at transitions and Poisson arrivals during sojourns, $\boldsymbol{R}(t)$ is the matrix whose ij'th element is the expected reward accumulated during $[t,t+\mathrm{d}t)$ upon transition from i to j, or during a sojourn in state i if $i=j$. The $\boldsymbol{C}^{(k)}(t)$ matrix is more technical to be used when dealing with higher-order moments.

The interest rate is supposed to be Markovian, $r_{X(t)}(t)$, also generated by the same Markov process $\{X(t)\}_{t\geq 0}$.

In the following example, we consider the simplifications in the representations when assuming independence.

Example 4.2 (Independence)**:** If the processes X_b and X_r are independent, then the transition intensity matrix of X is of the form

$$\boldsymbol{\Lambda}(t) = \boldsymbol{\Lambda}_b(t)\oplus\boldsymbol{\Lambda}_r(t) = \boldsymbol{\Lambda}_b(t)\otimes\boldsymbol{I}_p + \boldsymbol{I}_q\otimes\boldsymbol{\Lambda}_r(t),$$

where $\oplus$ denotes the Kronecker sum, and where $\boldsymbol{I}_n$ denotes the identity matrix of dimension $n\times n$. We recall that the Kronecker product, $\otimes$, is defined by $\boldsymbol{A}\otimes\boldsymbol{B} = \{a_{ij}\boldsymbol{B}\}$, where $\boldsymbol{A} = \{a_{ij}\}$. The interest rates are not influenced by the biometric states, so

$$\boldsymbol{r}(t) = \boldsymbol{e}\otimes(r_1(t),...,r_p(t)),$$

where $\boldsymbol{e} = (1,1,...,1)'$. If the payment process is independent of the interest rate process X_r, then

$$\boldsymbol{B}(t) = \boldsymbol{B}^b(t) \otimes \boldsymbol{I}, \quad \boldsymbol{b}(t) = \boldsymbol{b}^b(t) \otimes \boldsymbol{e}$$

for some

$$\boldsymbol{b}^b(t) = \left(b_1^b(t), \ldots, b_q^b(t)\right)' \qquad \text{and} \qquad \boldsymbol{B}(t) = \left\{b_{ij}^b(t)\right\}_{i,j \in E_b}.$$

Example 4.3: In principle, the model allows for dependency between the interest rate and the biometric processes. This can be achieved simply by definining processes on the same state–space and being generated by the same underlying Markov jump process, in which transition in the interest rates and other rates may e.g. take place simultaneously.

3. Reserves under Markovian interest rates

Introduce the matrix of partial state–wise prospective reserves,

$$\boldsymbol{V}(s,t) = \left\{V_{ij}(s,t)\right\}_{i,j \in E},$$

$$V_{ij}(s,t) = \mathbb{E}\left(1\{X(t) = j\} \int_s^t \mathrm{e}^{-\int_s^x r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}B(x) \,\middle|\, X(s) = i\right).$$

With $\boldsymbol{D}(s,t)$, introduced in (23), modified to the setup of this section as

$$\boldsymbol{D}(s,t) = \prod_s^t \left(\boldsymbol{I} + [\boldsymbol{\Lambda}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u))]\,\mathrm{d}u\right).$$

We need the following theorem on the precise structure of $\boldsymbol{D}(s,t)$.

Theorem 4.4. *The elements of the matrix $\boldsymbol{D}(s,t) = \{d_{ij}(s,t)\}_{i,j \in E}$ satisfy*

$$d_{ij}(s,t) = \mathbb{E}\left(1\{X(t) = j\} \exp\left(-\int_s^t r_{X(u)}(u)\,\mathrm{d}u\right) \,\middle|\, X(s) = i\right), \quad s \le t.$$

for all $i,j \in E$.

Proof. We show that the matrix defined by the proposed formula for $d_{ij}(s,t)$ coincides with the product integral.

Conditioning on the state of $s + \mathrm{d}s$, we get that

$$\begin{aligned} & d_{ij}(s,t) \\ &= (1 + \lambda_{ii}(s)\,\mathrm{d}s) d_{ij}(s + \mathrm{d}s, t)(1 - r_i(s)\,\mathrm{d}s) + \sum_{k \neq i} \lambda_{ik}(s)\,\mathrm{d}s\, d_{kj}(s + \mathrm{d}s, t)(1 - r_i(s)\,\mathrm{d}s) \end{aligned}$$

$$= \quad d_{ij}(s+\mathrm{d}s,t)(1-r_i(s)\,\mathrm{d}s)+\lambda_{ii}(s)\,\mathrm{d}s\,d_{ij}(s+\mathrm{d}s,t)+\sum_{k\neq i}\lambda_{ik}(s)\,\mathrm{d}s\,d_{kj}(s+\mathrm{d}s,t)$$

so that

$$-\frac{\partial}{\partial s}d_{ij}(s,t) \quad = \quad -r_i(t)d_{ij}(s,t)+\sum_k \lambda_{ik}(t)d_{kj}(s,t). \tag{39}$$

In matrix form, this amounts to

$$\frac{\partial}{\partial s}\boldsymbol{D}(s,t) = -\left(\boldsymbol{\Lambda}(s)-\boldsymbol{\Delta}(\boldsymbol{r}(s))\right)\boldsymbol{D}(s,t). \tag{40}$$

Noting that $\boldsymbol{D}(t,t)=\boldsymbol{P}(t,t)=\boldsymbol{I}$, we hence conclude that result holds. □

Theorem 4.5. *The matrix of partial state-wise prospective reserves $\boldsymbol{V}(s,t)$ has the following integral representation:*

$$\boldsymbol{V}(s,t)=\int_s^t \boldsymbol{D}(s,x)\boldsymbol{R}(x)\boldsymbol{P}(x,t)\,\mathrm{d}x. \tag{41}$$

Proof. First, we notice that, for $i,j\in E$,

$$\begin{aligned}
V_{ij}(s,t) \quad &= \quad \sum_{k\in E}\mathbb{E}\left(1\{X(t)=j\}\int_s^t 1\{X(x)=k\}\mathrm{e}^{-\int_s^x r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}B(x)\middle| X(s)=i\right)\\
&= \sum_{k\in E}\int_s^t \mathbb{E}\left(1\{X(t)=j\}1\{X(x)=k\}\mathrm{e}^{-\int_s^x r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}B(x)\middle| X(s)=i\right)\\
&= \sum_{k\in E}\int_s^t \mathbb{E}\left(1\{X(x)=k\}\mathrm{e}^{-\int_s^x r_{X(u)}(u)\,\mathrm{d}u}\mathbb{E}\left(1\{X(t)=j\}\,\mathrm{d}B(x)|\,\mathcal{F}_{s,x}\right)\middle| X(s)=i\right).\\
&= \sum_{k\in E}\int_s^t \mathbb{E}\left(1\{X(x)=k\}\mathrm{e}^{-\int_s^x r_{X(u)}(u)\,\mathrm{d}u}\mathbb{E}\left(1\{X(t)=j\}\,\mathrm{d}B(x)|\,X(x)=k\right)\middle| X(s)=i\right)\\
&= \sum_{k\in E}\int_s^t d_{ik}(s,x)\mathbb{E}\left(1\{X(t)=j\}\,\mathrm{d}B(x)|\,X(x)=k\right),
\end{aligned}$$

where $\mathcal{F}_{s,x}=\sigma(X_u : s\le u\le x)$. But

$$\mathbb{E}\left(1\{X(t)=j\}\,\mathrm{d}B(x)|\,X(x)=k\right) \quad = \quad b_k(x)\,\mathrm{d}x\;p_{kj}(x,t)+\sum_{\ell\in E}b_{k\ell}(x)\lambda^1_{k\ell}(x)\,\mathrm{d}x\;p_{\ell j}(x,t)$$

so

$$\begin{aligned}
V_{ij}(s,t) \quad &= \quad \int_s^t\sum_{k\in E}d_{ik}(s,x)b_k(x)p_{kj}(x,t)\,\mathrm{d}x+\int_s^t\sum_{k,\ell\in E}d_{ik}(s,x)b_{k\ell}(x)\lambda^1_{k\ell}(x)p_{\ell j}(x,t)\,\mathrm{d}x\\
&= \quad \int_s^t\sum_{k,\ell\in E}d_{ik}(s,x)\delta_{k\ell}b_k(x)p_{\ell j}(x,t)\,\mathrm{d}x+\int_s^t\sum_{k,\ell\in E}d_{ik}(s,x)b_{k\ell}(x)\lambda^1_{k\ell}(x)p_{\ell j}(x,t)\,\mathrm{d}x.
\end{aligned}$$

In matrix form, this amounts to (41). □

The actual computation of the reserves can be effectively executed using the following Van–Loan formula, which avoids integration.

Corollary 4.6. *$\boldsymbol{V}(s,t)$ can be extracted from the relation*

$$\prod_s^t \left(\boldsymbol{I} + \begin{pmatrix} \boldsymbol{\Lambda}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u)) & \boldsymbol{R}(u) \\ \boldsymbol{0} & \boldsymbol{\Lambda}(u) \end{pmatrix} \mathrm{d}u \right) = \begin{pmatrix} \boldsymbol{D}(s,t) & \boldsymbol{V}(s,t) \\ \boldsymbol{0} & \boldsymbol{P}(s,t) \end{pmatrix}.$$

Remark 4.7: Consider the model where the interest rate Markov process and the biometric Markov process are independent. If the interest rate model corresponds to a phase–type distribution with initial distribution $\boldsymbol{\pi}$ and subintensity matrix $\boldsymbol{\Lambda}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u)$, then the classical state–wise reserve, given initiation in state i, is obtained by

$$(\boldsymbol{e}_i \otimes \boldsymbol{\pi}) \boldsymbol{V}(0,T) \boldsymbol{e}.$$

Finally, we state and prove Thiele's differential equations for partial reserves with stochastic interest rates.

Theorem 4.8 (Thiele)**.**

$$\frac{\partial}{\partial s} \boldsymbol{V}(s,t) = -\left[\boldsymbol{\Lambda}(s) - \boldsymbol{\Delta}(\boldsymbol{r}(s))\right] \boldsymbol{V}(s,t) - \boldsymbol{R}(s) \boldsymbol{P}(s,t),$$

where $\boldsymbol{V}(t,t) = \boldsymbol{0}$. For the conventional state–wise prospective reserves, $\boldsymbol{V}^{\mathrm{th}}(t) = \boldsymbol{V}(t,T)\boldsymbol{e}$, this has the form

$$\frac{\partial}{\partial t} \boldsymbol{V}^{\mathrm{th}}(t) = \boldsymbol{\Delta}(\boldsymbol{r}(t)) \boldsymbol{V}^{\mathrm{th}}(t) - \boldsymbol{\Lambda}(t) \boldsymbol{V}^{\mathrm{th}}(t) - \boldsymbol{R}(t) \boldsymbol{e},$$

where $\boldsymbol{V}^{\mathrm{th}}(T) = \boldsymbol{0}$.

Proof. Using that the product integral satisfies Kolmogorov's forward and backward equations, we get that

$$\begin{aligned} \begin{pmatrix} \frac{\partial}{\partial s} \boldsymbol{D}(s,t) & \frac{\partial}{\partial s} \boldsymbol{V}(s,t) \\ \boldsymbol{0} & \frac{\partial}{\partial s} \boldsymbol{P}(s,t) \end{pmatrix} &= \frac{\partial}{\partial s} \prod_s^t \left(\boldsymbol{I} + \begin{pmatrix} \boldsymbol{\Lambda}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u)) & \boldsymbol{R}(u) \\ \boldsymbol{0} & \boldsymbol{\Lambda}(u) \end{pmatrix} \mathrm{d}u \right) \\ &= -\begin{pmatrix} \boldsymbol{\Lambda}(s) - \boldsymbol{\Delta}(\boldsymbol{r}(s)) & \boldsymbol{R}(s) \\ \boldsymbol{0} & \boldsymbol{\Lambda}(s) \end{pmatrix} \begin{pmatrix} \boldsymbol{D}(s,t) & \boldsymbol{V}(s,t) \\ \boldsymbol{0} & \boldsymbol{P}(s,t) \end{pmatrix}, \end{aligned}$$

from which Thiele's differential equation can be pulled out from the upper right corner of each side of the equation. □

Example 4.9 (Relation to classical Thiele with stochastic interest rate)**:** In the classical situation, there are possibilities for assigning probabilities to lump sums upon transitions. Either they are zero or one. Therefore there is no need for a split of $\boldsymbol{\Lambda}(s)$ into $\boldsymbol{\Lambda}^0$ and $\boldsymbol{\Lambda}^1(s)$.

Hence, we only consider $\boldsymbol{\Lambda}(s)$ as a whole. There are no Poisson arrivals, so $b_{ii}(s) = 0$ for all i.

Here we consider

$$\boldsymbol{V}^{\text{th}}(s) = \boldsymbol{V}(s,T)\boldsymbol{e}.$$

From

$$\frac{\partial}{\partial s}\boldsymbol{V}(s,t) = -\left[\boldsymbol{\Lambda}(s) - \boldsymbol{\Delta}(\boldsymbol{r}(s))\right]\boldsymbol{V}(s,t) - \boldsymbol{R}(s)\boldsymbol{P}(s,t),$$

we immediately get that

$$\frac{\partial}{\partial s}\boldsymbol{V}^{\text{th}}(s) \quad = \quad -\left[\boldsymbol{\Lambda}(s) - \boldsymbol{\Delta}(\boldsymbol{r}(s))\right]\boldsymbol{V}^{\text{th}}(s) - \boldsymbol{R}(s)\boldsymbol{e}$$

□

Remark 4.10: We recall Remark 4.7. In case of independence between the Markovian interest rates and the biometric process, the state–wise differential equations above should be used with care. If one wishes to compute the state–wise reserves relative to the biometric states only (which is reasonable since the states of the underlying interest rate process have no interpretation), one should sum out $\mathrm{d}V_i(s)$ with weights being π_i over all states in the state–space corresponding to the interest rate model.

Example 4.11 (Independence continued)**:** We assume indpendence of the biometric and interest rate processes. Let the initial biometric state be $i \in E_b$, the terminal $j \in E_b$, and assume that the initial distribution of the fitted interest rate phase–type distribution is $\boldsymbol{\pi}$. Then

$$V_{ij}(0,T) = (\boldsymbol{e}_i' \otimes \boldsymbol{\pi}) \int_0^T \left(\prod_0^x (\boldsymbol{I} + (\boldsymbol{\Lambda}_b(u) \oplus \boldsymbol{\Lambda}_r(u) - \boldsymbol{I} \otimes \boldsymbol{\Delta}(\boldsymbol{r}(u))\,\mathrm{d}u) \right) (\boldsymbol{R}(x) \otimes \boldsymbol{I})$$
$$\times \left(\prod_x^T (\boldsymbol{I} + \boldsymbol{\Lambda}_b(u) \oplus \boldsymbol{\Lambda}_r(u)\,\mathrm{d}u) \right) \mathrm{d}x\,(\boldsymbol{e}_j \otimes \boldsymbol{e})$$

This integral is then computed using Van Loan as follows.

$$\prod_0^T \left(\boldsymbol{I} + \begin{pmatrix} \boldsymbol{\Lambda}_b(u) \oplus \boldsymbol{\Lambda}_r(u) - \boldsymbol{I} \otimes \boldsymbol{\Delta}(\boldsymbol{r}(u)) & \boldsymbol{R}(u) \otimes \boldsymbol{I} \\ \boldsymbol{0} & \boldsymbol{\Lambda}_b(u) \oplus \boldsymbol{\Lambda}_r(u) \end{pmatrix} \right) = \begin{pmatrix} * & \tilde{\boldsymbol{V}}(0,T) \\ \boldsymbol{0} & * \end{pmatrix}$$

and

$$V_{ij}(0,T) = (\boldsymbol{e}_i' \otimes \boldsymbol{\pi})\tilde{\boldsymbol{V}}(0,T)(\boldsymbol{e}_j \otimes \boldsymbol{e}).$$

For a general time $t < T$,

$$V_{ij}(0,T) = (\boldsymbol{e}_i' \otimes \boldsymbol{\pi}_t) \int_t^T \left(\prod_t^x (\boldsymbol{I} + (\boldsymbol{\Lambda}_b(u) \oplus \boldsymbol{\Lambda}_r(u) - \boldsymbol{I} \otimes \boldsymbol{\Delta}(\boldsymbol{r}(u))\,\mathrm{d}u) \right) (\boldsymbol{R}(x) \otimes \boldsymbol{I})$$

$$\times \left(\prod_x^T (\boldsymbol{I} + \boldsymbol{\Lambda}_b(u) \oplus \boldsymbol{\Lambda}_r(u)\,\mathrm{d}u) \right) \mathrm{d}x\,(\boldsymbol{e}_j \otimes \boldsymbol{e})$$

where

$$\boldsymbol{\pi}_t = \boldsymbol{\pi}\boldsymbol{P}_r(0,t)$$

is the distribution of $X_r(t)$. The Van Loan is then applied as above.

4. Equivalence premium

The arguments are essentially identical to the case of deterministic interest rates. It is assumed that $\boldsymbol{R}(t) = \boldsymbol{R}(t;\theta)$ such that θ is a parameter of either $\boldsymbol{B}(t)$ and/or $\boldsymbol{\Delta}(\boldsymbol{b}(t))$ only, and we write $\boldsymbol{V}(t) = \boldsymbol{V}(t;\theta)$. Then

$$\boldsymbol{V}(t;\theta) = \int_t^T \prod_t^u (\boldsymbol{I} + [\boldsymbol{\Lambda}(s) - \boldsymbol{\Delta}(\boldsymbol{r}(s))]\mathrm{d}s)\boldsymbol{R}(u;\theta) \prod_u^T (\boldsymbol{I} + \boldsymbol{\Lambda}(s)\mathrm{d}s)\mathrm{d}u.$$

If the interest rates satisfy $\boldsymbol{\Delta}(\boldsymbol{r}(s)) \geq \boldsymbol{0}$, then $\boldsymbol{\Lambda}(s) - \boldsymbol{\Delta}(\boldsymbol{r}(s))$ is a sub–intensity matrix, so that $\prod_t^u (\boldsymbol{I} + [\boldsymbol{\Lambda}(s) - \boldsymbol{\Delta}(\boldsymbol{r}(s))]\mathrm{d}s)$ is a sub–probability matrix, i.e.

$$0 \leq \prod_t^u (\boldsymbol{I} + [\boldsymbol{\Lambda}(s) - \boldsymbol{\Delta}(\boldsymbol{r}(s))]\mathrm{d}s)\boldsymbol{e} \leq \boldsymbol{e}.$$

If $\boldsymbol{R}(\cdot;\theta)$ is continuously differentiable and $\boldsymbol{\Lambda}$ and $\boldsymbol{r}$ are continuous, then by Leibniz' integral rule

$$\frac{\partial}{\partial\theta}\boldsymbol{V}(t;\theta) = \int_t^T \prod_t^u (\boldsymbol{I} + [\boldsymbol{\Lambda}(s) - \boldsymbol{\Delta}(\boldsymbol{r}(s))]\mathrm{d}s) \frac{\partial}{\partial\theta}\boldsymbol{R}(u;\theta) \prod_u^T (\boldsymbol{I} + \boldsymbol{\Lambda}(s)\mathrm{d}s)\mathrm{d}u. \tag{42}$$

Hence, we get from the Van Loan formula,

$$\prod_t^T \left(\boldsymbol{I} + \begin{pmatrix} \boldsymbol{\Lambda}(u) - \boldsymbol{\Delta}(\boldsymbol{r}(u)) & \frac{\partial}{\partial\theta}\boldsymbol{R}(u;\theta) \\ \boldsymbol{0} & \boldsymbol{\Lambda}(u) \end{pmatrix} \mathrm{d}u \right) = \begin{pmatrix} \boldsymbol{D}(t,T) & \frac{\partial}{\partial\theta}\boldsymbol{V}(t;\theta) \\ \boldsymbol{0} & \boldsymbol{P}(t,T) \end{pmatrix}. \tag{43}$$

If state $i \in E$ is the starting state, we can formulate the equivalence principle as the task of finding the θ that solves

$$\boldsymbol{e}_i'\boldsymbol{V}(0;\theta)\boldsymbol{e} = 0.$$

This can, for example, be achieved by the use of Newton's method,

$$\theta_{n+1} = \theta_n - \frac{\boldsymbol{e}_i'\boldsymbol{V}(0;\theta)\boldsymbol{e}}{\boldsymbol{e}_i'\boldsymbol{V}_\theta(0;\theta)\boldsymbol{e}},$$

where $\boldsymbol{V}_\theta$ denotes the partial derivative w.r.t. θ.

In many cases, θ is simply the premium rate in a certain state, in which case

$$R_\theta(t;\theta) = \frac{\partial}{\partial\theta}\boldsymbol{R}(t;\theta) = \boldsymbol{A}(t),$$

i.e. a matrix function not depending on θ. Then by (42), $\boldsymbol{V}_\theta(t;\theta) = \boldsymbol{V}_\theta(t)$ (partial derivatives w.r.t. θ) will not depend on θ either. But then the map $\theta \mapsto \boldsymbol{e}_i'\boldsymbol{V}(t;\theta)\boldsymbol{e}$ must be linear (for fixed t), so that

$$\boldsymbol{e}_i'\boldsymbol{V}(0;\theta)\boldsymbol{e} = a\theta + b,$$

for some constants a, b. Here

$$\begin{aligned} b &= \boldsymbol{e}_i'\boldsymbol{V}(0;0)\boldsymbol{e} \\ a &= \boldsymbol{e}_i'\boldsymbol{V}_\theta(0;0)\boldsymbol{e}. \end{aligned}$$

and the equivalence premium is hence given by

$$\theta = -\frac{b}{a} = -\frac{\boldsymbol{e}_i'\boldsymbol{V}(0;0)\boldsymbol{e}}{\boldsymbol{e}_i'\boldsymbol{V}_\theta(0;0)\boldsymbol{e}}. \tag{44}$$

5. Higher order moments with stochastic interest rates

Consider the matrix of partial state-wise higher order moments of future payments, given by, for $k \in \mathbb{N}$,

$$\boldsymbol{V}^{(k)}(t,T) = \left\{ V_{ij}^{(k)}(t,T) \right\}_{i,j\in E},$$

$$V_{ij}^{(k)}(t,T) = \mathbb{E}\left(1\{X(T)=j\} \left(\int_t^T \mathrm{e}^{-\int_t^x r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}B(x) \right)^k \middle| X(t)=i \right),$$

and introduce what we shall term the *reduced* partial state-wise higher order moments:

$$\boldsymbol{V}_r^{(k)}(t,T) = \frac{\boldsymbol{V}^{(k)}(t,T)}{k!}.$$

Denoting

$$\boldsymbol{D}^{(m)}(s,t) = \prod_s^t (\boldsymbol{I} + [\boldsymbol{\Lambda}(u) - m\boldsymbol{\Delta}(\boldsymbol{r}(u))]\,\mathrm{d}u), \quad m \in \mathbb{N},$$

we then obtain the following theorem for partial reserves with a stochastic interest rate.

Theorem 4.12. *The matrix of reduced partial state-wise higher order moments satisfies the integral equation, for $k \in \mathbb{N}_0$,*

$$\begin{aligned} \boldsymbol{V}_r^{(k)}(t,T) &= \int_t^T \boldsymbol{D}^{(k)}(t,x)\boldsymbol{R}(x)\boldsymbol{V}_r^{(k-1)}(x,T)\,\mathrm{d}x \\ &\quad + \sum_{m=2}^k \int_t^T \boldsymbol{D}^{(k)}(t,x)\boldsymbol{C}_r^{(m)}(x)\boldsymbol{V}_r^{(k-m)}(x,T)\,\mathrm{d}x. \end{aligned}$$

Proof. Write

$$\left(\int_t^T \mathrm{e}^{-\int_t^x r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}B(x) \right)^k$$

$$= \int_t^T \cdots \int_t^T \mathrm{e}^{-\int_t^{x_1} r_{X(u)}(u)\,\mathrm{d}u} \cdots \mathrm{e}^{-\int_t^{x_k} r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}B(x_k)\cdots\mathrm{d}B(x_1). \tag{45}$$

Now

$$\mathrm{d}B(t) = b_{X(t)}(t)\,\mathrm{d}t + \sum_j b_{X(t-)j}(t)\,\mathrm{d}N_{X(t-)j}(t),$$

and assume that $s \in [t,T]$ is a point of increase for the counting process $x \to N_{ab}(x)$ which triggers a lump sum payment. Then, in the computation of the above integral, there will be jump contributions at time s, where any number $m \in \{1,2,...,k\}$ of the variables $x_1,...,x_k$ may be equal to s, say $x_{i_1} = \cdots x_{i_m} = s$. We can pick m out of the k variables in

$$\binom{k}{m} = \frac{k!}{m!(k-m)!}$$

ways. If m variables coincide at the jump time s, then a contribution of $b_{ab}(s)^m$ is added. Hence only looking at jump *coincidences*, i.e. $m \geq 2$, the contribution to the integral (45) is

$$\sum_{m=2}^{k} \binom{k}{m} \int_t^T \mathrm{e}^{-m\int_t^s r_{X(u)}(u)\,\mathrm{d}u} b_{ab}(s)^m$$
$$\times \left(\int_s^T \cdots \int_s^T \mathrm{e}^{-\int_t^{x_{m+1}} r_{X(u)}(u)\,\mathrm{d}u} \cdots \mathrm{e}^{-\int_t^{x_k} r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}A(x_k)\cdots\mathrm{d}A(x_{m+1}) \right) \mathrm{d}N_{ab}(s).$$

Indeed, since there are precisely m coincidences, the remaining integrals must start from $s+ = s$; otherwise, the integration intervals would contain s as well. Changing the lower limits of the integrals appearing in the exponentials, we can further rewrite the expression as

$$\sum_{m=2}^{k} \binom{k}{m} \int_t^T \mathrm{e}^{-k\int_t^s r_{X(u)}(u)\,\mathrm{d}u} b_{ab}(s)^m$$
$$\times \left(\int_s^T \cdots \int_s^T \mathrm{e}^{-\int_s^{x_{m+1}} r_{X(u)}(u)\,\mathrm{d}u} \cdots \mathrm{e}^{-\int_s^{x_k} r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}A(x_k)\cdots\mathrm{d}A(x_{m+1}) \right) \mathrm{d}N_{ab}(s). \tag{46}$$

Taking conditional expectation $\mathbb{E}(1\{X(T)=j\}\cdot|X(t)=i)$ of (46), we get

$$\sum_{m=2}^{k} \binom{k}{m} \int_t^T \mathbb{E}\Bigg(1\{X(T)=j\}\mathrm{e}^{-k\int_t^s r_{X(u)}(u)\,\mathrm{d}u} b_{ab}(s)^m$$
$$\times \left(\int_s^T \cdots \int_s^T \mathrm{e}^{-\int_s^{x_{m+1}} r_{X(u)}(u)\,\mathrm{d}u} \cdots \mathrm{e}^{-\int_s^{x_k} r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}A(x_k)\cdots\mathrm{d}A(x_{m+1}) \right) \mathrm{d}N_{ab}(s) \Bigg| X(t)=i \Bigg)$$
$$= \sum_{m=2}^{k} \binom{k}{m} \int_t^T \mathbb{E}\left(1\{X(T)=j\} Y\, Z\, \mathrm{d}N_{ab}(s) | \, X(t)=i \right), \tag{47}$$

where $Y = \mathrm{e}^{-k\int_t^s r_{X(u)}(u)\,\mathrm{d}u} b_{ab}(s)^m$ and

$$Z = \int_s^T \cdots \int_s^T \mathrm{e}^{-\int_s^{x_{m+1}} r_{X(u)}(u)\,\mathrm{d}u} \cdots \mathrm{e}^{-\int_s^{x_k} r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}A(x_k)\cdots \mathrm{d}A(x_{m+1}).$$

Further conditioning on a lump sum triggering event at time s, caused by $\{X(s)=b, X(s-)=a\}$ and the probability of which is $\lambda^1_{ab}(s)\,\mathrm{d}s$, and using that $1\{X(T)=j\}Z$ and $1\{X(s-)=a\}Y$ are conditionally independent given $X(s)=b$, (47) reduces to

$$\begin{aligned}
&\sum_{m=2}^{k}\binom{k}{m}\int_t^T \mathbb{E}\left(1\{X(T)=j\}Z\middle|X(s)=b\right)\mathbb{E}\left(1\{X(s)=a\}Y\middle|X(t)=i\right)\lambda^1_{ab}(s)\,\mathrm{d}s \\
= &\sum_{m=2}^{k}\binom{k}{m}\int_t^T D^{(k)}_{ia}(t,s)\lambda^1_{ab}(s)b_{ab}(s)^m V^{(k-m)}_{bj}(s,T)\,\mathrm{d}s. \qquad (48)
\end{aligned}$$

Summing over a and b, and putting (48) on matrix form (in i,j) this amounts to

$$\sum_{m=2}^{k}\binom{k}{m}\int_t^T \boldsymbol{D}^{(k)}(t,s)\boldsymbol{C}^{(m)}(s)\boldsymbol{V}^{(k-m)}(s,T)\,\mathrm{d}s. \qquad (49)$$

Now we consider the integral when there are no coincidences. To this end, we rewrite

$$\begin{aligned}
\left(\int_t^T \mathrm{e}^{-\int_t^x r_{X(u)}(u)\,\mathrm{d}u}\mathrm{d}A(x)\right)^k &= k\int_t^T \mathrm{e}^{-\int_t^x r_{X(u)}(u)\,\mathrm{d}u}\left(\int_x^T \mathrm{e}^{-\int_t^y r_{X(u)}(u)\,\mathrm{d}u}\mathrm{d}A(y)\right)^{k-1}\mathrm{d}A(x) \\
&= k\int_t^T \mathrm{e}^{-k\int_t^x r_{X(u)}(u)\,\mathrm{d}u}\left(\int_x^T \mathrm{e}^{-\int_x^y r_{X(u)}(u)\,\mathrm{d}u}\mathrm{d}A(y)\right)^{k-1}\mathrm{d}A(x)
\end{aligned}$$

Then

$$\begin{aligned}
V^{(k)}_{ij}(t,T) &= \mathbb{E}\left(1\{X(T)=j\}\left(\int_t^T \mathrm{e}^{-\int_t^x r_{X(u)}(u)\,\mathrm{d}u}\mathrm{d}A(x)\right)^k\middle|X(t)=i\right) \\
&= k\int_t^T \mathbb{E}\left(1\{X(T)=j\}\mathrm{e}^{-k\int_t^x r_{X(u)}(u)\,\mathrm{d}u}\left(\int_x^T \mathrm{e}^{-\int_x^y r_{X(u)}(u)\,\mathrm{d}u}\mathrm{d}A(y)\right)^{k-1}\mathrm{d}A(x)\middle|X(t)=i\right) \\
&= k\sum_\ell \int_t^T \mathbb{E}\Bigg(1\{X(x)=\ell\}\mathrm{e}^{-k\int_t^x r_{X(u)}(u)\,\mathrm{d}u}1\{X(T)=j\} \\
&\quad \times\mathbb{E}\left(1\{X(T)=j\}\left(\int_x^T \mathrm{e}^{-\int_x^y r_{X(u)}(u)\,\mathrm{d}u}\mathrm{d}A(y)\right)^{k-1}\mathrm{d}A(x)\middle|X(x)=\ell\right)\Bigg|X(t)=i\Bigg). \qquad (50)
\end{aligned}$$

On the event that $X(x)=\ell$, the contribution to the expectation of the above integral (50), where no coincidences are allowed (i.e., the reward at time x from at most one jump and benefit rates), amounts to

$$\mathbb{E}\left(1\{X(T)=j\}\left(\int_x^T \mathrm{e}^{-\int_x^y r_{X(u)}(u)\,\mathrm{d}u}\mathrm{d}A(y)\right)^{k-1}\middle|X(x)=\ell\right)b_\ell(x)\,\mathrm{d}x$$

$$
\begin{aligned}
&+\sum_m \mathbb{E}\left(1\{X(T)=j\}\left(\int_x^T \mathrm{e}^{-\int_x^y r_{X(u)}(u)\,\mathrm{d}u}\,\mathrm{d}A(y)\right)^{k-1}\middle| X(x)=m\right)\lambda^1_{\ell m}(x)b_{\ell m}(x)\,\mathrm{d}x\\
=\;& b_\ell(x)V^{(k-1)}_{\ell j}(x,T)\,\mathrm{d}x+\sum_m \lambda^1_{\ell m}(x)b_{\ell m}(x)V^{(k-1)}_{mj}(x,T)\,\mathrm{d}x,
\end{aligned}
$$

and the integral (50) then equals

$$
\begin{aligned}
&k\sum_\ell \int_t^T \mathbb{E}\left(1\{X(x)=\ell\}\mathrm{e}^{-k\int_t^x r_{X(u)}(u)\,\mathrm{d}u}\middle| X(t)=i\right)\\
&\qquad\times\left(b_\ell(x)V^{(k-1)}_{\ell j}(x,T)+\sum_m \lambda^1_{\ell m}(x)b_{\ell m}(x)V^{(k-1)}_{mj}(x,T)\right)\mathrm{d}x
\end{aligned}
$$

which in matrix form amounts to

$$
k\int_t^T \boldsymbol{D}^{(k)}(t,x)\boldsymbol{R}(x)\boldsymbol{V}^{(k-1)}(x,T)\,\mathrm{d}x. \tag{51}
$$

Adding (49) and (51) then prove the result. □

Defining

$$
\boldsymbol{F}^{(k)}_U(x)=\begin{pmatrix}
\boldsymbol{\Lambda}(x)-k\boldsymbol{\Delta}(\boldsymbol{r}(x)) & \boldsymbol{R}(x) & \boldsymbol{C}^{(2)}_r(x)\cdots & \boldsymbol{C}^{(k-1)}_r(x) & \boldsymbol{C}^{(k)}_r(x)\\
\boldsymbol{0} & \boldsymbol{\Lambda}(x)-(k-1)\boldsymbol{\Delta}(\boldsymbol{r}(x)) & \boldsymbol{R}(x)\;\cdots & \boldsymbol{C}^{(k-2)}_r(x) & \boldsymbol{C}^{(k-1)}_r(x)\\
\vdots & \vdots & \vdots\;\vdots & \vdots & \vdots\\
\boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0}\;\cdots & \boldsymbol{\Lambda}(x)-\boldsymbol{\Delta}(\boldsymbol{r}(x)) & \boldsymbol{R}(x)\\
\boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0}\;\cdots & \boldsymbol{0} & \boldsymbol{\Lambda}(x)
\end{pmatrix}
$$

we get by Van Loan that

$$
\prod_t^T(\boldsymbol{I}+\boldsymbol{F}^{(k)}_U(x)\,\mathrm{d}x)=\begin{pmatrix}
* & * & * & * & \cdots & * & \boldsymbol{V}^{(k)}_r(t)\\
* & * & * & * & \cdots & * & \boldsymbol{V}^{(k-1)}_r(t)\\
* & * & * & * & \cdots & * & \boldsymbol{V}^{(k-2)}_r(t)\\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots\\
* & * & * & * & \cdots & * & \boldsymbol{V}^{(1)}_r(t)\\
* & * & * & * & \cdots & * & \boldsymbol{P}(t,T)
\end{pmatrix}. \tag{52}
$$

From these results, we can derive a number of classical results. Differentiation of (52) gives

$$\begin{pmatrix} * & * & * & * & \cdots & * & \frac{\partial}{\partial t}\boldsymbol{V}_r^{(k)}(t) \\ * & * & * & * & \cdots & * & \frac{\partial}{\partial t}\boldsymbol{V}_r^{(k-1)}(t) \\ * & * & * & * & \cdots & * & \frac{\partial}{\partial t}\boldsymbol{V}_r^{(k-2)}(t) \\ \vdots & \vdots & \vdots & \vdots & \vdots\vdots\vdots & \vdots & \vdots \\ * & * & * & * & \cdots & * & \frac{\partial}{\partial t}\boldsymbol{V}_r^{(1)}(t) \\ * & * & * & * & \cdots & * & \frac{\partial}{\partial t}\boldsymbol{P}(t,T) \end{pmatrix} = -\boldsymbol{F}_U^{(k)}(t)\prod_t^T(\boldsymbol{I}+\boldsymbol{F}_U^{(k)}(x)\,\mathrm{d}x)$$

$$= -\begin{pmatrix} \boldsymbol{\Lambda}(t)-k\boldsymbol{\Delta}(\boldsymbol{r}(t)) & \boldsymbol{R}(t) & \boldsymbol{C}_r^{(2)}(t) & \cdots & \boldsymbol{C}_r^{(k-1)}(t) & \boldsymbol{C}_r^{(k)}(t) \\ \boldsymbol{0} & \boldsymbol{\Lambda}(t)-(k-1)\boldsymbol{\Delta}(\boldsymbol{r}(t)) & \boldsymbol{R}(t) & \cdots & \boldsymbol{C}_r^{(k-2)}(t) & \boldsymbol{C}_r^{(k-1)}(t) \\ \vdots & \vdots & \vdots & \vdots\vdots\vdots & \vdots & \vdots \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{\Lambda}(t)-\boldsymbol{\Delta}(\boldsymbol{r}(t)) & \boldsymbol{R}(t) \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{0} & \boldsymbol{\Lambda}(t) \end{pmatrix}$$

$$\times\begin{pmatrix} * & * & * & * & \cdots & * & \boldsymbol{V}_r^{(k)}(t) \\ * & * & * & * & \cdots & * & \boldsymbol{V}_r^{(k-1)}(t) \\ * & * & * & * & \cdots & * & \boldsymbol{V}_r^{(k-2)}(t) \\ \vdots & \vdots & \vdots & \vdots & \vdots\vdots\vdots & \vdots & \vdots \\ * & * & * & * & \cdots & * & \boldsymbol{V}_r^{(1)}(t) \\ * & * & * & * & \cdots & * & \boldsymbol{P}(t,T) \end{pmatrix}$$

We then obtain the following version of Hattendorff's differential equation by only considering the first row times the last column.

Theorem 4.13. *The matrix of reduced partial state-wise higher order moments satisfies the system of differential equations, for $k \in \mathbb{N}_0$,*

$$\frac{\partial}{\partial s}\boldsymbol{V}_r^{(k)}(t) = (k\boldsymbol{\Delta}(\boldsymbol{r}(u)) - \boldsymbol{\Lambda}(t))\boldsymbol{V}_r^{(k)}(t) - \boldsymbol{R}(t)\boldsymbol{V}_r^{(k-1)}(t) - \sum_{i=2}^{k}\boldsymbol{C}_r^{(i)}(t)\boldsymbol{V}_r^{(k-i)}(t),$$

with terminal condition $\boldsymbol{V}_r^{(k)}(T) = 1_{(k=0)}\boldsymbol{I}$.

Let

$$\boldsymbol{C}(s,t) = \prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\otimes\boldsymbol{I}.$$

The Kronecker product between a $p_1\times q_1$ matrix $\boldsymbol{A}=\{a_{ij}\}$ and a $p_2\times q_2$ matrix $\boldsymbol{B}=\{b_{ij}\}$ is defined as the $p_1p_2\times q_1q_2$ matrix

$$\boldsymbol{A}\otimes\boldsymbol{B} = \{a_{ij}\boldsymbol{B}\}_{i=1,\ldots,p_1,j=1,\ldots,q_1} = \{a_{ij}b_{k\ell}\}.$$

Using that $(\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D}) = (\boldsymbol{AC})\otimes(\boldsymbol{BD})$, we get

$$\frac{\partial}{\partial t}\boldsymbol{C}(s,t) \quad = \quad \prod_s^t(\boldsymbol{I}+\boldsymbol{A}(x)\,\mathrm{d}x)\boldsymbol{A}(t)\otimes\boldsymbol{I}$$

$$
\begin{aligned}
&= \left(\prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) \otimes \boldsymbol{I}\right)(\boldsymbol{A}(t) \otimes \boldsymbol{I}) \\
&= \boldsymbol{C}(s,t)\,(\boldsymbol{A}(t) \otimes \boldsymbol{I}),
\end{aligned}
$$

and we conclude that

$$
\boldsymbol{C}(s,t) = \prod_s^t (\boldsymbol{I} + (\boldsymbol{A}(x) \otimes \boldsymbol{I})\,\mathrm{d}x). \tag{53}
$$

A similar argument gives that

$$
\boldsymbol{I} \otimes \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) = \prod_s^t (\boldsymbol{I} + (\boldsymbol{I} \otimes \boldsymbol{A}(x))\,\mathrm{d}x). \tag{54}
$$

Finally, if $\boldsymbol{A}(t)$ and $\boldsymbol{B}(t)$ are Riemann integrable matrix functions of dimensions $q \times q$ and $p \times p$ respectively, then

$$
\prod_s^t (\boldsymbol{I} + (\boldsymbol{A}(x) \oplus \boldsymbol{B}(x))\,\mathrm{d}x) = \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) \otimes \prod_s^t (\boldsymbol{I} + \boldsymbol{B}(x)\,\mathrm{d}x), \tag{55}
$$

where $\oplus$ denotes the Kronecker sum, defined by $\boldsymbol{A}(t) \oplus \boldsymbol{B}(t) = \boldsymbol{A} \otimes \boldsymbol{I} + \boldsymbol{I} \otimes \boldsymbol{B}(t)$, and where the first $\boldsymbol{I}$ has the dimension of $\boldsymbol{B}(t)$ and the second $\boldsymbol{I}$ has the dimension of $\boldsymbol{A}(t)$. To see this, we notice that $\boldsymbol{A}(t) \otimes \boldsymbol{I}$ and $\boldsymbol{I} \otimes \boldsymbol{B}(t)$ commute, so

$$
\begin{aligned}
\prod_s^t (\boldsymbol{I} + (\boldsymbol{A}(x) \oplus \boldsymbol{B}(x))\,\mathrm{d}x) &= \prod_s^t (\boldsymbol{I} + (\boldsymbol{A}(x) \otimes \boldsymbol{I})\,\mathrm{d}x) \prod_s^t (\boldsymbol{I} + (\boldsymbol{I} \otimes \boldsymbol{B}(x))\,\mathrm{d}x) \\
&= \left[\prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) \otimes \boldsymbol{I}\right] \left[\boldsymbol{I} \otimes \prod_s^t (\boldsymbol{I} + \boldsymbol{B}(x)\,\mathrm{d}x)\right] \\
&= \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) \otimes \prod_s^t (\boldsymbol{I} + \boldsymbol{B}(x)\,\mathrm{d}x).
\end{aligned}
$$

Example 4.14 (Independence continued)**:** We can continue our decompositions from the independence case of Example 4.2 to reserves and higher-order moments. Indeed, since

$$
\begin{aligned}
\boldsymbol{\Lambda}_b(u) \oplus \boldsymbol{\Lambda}_r(u) - k\boldsymbol{\Delta}(\boldsymbol{e} \otimes \boldsymbol{r}(u)) &= \boldsymbol{\Lambda}_b(u) \otimes \boldsymbol{I} + \boldsymbol{I} \otimes (\boldsymbol{\Lambda}_r(u) - k\boldsymbol{\Delta}(\boldsymbol{r}(u))) \\
&= \boldsymbol{\Lambda}_b(u) \oplus (\boldsymbol{\Lambda}_r(u) - k\boldsymbol{\Delta}(\boldsymbol{r}(u)),
\end{aligned}
$$

we get

$$
\begin{aligned}
&\prod_s^t \Big(\boldsymbol{I} + (\boldsymbol{\Lambda}_b(u) \oplus \boldsymbol{\Lambda}_r(u) - k\boldsymbol{\Delta}(\boldsymbol{e} \otimes \boldsymbol{r}(u))\,\mathrm{d}u)\Big) \\
&\quad = \prod_s^t (\boldsymbol{I} + \boldsymbol{\Lambda}_b(u)\,\mathrm{d}u) \otimes \prod_s^t (\boldsymbol{I} + (\boldsymbol{\Lambda}_r(u) - k\boldsymbol{\Delta}(\boldsymbol{r}(u))\,\mathrm{d}u) \\
&\quad = \prod_s^t (\boldsymbol{I} + \boldsymbol{\Lambda}_b(u)\,\mathrm{d}u) \otimes \boldsymbol{D}^{(k)}(s,t)
\end{aligned}
$$

Thus, each diagonal block element can be computed using these representations when setting up the matrix $\boldsymbol{F}_U$ to compute these higher–order moments.

□

CHAPTER 5

Semi–Markov models

1. Basic properties

Let $\{Z(t)\}_{t\geq 0}$ denote a stochastic process taking values in $E = \{1,2,...,p\}$. Then the sample paths $t \to Z(t)$ are piecewise constant. The time the process spends in each state before jumping to another is called a sojourn time. The duration process $\{U(t)\}_{t\geq 0}$ related to $\{Z(t)\}_{t\geq 0}$ is defined by

$$U(t) = \inf\{u \geq 0 : Z((t-u)-) \neq Z(t)\},$$

i.e. the time eleapsed since the last jump into the current state, which equals the value of $Z(t)$, or the time since initiation, which equals t, if there has been no previous jump.

Definition 5.1: $\{Z(t)\}_{t\geq 0}$ is said to be a semi–Markov process if $\{(Z(t),U(t))\}_{t\geq 0}$ is a Markov process.

The state–space for $\{(Z(t),U(t))\}_{t\geq 0}$ is $E \times \mathbb{R}_+$, and the Markov property then amounts to

$$\mathbb{P}(Z(t) = j, U(t) \leq v | \mathcal{F}_s) = \mathbb{P}(Z(t) = j, U(t) \leq v | Z(s), U(s)),$$

where $\mathcal{F}_s = \sigma((Z(x),U(x)) : x \in [0,s])$.

For a semi–Markov process we define transition probabilities as

$$P_{ij}(s,t,u,v) = \mathbb{P}(Z(t) = j, U(t) \leq v \,|\, Z(s) = i, U(s) = u), \tag{56}$$

and we will assume the existence of the corresponding (defective) transition densities

$$p_{ij}(s,t,u,v) = \frac{\partial}{\partial v} P_{ij}(s,t,u,v)$$

which has the probabilistic interpretation

$$p_{ik}(s,t,u,v)dv = \mathbb{P}(Z(t) = k, U(t) \in (v, v+\mathrm{d}v] | Z(s) = i, U(s) = u).$$

We form transition probability and transition density matrices

$$\boldsymbol{P}(s,t,u,v) = \{P_{ij}(s,t,u,v)\}_{i,j=1,\ldots,p}. \tag{57}$$

$$\boldsymbol{p}(s,t,u,v) = \{p_{ij}(s,t,u,v)\}_{i,j=1,\ldots,p} \tag{58}$$

For $i \neq k$, we assume the existence of the limits

$$\mu_{ik}(t,u) = \lim_{h\downarrow 0} \frac{\mathbb{P}(Z(t+h) = k \,|\, Z(t) = i, U(t) = u)}{h}$$

$$
\begin{aligned}
&= \lim_{h\downarrow 0} \frac{P_{ik}(t,t+h,u,+\infty)}{h} \\
&= \frac{\partial}{\partial t} P_{ik}(s,t,u,+\infty)\Big|_{t=s}.
\end{aligned}
$$

The interpreation of $\mu_{ik}(t,u)$ is similar to the Markov case as

$$\mu_{ik}(t,u)\,\mathrm{d}t = \mathbb{P}(Z(t+\mathrm{d}t)=k|Z(t)=i,U(t)=u),$$

i.e. the probability of jumping from i to j at time t (and with duration u). Here, the target duration just after the jump is irrelevant since it will be zero.

Let

$$\mu_{ii}(t,u) = -\mu_i(t,u) = -\sum_{k\neq i}\mu_{ik}(t,u).$$

and define the intensity matrices

$$\boldsymbol{M}(t,u) = \{\mu_{ij}(t,u)\}_{i,j\in E}.$$

We assume, unless otherwise stated, that $(t,u)\to\boldsymbol{M}(t,u)$ is continuous, which simply means that each of it elements is a continuous function in (t,u).

Definition 5.2: For a matrix $\boldsymbol{A}=\{a_{ij}\}_{i,j=1,..,p}$ we let $\Delta\boldsymbol{A}$ denote the diagonal matrix whose diagonal is that of $\boldsymbol{A}$, and $\nabla\boldsymbol{A}=\boldsymbol{A}-\Delta\boldsymbol{A}$ the remaining part of $\boldsymbol{A}$.

Then we have that

Theorem 5.3.

$$
\begin{aligned}
\boldsymbol{P}(s,t,u,v) = &\int_s^t \mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}\nabla\boldsymbol{M}(w,u+w-s)\boldsymbol{P}(w,t,0,v)\,\mathrm{d}w \\
&+\mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}1\{u+t-s\le v\}. \qquad (59)
\end{aligned}
$$

Proof. Consider $P_{ij}(s,t,u,v)$. If there are no jumps during $[s,t]$, which happens with probability

$$\mathrm{e}^{-\int_s^t \mu_i(x,u+x-s)\,\mathrm{d}x},$$

then the final state will be i and the duration will be $u+t-s$. Hence, the contribution from this case will be

$$\mathrm{e}^{-\int_s^t \mu_i(x,u+x-s)\,\mathrm{d}x}\delta_{ij}1\{u+t-s\le v\}.$$

Here δ_{ij} denotes the Kronecker–δ, which is one for $i=j$ and zero otherwise.

On the complementary event, that there is at least one jump during $[s,t]$, we condition on the first jump at time w, and that is jumps to state k. The density of this is

$$\mathrm{e}^{-\int_s^w \mu_i(x,u+x-s)\,\mathrm{d}x}\mu_{ik}(w,u+w-s).$$

After jumping to k at time w, the duration is reset to zero, and the remaining probability of being in state j at time t and with a duration that is less than v then amounts to $P_{kj}(w,t,0,v)$.

Integrating up and putting it in matrix form yields the result. □

Once $\boldsymbol{P}(w,t,0,v)$ is known, $\boldsymbol{P}(s,t,u,v)$ can then be obtained by integration in (60).

In order to find $\boldsymbol{P}(w,t,0,v)$, we see that it satisfies the equation (set $u=0$ in (60))

$$\begin{aligned}\boldsymbol{P}(s,t,0,v) &= \int_s^t \mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}\nabla\boldsymbol{M}(w,w-s)\boldsymbol{P}(w,t,0,v)\,\mathrm{d}w \\ &\quad + \mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}1\{t-s\le v\}. \end{aligned} \tag{60}$$

Define

$$\begin{aligned}\boldsymbol{P}^{(0)}(s,t,0,v) &= \mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}1\{t-s\le v\}, \\ \boldsymbol{P}^{(n+1)}(s,t,0,v) &= \int_s^t \mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}\nabla\boldsymbol{M}(w,w-s)\boldsymbol{P}^{(n)}(w,t,0,v)\,\mathrm{d}w\end{aligned}$$

Then $\boldsymbol{P}^{(0)}(s,t,0,v)$ is the (defective, due to $U(t)\le v$) transition probability matrix for the case of no jumps. Similarly, $\boldsymbol{P}^{(n)}(s,t,0,v)$ is the defective transition probaility matrix for the case of exactly n transitions. Thus we can obtain $\boldsymbol{P}(s,t,0,v)$ from

$$\boldsymbol{P}(s,t,0,v) = \sum_{n=0}^{\infty}\boldsymbol{P}^{(n)}(s,t,0,v).$$

Define

$$\boldsymbol{Q}(s,t) = \mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}\nabla\boldsymbol{M}(t,t-s), \quad \boldsymbol{Q}(s,t) = \{q_{ij}(s,t)\}_{i,j\in E}. \tag{61}$$

Then $q_{ij}(s,t)$ is the defective density function of a sojourn time in a semi–Markov process that enters state i at time s and terminates at time t by a jump to state j, Thus $\sum_j q_{ij}(s,t)$ is the (non–defective) density for the sojourn time in a semi–Markov process that enters state i at time s and lasts until time t.

With this notation,

$$\boldsymbol{P}(s,t,0,v) = \int_s^t \boldsymbol{Q}(s,w)\boldsymbol{P}(w,t,0,v)\,\mathrm{d}w + \mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}1\{t-s\le v\}. \tag{62}$$

Assuming differentiability of $v\to\boldsymbol{P}(s,t,0,v)$, by definition we obtain $\boldsymbol{p}(s,t,0,v)$ as its derivative w.r.t. v. There is an issue, however, which is the indicator function $v\to 1\{t-s\le v\}$ which is not differentiable w.r.t. v in the classical sense. The derivation of the indicator function formally falls into the field of distributions, also known as generalised functions. Without diving further into this theory, the derivative $x\to 1\{x\ge y\}$ will be defined as a (generalised) function $\delta_y(x)$ which satisfies

$$\int_{-\infty}^x \delta_y(u)\,\mathrm{d}u = 1\{x\ge y\}.$$

This function is referred to as the Dirac–δ function.

Remark 5.4: We briefly comment on the Dirac–δ function, δ_y, and its construction. The function used throughout in the notes is δ_0, which uses $y=0$. In general, $\delta_y(x)=\delta_0(x-y)$.

Let $X_\sigma \sim \mathrm{N}(y, \sigma^2)$. Then X_σ has density

$$f_{X_\sigma}(x) = \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-(x-y)^2/2\sigma^2}$$

and with mean $\mathbb{E}(X_\sigma) = y$ and variance $\mathbb{E}((X_\sigma - y)^2) = \sigma^2$.

Then for any σ

$$\int_{-\infty}^{\infty} f_{X_\sigma}(x)\,\mathrm{d}x = 1.$$

We formally define a symbol (think of it as a function), $\delta_y(x)$, such that

$$\int_{-\infty}^{\infty} \delta_y(x)\,\mathrm{d}x = \lim_{\sigma\to 0} \int_{-\infty}^{\infty} f_{X_\sigma}(x)\,\mathrm{d}x = 1.$$

We think of $\delta_y(x)$ as the limit of $f_{X_\sigma}(x)$ as $\sigma \to 0$.

By Markov's inequality,

$$\mathbb{P}\left(|X_\sigma - y| > a\right) \leq \frac{\mathbb{E}((X_\sigma - y)^2)}{a^2} = \frac{\sigma^2}{a^2} \to 0$$

as $\sigma \to 0$. Hence $X_\sigma \to y$ in probability as $\sigma \to 0$. Since the limit, y, is degenerated (constant), then X_σ also converges to y in distribution. Hence, for any bounded and continuous function, f, we have that

$$\mathbb{E}(f(X_\sigma)) \to \mathbb{E}(f(y)) = f(y).$$

But then

$$\int_{-\infty}^{\infty} f(x)\delta_y(x)\,\mathrm{d}x = f(y),$$

which is a key property of $\delta_y(x)$.

This property then implies that

$$\begin{aligned}
\int_{-\infty}^{v} \delta_0(x-y)\,\mathrm{d}x &= \int_{-\infty}^{\infty} 1_{(-\infty,v]}(x)\delta_0(x-y)\,\mathrm{d}x \\
&= \int_{-\infty}^{\infty} 1_{(-\infty,v]}(x)\delta_y(x)\,\mathrm{d}x \\
&= 1_{(-\infty,v]}(y) \\
&= 1\{y \leq v\}
\end{aligned}$$

as it should. □

After this small intermezzo, we may now, with a clear conscience, differentiate through in (60) and (62) to get

$$\begin{aligned}
\boldsymbol{p}(s,t,u,v) = &\int_s^t \mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y} \nabla\boldsymbol{M}(w,u+w-s)\boldsymbol{p}(w,t,0,v)\,\mathrm{d}w \\
&+ \mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y} \delta_{u+t-s}(v) \qquad (63)
\end{aligned}$$

$$\boldsymbol{p}(s,t,0,v) = \int_s^t \boldsymbol{Q}(s,w)\boldsymbol{p}(w,t,0,v)\,\mathrm{d}w + \mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}\delta_0(v-(t-s)). \qquad (64)$$

Remark 5.5: The inclusion of a Dirac–δ function in an expression for a density function happens frequently, for example, if the distribution of a random variable X, say, consists of an absolutely continuous part, and an atom at some point. To be precise, we let

$$X = \begin{cases} Z & \text{w.p. } 1-p \\ a & \text{w.p. } p \end{cases},$$

where a is a constant and Z and absolute continuous random variable. Hence, X takes the constant value a with probability p and, with probability $1-p$, takes the value of a random variable Z with a (defective) density function.

Let f_Z denote the density function for Z and let f_X be the density function for X. The latter cannot be expressed in the usual way due to the atom at a. But we can write

$$f_X(x) = (1-p)f_Z(x) + p\delta_a(x).$$

Indeed, this expression implies that for their corresponding c.d.f. (cumulative distribution function)

$$F_X(x) = (1-p)F_Z(x) + p\int_{\infty}^{x} \delta_a(u)\,\mathrm{d}u = (1-p)F_Z(x) + p1_{(-\infty,x]}(a),$$

where

$$1_{(-\infty,x]}(a) = 1\{x \geq a\}$$

is exactly the c.d.f. of a point mass at a.

Therefore, in (63), there are a point masses of sizes (matrix)

$$\mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}$$

at $t-s$, which occurs if there are no jumps.

Now let us consider (63) a little more closely. We can obtain an expression for the (defective) density function $\boldsymbol{p}(s,t,0,v)$ in the following way. If $U(t) = v$ should hold true, then there must be a transition exactly at time $t-v$ and no further transition during $[t-v,t]$. Also, $v \in [0,t-s]$, otherwise the probability will be zero.

Let $p_{ij}^{(n)}(s,t,0,v)$ denote the defective probability density of transition from $(Z(s) = i, U(s) = 0)$ to $(Z(t) = j, U(t) = v)$ using exactly n transitions, $n \geq 0$.

For $n = 0$, we clearly have

$$\boldsymbol{p}^{(0)}(s,t,0,v) = \mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}\delta_0(v-(t-s)).$$

For $n = 1$, the only transition must be at time $t-v$, so that

$$\begin{aligned}\boldsymbol{p}^{(1)}(s,t,0,v) &= \mathrm{e}^{\int_s^{t-v} \Delta\boldsymbol{M}(y,y-s)\,\mathrm{d}y}\nabla\boldsymbol{M}(t-v,t-v-s)\mathrm{e}^{\int_{t-v}^t \Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y} \\ &= \boldsymbol{Q}(s,t-v)\mathrm{e}^{\int_{t-v}^t \Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y}.\end{aligned}$$

For $n > 1$, we must have a transition exactly at $t-v$ and another $n-1$ previous to this one. Conditioning out on the latter, we get

$$\boldsymbol{p}^{(n)}(s,t,0,v)$$
$$= \int_s^{t-v} \int_{w_1}^{t-v} \cdots \int_{w_{n-2}}^{t-v} \boldsymbol{Q}(s,w_1)\boldsymbol{Q}(w_1,w_2)\cdots\boldsymbol{Q}(w_{n-1},t-v)\,\mathrm{d}w_{n-1}\cdots\mathrm{d}w_1\,\mathrm{e}^{\int_{t-v}^t \Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y}.$$

Rewrite

$$\begin{aligned}\boldsymbol{p}^{(0)}(s,t,0,v) &= \mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y}\delta_0(s-(t-v))\\ &= \mathrm{e}^{\int_{t-v}^t \Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y}\delta_0(s-(t-v)).\end{aligned}$$

Then define a nonhomogeneous convolution operator $*$ by

$$(\boldsymbol{Q}_1 * \boldsymbol{Q}_2)(s,t) = \int_s^t \boldsymbol{Q}_1(s,u)\boldsymbol{Q}_2(u,t)\,\mathrm{d}u$$

and let

$$\begin{aligned}\boldsymbol{Q}^{*0}(s,t) &= \delta_0(t-s)\\ \boldsymbol{Q}^{*1}(s,t) &= \boldsymbol{Q}(s,t)\\ \boldsymbol{Q}^{*(n+1)}(s,t) &= \left(\boldsymbol{Q}^{*n} * \boldsymbol{Q}\right)(s,t).\end{aligned}$$

Then

$$\begin{aligned}\boldsymbol{p}(s,t,0,v) &= \sum_{n=0}^{\infty} \boldsymbol{p}^{(n)}(s,t,0,v)\\ &= \left(\sum_{n=0}^{\infty} \boldsymbol{Q}^{*n}(s,t-v)\right)\mathrm{e}^{\int_{t-v}^t \Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y}.\end{aligned}$$

Define

$$\boldsymbol{u}(s,t) = \sum_{n=0}^{\infty} \boldsymbol{Q}^{*n}(s,t), \quad \boldsymbol{u}(s,t) = \{u_{ij}(s,t)\}_{i,j\in E}.$$

Then

$$u_{ij}(s,t) = \text{probability of a jump to state } j \text{ in } [t,t+\mathrm{d}t) \text{ given } Z(s)=i, Z(s-)\neq i..$$

This is a so–called time–inhomogeneous Markov renewal density. We shall not delve into the theory of Markov renewal processes, but just mention that it exists and is a well–developed and extensive theory in the time–homogeneous case. The theory is less developed in the inhomogeneous setting.

For a general initial duration, u, we proceed as follows. For $n=0$, we clearly have

$$\boldsymbol{p}^{(0)}(s,t,u,v) = \mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}\delta_0(v-(u+t-s)).$$

For $n=1$, the only transition must be at time $t-v$, so that

$$\boldsymbol{p}^{(1)}(s,t,u,v) = \mathrm{e}^{\int_s^{t-v} \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}\nabla\boldsymbol{M}(t-v,u+t-v-s)\mathrm{e}^{\int_{t-v}^t \Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y}$$

$$= \boldsymbol{Q}(s,u,t-v)\mathrm{e}^{\int_{t-v}^{t}\Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y},$$

where

$$\boldsymbol{Q}(s,u,t) = \mathrm{e}^{\int_{s}^{t}\Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}\nabla\boldsymbol{M}(t,u+t-s)$$

are the defective densities for transitions at time t given initial duration of u at time s.

For $n>1$, we get as for the case of $u=0$ that

$$\boldsymbol{p}^{(n)}(s,t,u,v)$$
$$= \int_{s}^{t-v}\int_{w_1}^{t-v}\cdots\int_{w_{n-2}}^{t-v}\boldsymbol{Q}(s,u,w_1)\boldsymbol{Q}(w_1,w_2)\cdots\boldsymbol{Q}(w_{n-1},t-v)\,\mathrm{d}w_{n-1}\cdots\mathrm{d}w_1\,\mathrm{e}^{\int_{t-v}^{t}\Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y}.$$

Then we obtain what could be named a delayed time–inhomogeuous Markov renewal density, $\boldsymbol{u}(s,u,t)$, by

$$\boldsymbol{u}(s,u,t) = \boldsymbol{Q}(s,u,t)*\boldsymbol{u}(s,t),$$

and

$$\begin{aligned}\boldsymbol{p}(s,t,u,v) &= \sum_{n=0}^{\infty}\boldsymbol{p}^{(n)}(s,t,u,v)\\ &= \delta_0(v-(u+t-s))\mathrm{e}^{\int_{s}^{t}\Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y} + \boldsymbol{u}(s,u,t-v)\mathrm{e}^{\int_{t-v}^{t}\Delta\boldsymbol{M}(y,y-(t-v))\,\mathrm{d}y}.\end{aligned}$$

This equation implies immediately that

$$\left(\frac{\partial}{\partial t}+\frac{\partial}{\partial v}\right)\boldsymbol{p}(s,t,u,v) = \boldsymbol{p}(s,t,u,v)\Delta\boldsymbol{M}(t,v).$$

Regarding the initial condtion, consider

$$\begin{aligned}\boldsymbol{u}(s,u,t) &= \boldsymbol{Q}(s,t,u)*\boldsymbol{u}(s,t)\\ &= \int_{s}^{t}\boldsymbol{Q}(s,u,w)\boldsymbol{u}(w,t)\,\mathrm{d}w\\ &\to \nabla\boldsymbol{M}(s,u)\end{aligned}$$

as $s\to t$ due to the term $\delta_0(t-w)$ in $\boldsymbol{u}(w,t)$. Thus[1]

$$\boldsymbol{p}(s,s,u,v) = \delta_0(v-u)\boldsymbol{I} + \nabla\boldsymbol{M}(s,u)\delta_0(v).$$

2. Delayed partial differential equations for the transitions

In this section, we derived what corresponds to the Kolmogorov backward differential equations for the transition probabilities and densities.

We start with $\boldsymbol{P}(s,t,u,v)$.

[1] check carefully, in particular the Dirac delta

Theorem 5.6.

$$-\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{P}(s,t,u,v)=\Delta\boldsymbol{M}(s,u)\boldsymbol{P}(s,t,u,v)+\nabla\boldsymbol{M}(s,u)\boldsymbol{P}(s,t,0,v)$$

with boundary condition $\boldsymbol{P}(t,t,u,v)=\boldsymbol{I}1\{v\geq u\}$.

Proof. From (60) we have that

$$\begin{aligned}\boldsymbol{P}(s,t,u,v)=&\int_s^t \mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}\nabla\boldsymbol{M}(w,u+w-s)\boldsymbol{P}(w,t,0,v)\,\mathrm{d}w\\&+\mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}1\{u+t-s\leq v\}.\end{aligned}$$

Due to the different signs of s and u, we first notice that

$$\begin{aligned}\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)1\{u+t-s\leq v\}&=\mathbf{0},\\\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\nabla\boldsymbol{M}(w,u+w-s)&=\mathbf{0},\\\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\Delta\boldsymbol{M}(w,u+w-s)&=\mathbf{0},\end{aligned}$$

so that

$$\begin{aligned}&\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}=\mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\int_s^w \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y\\&=\mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}\left(\int_s^w \frac{\partial}{\partial s}\Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y-\Delta\boldsymbol{M}(s,u)+\int_s^w \frac{\partial}{\partial u}\Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y\right)\\&=-\Delta\boldsymbol{M}(s,u)\mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}.\end{aligned}$$

Then we get using Leibniz's rule for integrals, which only applies to the $\frac{\partial}{\partial s}$ part,

$$\begin{aligned}&\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{P}(s,t,u,v)\\&\quad=\int_s^t\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\left(\mathrm{e}^{\int_s^w \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}\nabla\boldsymbol{M}(w,u+w-s)\boldsymbol{P}(w,t,0,v)\right)\mathrm{d}w\\&\qquad-\nabla\boldsymbol{M}(s,u)\boldsymbol{P}(s,t,0,v)\\&\qquad+\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\left(\mathrm{e}^{\int_s^t \Delta\boldsymbol{M}(y,u+y-s)\,\mathrm{d}y}1\{u+t-s\leq v\}\right)\\&\quad=-\Delta\boldsymbol{M}(s,u)\boldsymbol{P}(s,t,u,v)-\nabla\boldsymbol{M}(s,u)\boldsymbol{P}(s,t,0,v).\end{aligned}$$

The initial condition amounts to $\boldsymbol{P}(t,t,u,v)=\boldsymbol{I}1\{v\geq u\}$ follows directly from (60). □

Differentiating through with v on both sides gives us the PDE for densities:

Theorem 5.7.

$$\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{p}(s,t,u,v)=-\Delta\boldsymbol{M}(s,u)\boldsymbol{p}(s,t,u,v)-\nabla\boldsymbol{M}(s,u)\boldsymbol{p}(s,t,0,v)$$

with boundary condition $\boldsymbol{p}(t,t,u,v)=\boldsymbol{I}\delta_0(v-u)$.

So we see that the PDEs are the same, but with different boundary conditions.

3. The life–insurance model

We consider a life insurance contract, stipulating that the insured can be in one of p different (biometric) states at any time. This could, for example, be a simple "alive–death model ($p=2$), "active–disabled–death" model or "active–disabled–lapsing–death" with $p=2$. There, the original "alive"–state has been refined into subconditions, which are of importance for differentiating between payments.

The state of the insured is assumed to follow a semi–Markov process $\{Z(t)\}_{t\geq 0}$ with state–space $E=\{1,2,...,p\}$ and duration process $\{U(t)\}_{u\geq 0}$. Let $\boldsymbol{M}(t,u)=\{\mu_{ij}(t,u)\}_{i,j\in E}$ denote its intensity matrix.

Given that $Z(t)=i, U(t)=u$, there is a continuous benefit payment with rate $b_i(t,u)$ during sojourns. Negative values are premium rates. We assume the matrix is split into

$$\boldsymbol{M}(t,u)=\boldsymbol{M}^0(t,u)+\boldsymbol{M}^1(t,u),$$

with corresponding intensities $\boldsymbol{M}^0(t,u)=\{\mu^0_{ij}(t,u)\}_{i,j\in E}$ and $\boldsymbol{M}^1(t,u)=\{\mu^1_{ij}(t,u)\}_{i,j\in E}$. With intensity $\mu^0_{ij}(t,u)$, $i\neq j$, there is a transition from i to j which does not result in a lump sum payment, while $\mu^1_{ij}(t,u)$ denotes the contrary, i.e. the intensity for a transition triggering a (deterministic) lump sum payment of size $b_{ij}(t,u)$, say. Furthermore, at a rate of $\mu^1_{ii}(t,u)$, there will be a lump sum payment of size $b_{ii}(t,u)$ during sojourns. In this case,

$$\Delta\boldsymbol{M}^0(s,u)=\Delta\boldsymbol{M}(s,u)-\boldsymbol{\Delta}(\boldsymbol{\mu}^1(s,u)),$$

where $\boldsymbol{\mu}^1(s,u)=(\mu^1_{11}(s,u),...,\mu^1_{pp}(s,u))$.

We arrange the lumpsum payments into a matrix $\boldsymbol{B}(t,u)=\{b_{ij}(t,u)\}_{i,j\in E}$.

In many cases, we may let $\boldsymbol{M}^0(t,u)=\Delta\boldsymbol{M}(t,u)$ and $\boldsymbol{M}^0(t,u)=\nabla\boldsymbol{M}(t,u)$ by letting $b_{ij}(t,u)=0$ in appropriate places. If there is an $i\neq j$ such that both $\mu^0_{ij}(t,u)>0$ and $\mu^1_{ij}(t,u)>0$, then the probability of a lump sum payment upon a transition from i to j equals

$$q_{ij}(t,u)=\frac{\mu^1_{ij}(t,u)}{\mu^0_{ij}(t,u)+\mu^1_{ij}(t,u)}.$$

Such a mechanism could be present in practice if, for example, disability lump-sum compensation depends on where and how the accident occurred.

Remark 5.8: The lump sum payments during the sojourn can be used to control a variety of additional features. We can even use it to specify lump sum payments at a deterministic time S (for example at termination) by letting $\mu_{ii}^1(t,u) = \delta_0(S-t)$ (with the convention that $+\infty \cdot 0 = 0$).

Let $N_{jk}(t)$ denote the counting process that counts the number of lump sum paying transitions (or lump sum during sojourn if $j = k$), and define the payment process by

$$\mathrm{d}B(t) = \sum_j 1\{Z(t) = j\} \left(b_j(t, U(t)) + \sum_k b_{jk}(t, U(t-)) \mathrm{d}N_{jk}(t) \right).$$

The undiscounted partial reward densities are then defined as

$$v_{ij}^0(s,t,u,v) = \mathbb{E}\left(1\{Z(t) = j, U(t) \in (v, v+\mathrm{d}v)\} \int_s^t \mathrm{d}B(x) \middle| Z(s) = i, U(s) = u \right).$$

Let $\boldsymbol{v}^0(s,t,u,v) = \{v_{ij}^0(s,t,u,v)\}$. The superscript zero indicates that the reserve is not discounted by an interest rate.

The rewards are organised in

$$\boldsymbol{R}(s,u) = \boldsymbol{R}^0(s,u) + \boldsymbol{R}^1(s,u),$$

where

$$\begin{aligned} \boldsymbol{R}^0(s,u) &= \boldsymbol{\Delta}(\boldsymbol{b}(s,u)) + \Delta \boldsymbol{M}^1(s,u) \bullet \boldsymbol{B}(s,u) \\ \boldsymbol{R}^1(s,u) &= \nabla \boldsymbol{M}^1(s,u) \bullet \boldsymbol{B}(s,u). \end{aligned}$$

The elements of the diagonal of $\boldsymbol{M}^1(s,u)$ correspond to the arrival of lump sum payments during a sojourn, and hence the duration will not reset for these events.

Then

Theorem 5.9.

$$\begin{aligned} \boldsymbol{v}^0(s,t,u,v) = &\int_s^t \int_0^{u+x-s} \boldsymbol{p}(s,x,u,w) \boldsymbol{R}^0(x,w) \boldsymbol{p}(x,t,w,v) \,\mathrm{d}w \,\mathrm{d}x \\ &+ \int_s^t \int_0^{u+x-s} \boldsymbol{p}(s,x,u,w) \boldsymbol{R}^1(x,w) \boldsymbol{p}(x,t,0,v) \,\mathrm{d}w \,\mathrm{d}x. \end{aligned} \tag{65}$$

Proof.

$$\begin{aligned} v_{ij}^0(s,t,u,v) &= \mathbb{E}\left(1\{Z(t) = j, U(t) \in (v, v+\mathrm{d}v)\} \int_s^t \mathrm{d}B(x) \middle| Z(s) = i, U(s) = u \right) \\ &= \int_s^t \mathbb{E}\left(1\{Z(t) = j, U(t) \in (v, v+\mathrm{d}v)\} \,\mathrm{d}B(x) \middle| Z(s) = i, U(s) = u \right) \end{aligned}$$

For a time $x \in [s,t]$, $Z(x) \in E$ and $U(x) \in [0, u+(x-s)]$, so by the Markov property of $(Z(t), U(t))$, we get

$$\begin{aligned} v^0_{ij}(s,t,u,v) &= \sum_k \int_s^t \int_0^{u+x-s} \mathbb{E}(1\{Z(t)=j, U(t) \in (v, v+\mathrm{d}v)\} \times \\ &\qquad 1\{Z(x)=k, U(x) \in (w, w+\mathrm{d}w)\}\,\mathrm{d}B(x) | Z(s)=i, U(s)=u)\,\mathrm{d}w \\ &= \sum_k \int_s^t \int_0^{u+x-s} \mathbb{E}(1\{Z(t)=j, U(t) \in (v, v+\mathrm{d}v)\}\,\mathrm{d}B(x) | Z(x)=k, U(x)=w) \times \\ &\qquad \mathbb{E}(1\{Z(x)=k, U(x) \in (w, w+\mathrm{d}w)\} | Z(s)=i, U(s)=u)\,\mathrm{d}w. \end{aligned}$$

First we notice that

$$\mathbb{E}(1\{Z(x)=k, U(x) \in (w, w+\mathrm{d}w)\} | Z(s)=i, U(s)=u) = p_{ik}(s,x,u,w).$$

Secondly,

$$\begin{aligned} &\mathbb{E}(1\{Z(t)=j, U(t) \in (v, v+\mathrm{d}v)\}\,\mathrm{d}B(x) | Z(x)=k, U(x)=w) \\ &\quad = b_k(x,w)\,\mathrm{d}x\ p_{kj}(x,t,w,v) + \sum_{\ell \neq k} \mu^1_{k\ell}(x,w)\,\mathrm{d}x\ b_{k\ell}(x,w) p_{\ell j}(x,t,0,v) \\ &\qquad + \mu^1_{kk}(x,w)\,\mathrm{d}x\ b_{kk}(x,w) p_{\ell j}(x,t,w,v) \\ &\quad = (b_k(x,w) + \mu^1_{kk}(w,w) b_{kk}(x,w))\,\mathrm{d}x p_{\ell j}(x,t,w,v) + \sum_{\ell \neq k} \mu^1_{k\ell}(x,w)\,\mathrm{d}x\ b_{k\ell}(x,w) p_{\ell j}(x,t,0,v) \\ &\quad = \boldsymbol{R}^0_{kk}(x,w)\,\mathrm{d}x\ p_{kj}(x,t,w,v) + \sum_{\ell \neq k} \boldsymbol{R}^1_{k\ell}(x,w)\,\mathrm{d}x\ p_{\ell j}(x,t,0,v) \\ &\quad = \boldsymbol{R}^0_{kk}(x,w)\,\mathrm{d}x\ p_{kj}(x,t,w,v) + \sum_{\ell} \boldsymbol{R}^1_{k\ell}(x,w)\,\mathrm{d}x\ p_{\ell j}(x,t,0,v), \end{aligned}$$

since the diagonal elements of $\boldsymbol{R}^1(x,w)$ are zero.

Then we obtain

$$\begin{aligned} v^0_{ij}(s,t,u,v) &= \sum_k \int_s^t \int_0^{u+x-s} p_{ik}(s,x,u,w) \boldsymbol{R}^0_{kk}(x,w) p_{kj}(x,t,w,v)\,\mathrm{d}x\,\mathrm{d}w \\ &\quad + \sum_{k,\ell} \int_s^t \int_0^{u+x-s} p_{ik}(s,x,u,w) \boldsymbol{R}^1_{k\ell}(x,w)\,\mathrm{d}x p_{\ell j}(x,t,0,v)\,\mathrm{d}w, \end{aligned}$$

which is exactly the (i,j) element of right hand side of (65). □

Lemma 5.10.

$$\begin{aligned} &\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \int_s^t \int_0^{u+x-s} g(s,x,u,w)\,\mathrm{d}w\,\mathrm{d}x \\ &\quad = \int_s^t \int_0^{u+x-s} \left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) g(s,x,u,w)\,\mathrm{d}w\,\mathrm{d}x - \int_0^u g(s,s,u,w)\,\mathrm{d}w. \end{aligned}$$

Proof. By Leibniz's rule,

$$\begin{aligned}
&\frac{\partial}{\partial s}\int_s^t\int_0^{u+x-s} g(s,x,u,w)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad=\int_s^t \frac{\partial}{\partial s}\int_0^{u+x-s} g(s,x,u,w)\,\mathrm{d}w\,\mathrm{d}x-\int_0^u g(s,x,u,w)\,\mathrm{d}w \\
&\quad=\int_s^t\left(\int_0^{u+x-s}\frac{\partial}{\partial s}g(s,x,u,w)\,\mathrm{d}w-g(s,x,u,u+x-s)\right)\mathrm{d}x-\int_0^u g(s,s,u,w)\,\mathrm{d}w
\end{aligned}$$

and

$$\begin{aligned}
&\frac{\partial}{\partial u}\int_s^t\int_0^{u+x-s} g(s,x,u,w)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad=\int_s^t \frac{\partial}{\partial u}\int_0^{u+x-s} g(s,x,u,w)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad=\int_s^t\left(\int_0^{u+x-s}\frac{\partial}{\partial s}g(s,x,u,w)\,\mathrm{d}w+g(s,x,u,u+x-s)\right)\mathrm{d}x
\end{aligned}$$

so

$$\begin{aligned}
&\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\int_s^t\int_0^{u+x-s} g(s,x,u,w)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad=\int_s^t\int_0^{u+x-s}\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)g(s,x,u,w)\,\mathrm{d}w\,\mathrm{d}x-\int_0^u g(s,s,u,w)\,\mathrm{d}w.
\end{aligned}$$

□

Definition 5.11: The partial reserve (density), discounted by a deterministic spot rate, $r(u)$, is defined as the matrix $\boldsymbol{v}(s,t,u,v)=\{v_{ij}(s,t,u,v)\}_{i,j\in E}$ with elements

$$v_{ij}(s,t,u,v)=\mathbb{E}\left(1\{Z(t)=j,U(t)\in(v,v+\mathrm{d}v)\}\int_s^t \mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\,\mathrm{d}B(x)\,\middle|\,Z(s)=i,U(s)=u\right).$$

Clearly the discounting factor $\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}$ can be multiplied onto the payment process (for fixed s), giving

$$\begin{aligned}
\boldsymbol{v}(s,t,u,v)=&\int_s^t \mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\int_0^{u+x-s}\boldsymbol{p}(s,x,u,w)\boldsymbol{R}^0(x,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x+ \\
&+\int_s^t \mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\int_0^{u+x-s}\boldsymbol{p}(s,x,u,w)\boldsymbol{R}^1(x,w)\boldsymbol{p}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x. \qquad (66)
\end{aligned}$$

Lemma 5.12. *Let* $\boldsymbol{Y}(s,x,u,w) = \mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w)$. *Then*

$$\begin{aligned}\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)&\boldsymbol{Y}(s,x,u,w)\\ &= -(\Delta\boldsymbol{M}(s,u)-r(s)\boldsymbol{I})\boldsymbol{Y}(s,x,u,w)-\nabla\boldsymbol{M}(s,u)\boldsymbol{Y}(s,x,0,w)\end{aligned}$$

with boundary condition $\boldsymbol{Y}(t,t,u,v) = \boldsymbol{I}\delta_0(v-u)$.

Proof. Differentiation of the product of functions,

$$\begin{aligned}-\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)&\boldsymbol{Y}(s,x,u,w)\\ &= -\frac{\partial}{\partial s}\left(\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w)\right)-\frac{\partial}{\partial u}\left(\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w)\right)\\ &= -r(s)\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w)-\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\frac{\partial}{\partial s}\boldsymbol{p}(s,x,u,w)\\ &\quad -\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\frac{\partial}{\partial u}\boldsymbol{p}(s,x,u,w)\\ &= -r(s)\boldsymbol{Y}(s,x,u,w)+\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}(\Delta\boldsymbol{M}(s,u)\boldsymbol{p}(s,x,u,w)+\nabla\boldsymbol{M}(s,u)\boldsymbol{p}(s,x,0,w))\\ &= -r(s)\boldsymbol{Y}(s,x,u,w)+\Delta\boldsymbol{M}(s,u)\boldsymbol{Y}(s,x,u,w)+\nabla\boldsymbol{M}(s,u)\boldsymbol{Y}(s,x,0,w)\\ &= (\Delta\boldsymbol{M}(s,u)-r(s)\boldsymbol{I})\boldsymbol{Y}(s,x,u,w)+\nabla\boldsymbol{M}(s,u)\boldsymbol{Y}(s,x,0,w).\end{aligned}$$

□

Theorem 5.13 (Thiele)**.** *The partial reserves densities satisfy the PDE*

$$\begin{aligned}-\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{v}(s,t,u,v) &= (\Delta\boldsymbol{M}(s,u)-r(s)\boldsymbol{I})\boldsymbol{v}(s,t,u,v)+\nabla\boldsymbol{M}(s,u)\boldsymbol{v}(s,t,0,v)\\ &\quad +\boldsymbol{R}^0(s,u)\boldsymbol{p}(s,t,u,v)+\boldsymbol{R}^1(s,u)\boldsymbol{p}(s,t,0,v)\end{aligned}$$

with boundary condition $\boldsymbol{v}(t,t,u,v) = \boldsymbol{0}$.

Proof. We differentiate (66),

$$\begin{aligned}-\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)&\boldsymbol{v}(s,t,u,v)\\ &= -\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\int_s^t\int_0^{u+x-s}\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w)\boldsymbol{R}^0(x,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x+\\ &\quad -\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\int_s^t\int_0^{u+x-s}\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w)\boldsymbol{R}^1(x,w)\boldsymbol{p}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x\end{aligned}$$

By Lemma 5.10, the first derivative amounts to

$$\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\int_s^t\int_0^{u+x-s}\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w)\boldsymbol{R}^0(x,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x$$

$$
\begin{aligned}
&= \int_s^t \int_0^{u+x-s} \left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \left(\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u} \boldsymbol{p}(s,x,u,w)\right) \boldsymbol{R}^0(x,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad - \int_0^u \mathrm{e}^{-\int_s^s r(u)\,\mathrm{d}u} \boldsymbol{p}(s,s,u,w)\boldsymbol{R}^0(s,w)\boldsymbol{p}(s,t,w,v)\,\mathrm{d}w \\
&= \int_s^t \int_0^{u+x-s} \left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \boldsymbol{Y}(s,x,u,w)\boldsymbol{R}^0(x,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad - \boldsymbol{R}^0(s,u)\boldsymbol{p}(s,t,u,v) \\
&= -(\Delta\boldsymbol{M}(s,u) - r(s)\boldsymbol{I}) \int_s^t \int_0^{u+x-s} \boldsymbol{Y}(s,x,u,w)\boldsymbol{R}^0(x,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad - \nabla\boldsymbol{M}(s,u) \int_s^t \int_0^{u+x-s} \boldsymbol{Y}(s,x,0,w)\boldsymbol{R}^0(x,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad - \boldsymbol{R}^0(s,u)\boldsymbol{p}(s,t,u,v) \\
&= -(\Delta\boldsymbol{M}(s,u) - r(s)\boldsymbol{I}) \int_s^t \int_0^{u+x-s} \boldsymbol{Y}(s,x,u,w)\boldsymbol{R}^0(x,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad - \nabla\boldsymbol{M}(s,u) \int_s^t \int_0^{x-s} \boldsymbol{Y}(s,x,0,w)\boldsymbol{R}^0(x,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad - \boldsymbol{R}^0(s,u)\boldsymbol{p}(s,t,u,v),
\end{aligned}
$$

and similarly for the second derivative

$$
\begin{aligned}
&\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \int_s^t \int_0^{u+x-s} \mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u} \boldsymbol{p}(s,x,u,w)\boldsymbol{R}^1(x,w)\boldsymbol{p}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x \\
&\quad = -(\Delta\boldsymbol{M}(s,u) - r(s)\boldsymbol{I}) \int_s^t \int_0^{u+x-s} \boldsymbol{Y}(s,x,u,w)\boldsymbol{R}^1(x,w)\boldsymbol{p}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x \\
&\qquad - \nabla\boldsymbol{M}(s,u) \int_s^t \int_0^{x-s} \boldsymbol{Y}(s,x,0,w)\boldsymbol{R}^1(x,w)\boldsymbol{p}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x \\
&\qquad - \boldsymbol{R}^1(s,u)\boldsymbol{p}(s,t,0,v).
\end{aligned}
$$

Collecting the factors to $\Delta\boldsymbol{M}$ and $\nabla\boldsymbol{M}$ then gives

$$
\begin{aligned}
-\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \boldsymbol{v}(s,t,u,v) &= (\Delta\boldsymbol{M}(s,u) - r(s)\boldsymbol{I})\boldsymbol{v}(s,t,u,v) \\
&\quad + \nabla\boldsymbol{M}\boldsymbol{v}(s,t,0,v) + \boldsymbol{R}^0(s,u)\boldsymbol{p}(s,t,u,v) + \boldsymbol{R}^1(s,u)\boldsymbol{p}(s,t,0,v).
\end{aligned}
$$

□

It looks like a simple PDE (delayed, though) with a simple initial condition. However, the PDE is coupled to another PDE that solves for the unknown $\boldsymbol{p}(s,t,0,v)$ (and $\boldsymbol{p}(s,t,u,v)$). In the case of Markov models, we have seen how product integration together with Van Loan's formula solves both transition probabilities, reserves and higher order moments. This will be the theme of the following section, where the product integration framework is extended to semi–Markov models.

4. Matrix form of the integral equation and its unique solution

Let $\boldsymbol{A}^0(s,u)$ and $\boldsymbol{A}^1(s,u)$ be arbitrary square matrices of the same dimension which vanish for $u > s$. While the interpretation of s and u is still time and duration, the matrices may contain other terms than transition rates, such as, e.g. reward sizes at transition or rates. This will be important later on when dealing with reserves and higher–order moments. Therefore the following theorem is still linked to the semi–Markov set–up in spite of the seemingly free choice of matrix functions.

Theorem 5.14. *Consider a fixed interval $[x_0,x]$ and let $s,t \in [x_0,x]$. Assume that $(x,y) \to \boldsymbol{A}^i(x,y)$, $i = 0,1$, are continuous matrix functions. Then the matrix integral equation*

$$\begin{aligned}\boldsymbol{X}(s,t,u,v) = &\int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,u+w-s)\boldsymbol{X}(w,t,0,v)\,\mathrm{d}w \\ &+ \prod_s^t (\boldsymbol{I} + \boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)1\{u+t-s \le v\} \qquad (67)\end{aligned}$$

has a unique solution $\boldsymbol{X}(s,t,u,v)$ for $s,t \in [x_0,t]$, $s \le t$.

Proof. Define

$$\boldsymbol{X}^{(0)}(s,t,0,v) = \prod_s^t (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)1\{t-s \le v\},$$

$$\boldsymbol{X}^{(n+1)}(s,t,0,v) = \int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s)\boldsymbol{X}^{(n)}(w,t,0,v)\,\mathrm{d}w,$$

and

$$\boldsymbol{X}(s,t,0,v) = \sum_{n=0}^{\infty} \boldsymbol{X}^{(n)}(s,t,0,v).$$

We show that there exists a unique solution $\boldsymbol{X}(s,t,0,v)$ to the integral matrix equations for $u = 0$. This implies that $\boldsymbol{X}(s,t,u,v)$ also exists and is unique.

By continuity, both $w \to \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)$ and $w \to \boldsymbol{A}^1(w,w-s)$ are bounded matrix functions on $[x_0,x]$, so there exists a matrix $\boldsymbol{K}$ such that[2]

$$\left|\prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\right| \le \boldsymbol{K} \quad \text{and} \quad \left|\boldsymbol{A}^1(w,w-s)\right| \le \boldsymbol{K}$$

for any $s,w \in [x_0,x]$. Then

$$\left|\boldsymbol{X}^{(0)}(s,t,0,v)\right| \le \boldsymbol{K}1\{t-s \le v\}$$

[2]We remember that the absolute value of a matrix is defined as the matrix of the absolute values of its elements.

and

$$\left|\boldsymbol{X}^{(1)}(s,t,0,v)\right| = \left|\int_s^t \prod_s^w (\boldsymbol{I}+\boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s)\boldsymbol{X}^{(0)}(w,t,0,v)\,\mathrm{d}w\right|$$
$$\leq \int_s^t \boldsymbol{K}^3 1\{w-s\leq v\}\,\mathrm{d}w \leq \boldsymbol{K}^3(t-s).$$

Similarly,

$$\left|\boldsymbol{X}^{(2)}(s,t,0,v)\right| = \left|\int_s^t \prod_s^w (\boldsymbol{I}+\boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s)\boldsymbol{X}^{(1)}(w,t,0,v)\,\mathrm{d}w\right|$$
$$\leq \int_s^t \boldsymbol{K}^5(w-s)\,\mathrm{d}w = \boldsymbol{K}^5\frac{1}{2}(t-s)^2.$$

In general,

$$\left|\boldsymbol{X}^{(n)}(s,t,0,v)\right| \leq \boldsymbol{K}^{2n+1}\frac{(t-s)^n}{n!} = \boldsymbol{K}\frac{\left[\boldsymbol{K}^2(t-s)\right]^n}{n!}.$$

Thus

$$\sum_{n=0}^{\infty}\left|\boldsymbol{X}^{(n)}(s,t,0,v)\right| \leq \boldsymbol{K}\exp\left(\boldsymbol{K}^2(t-s)\right) \leq \boldsymbol{K}\exp\left(\boldsymbol{K}^2(x-x_0)\right) < \infty,$$

proving that the series $\sum_{n=0}^{\infty}\boldsymbol{X}^{(n)}(s,t,0,v)$ converges absolutely and uniformly over $[x_0,x]$.

This justifies the interchange of integration and summation, so that

$$\begin{aligned}\boldsymbol{X}(s,t,0,v) &= \sum_{n=0}^{\infty}\boldsymbol{X}^{(n)}(s,t,0,v)\\ &= \boldsymbol{X}^{(0)}(s,t,0,v) + \sum_{n=1}^{\infty}\int_s^t\prod_s^w(\boldsymbol{I}+\boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s)\boldsymbol{X}^{(n-1)}(w,t,0,v)\,\mathrm{d}w\\ &= \boldsymbol{X}^{(0)}(s,t,0,v) + \int_s^t\prod_s^w(\boldsymbol{I}+\boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s)\boldsymbol{X}(w,t,0,v)\,\mathrm{d}w\end{aligned}$$

which shows that

$$\boldsymbol{X}(s,t,0,v) = \sum_{n=0}^{\infty}\boldsymbol{X}^{(n)}(s,t,0,v)$$

is indeed a solution, proving existence.

Regarding uniquess, assume that $\boldsymbol{X}_1(s,t,u,v)$ and $\boldsymbol{X}_2(s,t,u,v)$ both solve the integral equation. Then $\boldsymbol{Y}(s,t,u,v) = \boldsymbol{X}_1(s,t,u,v) - \boldsymbol{X}_2(s,t,u,v)$ satisfies

$$\boldsymbol{Y}(s,t,u,v) = \int_s^t\prod_s^w(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,u+w-s)\boldsymbol{Y}(w,t,0,v)\,\mathrm{d}w.$$

It is enough to prove that $\boldsymbol{Y}(w,t,0,v)\equiv\boldsymbol{0}$. This follows from

$$\begin{aligned}|\boldsymbol{Y}(s,t,0,v)| &\leq \int_s^t\left|\prod_s^w(\boldsymbol{I}+\boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\right|\left|\boldsymbol{A}^1(w,w-s)\right|\left|\boldsymbol{Y}(w,t,0,v)\right|\mathrm{d}w\\ &\leq \boldsymbol{K}_1^3(t-s),\end{aligned}$$

where $\boldsymbol{K}_1$ is now an upper bound for the previous $\boldsymbol{K}$ and $|\boldsymbol{Y}(w,t,0,v)|$. Repeating the argument shows that

$$|\boldsymbol{Y}(s,t,0,v)| \le \boldsymbol{K}_1 \frac{(\boldsymbol{K}_1^2)^n}{n!}(t-s)^n \le \boldsymbol{K}_1 \frac{(\boldsymbol{K}_1^2)^n}{n!}(x-x_0)^n \to \boldsymbol{0} \quad \text{as} \quad n \to \infty$$

since the series

$$\sum_{n=0}^{\infty} \frac{(\boldsymbol{K}_1^2)^n}{n!}(t-s)^n = \exp(\boldsymbol{K}_1^2(t-s)) < \infty$$

converges. □

There is a differential equation equivalent to (67), which may be convenient to work with.

Theorem 5.15. *The solution $\boldsymbol{X}(s,t,u,v)$ is the unique solution to*

$$-\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right)\boldsymbol{X}(s,t,u,v) = \boldsymbol{A}^0(s,u)\boldsymbol{X}(s,t,u,v) + \boldsymbol{A}^1(s,u)\boldsymbol{X}(s,t,0,v) \tag{68}$$

with initial condition

$$\boldsymbol{X}(t,t,u,v) = \boldsymbol{I}1\{u \le v\}.$$

Proof. By Leibniz's rule for product integrals,

$$\begin{aligned}\frac{\partial}{\partial s}\prod_s^t(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y) &= -\boldsymbol{A}^0(s,u)\prod_s^t(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\\ &\quad + \int_s^t \prod_s^x(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\frac{\partial}{\partial s}\boldsymbol{A}^0(x,u+x-s)\prod_x^t(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\,\mathrm{d}u\end{aligned}$$

and

$$\begin{aligned}&\frac{\partial}{\partial u}\prod_s^t(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\\ &\quad = \int_s^t \prod_s^x(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\frac{\partial}{\partial u}\boldsymbol{A}^0(s,u+x-s)\prod_u^t(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\,\mathrm{d}u.\end{aligned}$$

Since the derivatives under the integrals w.r.t. s and u are identical with opposite signs, we get

$$\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right)\prod_s^t(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y) = -\boldsymbol{A}^0(s,u)\prod_s^t(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y).$$

Then by Leibniz's rule for ordinary integrals,

$$\begin{aligned}&\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right)\boldsymbol{X}(s,t,u,v)\\ &\quad = -\boldsymbol{A}^0(s,u)\int_s^t \prod_s^w(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,u+w-s)\boldsymbol{X}(w,t,0,v)\,\mathrm{d}w\\ &\qquad - \boldsymbol{A}^1(s,u)\boldsymbol{X}(s,t,0,v) - \boldsymbol{A}^0(s,u)\prod_s^t(\boldsymbol{I}+\boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)1\{u+t-s \le v\}\end{aligned}$$

$$= -\boldsymbol{A}^0(s,u)\boldsymbol{X}(s,t,u,v) - \boldsymbol{A}^1(s,u)\boldsymbol{X}(s,t,0,v).$$

□

Assume that $v \to \boldsymbol{X}(s,t,u,v)$ is differentiable and let

$$\boldsymbol{x}(s,t,u,v) = \frac{\partial}{\partial v}\boldsymbol{X}(s,t,u,v).$$

Then we get a similar integral equation and PDE for $\boldsymbol{x}(s,t,u,v)$.

Theorem 5.16. *$\boldsymbol{x}(s,t,u,v)$ satisfies*

$$\begin{aligned}\boldsymbol{x}(s,t,u,v) = &\int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,u+w-s)\boldsymbol{x}(w,t,0,v)\,\mathrm{d}w \\ &+ \prod_s^t (\boldsymbol{I} + \boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\delta_0(v-u-t+s) \qquad (69)\end{aligned}$$

and

$$-\frac{\partial}{\partial s}\boldsymbol{x}(s,t,u,v) - \frac{\partial}{\partial u}\boldsymbol{x}(s,t,u,v) = \boldsymbol{A}^0(s,u)\boldsymbol{x}(s,t,u,v) + \boldsymbol{A}^1(s,u)\boldsymbol{x}(s,t,0,v) \qquad (70)$$

with initial condition

$$\boldsymbol{x}(t,t,u,v) = \boldsymbol{I}\delta_0(u-v),$$

where δ_0 denotes Dirac's δ–function.

Proof. Follows by differentiation w.r.t. v. We can differentiate under the integration sign since $\boldsymbol{A}^0$ and $\boldsymbol{A}^1$ are continuous functions, and hence bounded over $[s,t]$. □

The solution to (69) can obtained by

$$\begin{aligned}\boldsymbol{x}^{(0)}(s,t,0,v) &= \prod_s^t (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\delta_0\{v-(t-s)\} \\ \boldsymbol{x}^{(n+1)}(s,t,0,v) &= \int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s)\boldsymbol{x}^{(n)}(w,t,0,v)\,\mathrm{d}w \\ \boldsymbol{x}(s,t,0,v) &= \sum_{n=0}^{\infty} \boldsymbol{x}^{(n)}(s,t,0,v) \\ \boldsymbol{x}(s,t,u,v) &= \int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,u+w-s)\boldsymbol{x}(w,t,0,v)\,\mathrm{d}w \\ &\quad + \prod_s^t (\boldsymbol{I} + \boldsymbol{A}^0(y,u+y-s)\,\mathrm{d}y)\delta_0(v-u-t+s)\boldsymbol{x}(s,t,u,v).\end{aligned}$$

Let us carefully examine the first steps in the iteration scheme.

$$\begin{aligned}\boldsymbol{x}^{(1)}(s,t,0,v) &= \int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s)\boldsymbol{x}^{(0)}(w,t,0,v)\,\mathrm{d}w \\ &= \int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s)\prod_w^t (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\delta_0\{v-(t-w)\}\,\mathrm{d}w.\end{aligned}$$

If $v > t-s$, then $t-w < v$ for all $w \in [s,t]$ so $t-w$ will never reach v. In this case, the contribution is zero. Hence there can be a contribution only if $v \in [0,t-s]$, in which case we get

$$\begin{aligned}
&\boldsymbol{x}^{(1)}(s,t,0,v) \\
&\quad = \prod_{s}^{t-v} (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(t-v,t-v-s) \prod_{t-v}^{t} (\boldsymbol{I} + \boldsymbol{A}^0(y,y-(t-v))\,\mathrm{d}y) 1_{[0,t-s]}(v).
\end{aligned}$$

Thinking of it as transition matrices, $\boldsymbol{x}^{(1)}(s,t,0,v)$ is the transition density matrix for starting at time s with duration 0 and going to time t with end duration v with exactly one transition in $[0,t-s]$. The only way this can happen is by having a transition at time $t-v$ and no further transition afterwards.

Next iteration then amounts to

$$\begin{aligned}
\boldsymbol{x}^{(2)}(s,t,0,v) &= \int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s)\boldsymbol{x}^{(1)}(w,t,0,v)\,\mathrm{d}w \\
&= \int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(w,w-s) \\
&\quad \times \prod_{w}^{t-v} (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(t-v,t-v-w) \prod_{t-v}^{t} (\boldsymbol{I} + \boldsymbol{A}^0(y,y-(t-v))\,\mathrm{d}y) 1_{[0,t-w]}(v)\,\mathrm{d}w
\end{aligned}$$

Introduce

$$\boldsymbol{Q}(s,t) = \prod_s^t (\boldsymbol{I} + \boldsymbol{A}^0(y,y-s)\,\mathrm{d}y)\boldsymbol{A}^1(t,t-s).$$

Thinking of transition matrices again, this would be the transition density matrix for the first arrival since s occurring at time t. Then we can write

$$\begin{aligned}
\boldsymbol{x}^{(2)}(s,t,0,v) &= \int_s^t \boldsymbol{Q}(s,w)\boldsymbol{Q}(w,t-v) \prod_{t-v}^{t} (\boldsymbol{I} + \boldsymbol{A}^0(y,y-(t-v))\,\mathrm{d}y) 1_{[0,t-w]}(v)\,\mathrm{d}w \\
&= \int_s^{t-v} \boldsymbol{Q}(s,w)\boldsymbol{Q}(w,t-v) \prod_{t-v}^{t} (\boldsymbol{I} + \boldsymbol{A}^0(y,y-(t-v))\,\mathrm{d}y)\,\mathrm{d}w.
\end{aligned}$$

In general, for $n \geq 1$

$$\begin{aligned}
&\boldsymbol{x}^{(n)}(s,t,0,v) \\
&= \int_s^{t-v} \int_{w_1}^{t-v} \cdots \int_{w_{n-2}}^{t-v} \boldsymbol{Q}(s,w_1)\cdots\boldsymbol{Q}(w_{n-1},t-v) \prod_{t-v}^{t} (\boldsymbol{I} + \boldsymbol{A}^0(y,y-(t-v))\,\mathrm{d}y)\,\mathrm{d}w_{n-1}\cdots\mathrm{d}w_1 \\
&= \int_s^{t-v} \int_{w_1}^{t-v} \cdots \int_{w_{n-2}}^{t-v} \boldsymbol{Q}(s,w_1)\cdots\boldsymbol{Q}(w_{n-1},t-v)\,\mathrm{d}w_{n-1}\cdots\mathrm{d}w_1 \prod_{t-v}^{t} (\boldsymbol{I} + \boldsymbol{A}^0(y,y-(t-v))\,\mathrm{d}y)
\end{aligned}$$

5. An extended product integral

The unique solutions to the integral equations (67) and (69), or ditto delayed PDE (68) and (70), will be denoted by symbols that generalise the product integral for Markov processes.

Definition 5.17: Given matrices jointly continuous square matrix functions, $\boldsymbol{A}^0(t,u)$ and $\boldsymbol{A}^1(t,u)$, we denote the unique solution $\boldsymbol{x}(s,t,u,v)$ of (69) by

$$\boldsymbol{x}(s,t,u,v) = \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x,w)\,\mathrm{d}(x,w)),$$

and refer to it as the (extended) product integral of $\boldsymbol{A}(x,w) = (\boldsymbol{A}^0(t,u), \boldsymbol{A}^1(t,u))$. Solutions to (67) will be denoted

$$\boldsymbol{X}(s,t,u,v) = \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x,w)\,\mathrm{d}(x,w)).$$

Remark 5.18: Different choice of matrices $\boldsymbol{A}^0(t,u)$ and $\boldsymbol{A}^1(t,u)$ will result in different solutions. For the applications we have in mind, the matrices $\boldsymbol{A}^0(t,u)$ and $\boldsymbol{A}^1(t,u)$ will arise from a unique decomposition of a matrix

$$\boldsymbol{A}(t,u) = \boldsymbol{A}^0(t,u) + \boldsymbol{A}^1(t,u),$$

where the former matrix $\boldsymbol{A}^0(t,u)$ has to do with properties during sojourns while $\boldsymbol{A}^1(t,u)$ has to do with properties at transitions. For example, for intensity matrices $\boldsymbol{\Lambda}(t,u)$ we have the unique decomposition into a sum of its diagonal matrix $\Delta\boldsymbol{\Lambda}(t,u)$ and its off–diagonal matrix $\nabla\boldsymbol{\Lambda}(t,u)$.

In the case of a unique decomposition, will use the same notation for the product integrals, namely

$$\prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x,w)\,\mathrm{d}(x,w)) \quad \text{and} \quad \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x,w)\,\mathrm{d}(x,w)),$$

when $\boldsymbol{A}^0(t,u)$ and $\boldsymbol{A}^1(t,u)$ arise from a unique decomposition.

Lemma 5.19. *For a given differentiable function g,*

$$\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \int_0^{u+x-s} g(s,x,u,w)\,\mathrm{d}w = \int_0^{u+x-s} \left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) g(s,x,u,w)\,\mathrm{d}w.$$

Proof. Leiniz' rule gives that

$$\frac{\partial}{\partial s} \int_0^{u+x-s} g(s,x,u,w)\,\mathrm{d}w = \int_0^{u+x-s} \frac{\partial}{\partial s} g(s,x,u,w)\,\mathrm{d}w - g(s,x,u,u+x-s)$$
$$\frac{\partial}{\partial u} \int_0^{u+x-s} g(s,x,u,w)\,\mathrm{d}w = \int_0^{u+x-s} \frac{\partial}{\partial u} g(s,x,u,w)\,\mathrm{d}w + g(s,x,u,u+x-s)$$

from which the result follows. □

Theorem 5.20. *For $s \le x \le t$ we have that*

$$\prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)) = \int_0^{x+u-s} \prod_{s,u}^{x,w} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)) \prod_{x,w}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b))\,\mathrm{d}w \quad (71)$$

Proof. By definition the product integral solves

$$\begin{aligned} &-\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \prod_{s,u}^{x,w} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)) \\ &\quad = \boldsymbol{A}^0(s,u) \prod_{s,u}^{x,w} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)) + \boldsymbol{A}^1(s,u) \prod_{s,0}^{x,w} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)). \end{aligned}$$

Denote by $\boldsymbol{Z}(s,x,t,u,v)$ the right hand side of (71). From Lemma 5.19, we get

$$\begin{aligned} &-\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \boldsymbol{Z}(s,x,t,u,v) \\ &\quad = \int_0^{x+u-s} \left[\boldsymbol{A}^0(s,u) \prod_{s,u}^{x,w} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)) + \boldsymbol{A}^1(s,u) \prod_{s,0}^{x,w} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)) \right] \\ &\qquad \times \prod_{x,w}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b))\,\mathrm{d}w \\ &\quad = \boldsymbol{A}^0(s,u)\boldsymbol{Z}(s,x,t,u,v) + \boldsymbol{A}^1(s,u)\boldsymbol{Z}(s,x,t,0,v). \end{aligned}$$

Furthermore, initial conditions match since

$$\boldsymbol{Z}(s,s,s,u,v) = \int_0^u \boldsymbol{I}\delta_0(u-w)\boldsymbol{I}\delta_0(w-v)\,\mathrm{d}w = \boldsymbol{I}\delta_0(u-v).$$

Hence, for any $x \in [s,t]$

$$\boldsymbol{Z}(s,x,t,u,v) = \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)).$$

□

Corollary 5.21. *For $s \le x \le t$ we have that*

$$\prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)) = \int_0^{x+u-s} \prod_{s,u}^{x,w} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b)) \prod_{x,w}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b))\,\mathrm{d}w \quad (72)$$

Proof. Follows by integration of (71). □

We obtain the (well known) Chapman–Kolmogorov equations as follows.

Corollary 5.22 (Chapman–Kolmogorov)**.**

$$\boldsymbol{p}(s,t,u,v) = \int_0^{x+u-s} \boldsymbol{p}(s,x,u,w)\boldsymbol{p}(x,t,w,v)\,\mathrm{d}w$$

$$\boldsymbol{P}(s,t,u,v) = \int_0^{x+u-s} \boldsymbol{p}(s,x,u,w)\boldsymbol{P}(x,t,w,v)\,\mathrm{d}w = \int_0^{x+u-s} \boldsymbol{P}(s,x,u,\mathrm{d}w)\boldsymbol{P}(x,t,w,v)$$

6. The Van Loan formulae

Theorem 5.23 (Van Loan)**.** *Let* $\boldsymbol{A}(x,w) = (\boldsymbol{A}^0(x,w), \boldsymbol{A}^1(x,w))$, $\boldsymbol{B}(x,w) = (\boldsymbol{B}^0(x,w), \boldsymbol{B}^1(x,w))$ *and* $\boldsymbol{C}(x,w) = (\boldsymbol{C}^0(x,w), \boldsymbol{C}^1(x,w))$, *which all satisfy that they are the null matrix for* $w > x$. *Define*

$$\boldsymbol{Y}^i(s,u) = \begin{pmatrix} \boldsymbol{A}^i(s,u) & \boldsymbol{B}^i(s,u) \\ \boldsymbol{0} & \boldsymbol{C}^i(s,u) \end{pmatrix}, \quad i = 0,1$$

and $\boldsymbol{Y}(s,u) = (\boldsymbol{Y}^0(s,u), \boldsymbol{Y}^1(s,u))$. *Then*

$$\prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{Y}(x,w)\,\mathrm{d}(x,w)) = \boldsymbol{x}(s,t,u,v) = \begin{pmatrix} \boldsymbol{x}_{11}(s,t,u,v) & \boldsymbol{x}_{12}(s,t,u,v) \\ \boldsymbol{0} & \boldsymbol{x}_{22}(s,t,u,v) \end{pmatrix}$$

where

$$\boldsymbol{x}_{11}(s,t,u,v) = \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x,w)\,\mathrm{d}(x,w))$$

$$\boldsymbol{x}_{22}(s,t,u,v) = \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{C}(x,w)\,\mathrm{d}(x,w))$$

$$\begin{aligned} \boldsymbol{x}_{12}(s,t,u,v) = &\int_s^t \int_0^{u+s-x} \boldsymbol{x}_{11}(s,x,u,w)\boldsymbol{B}^0(x,w)\boldsymbol{x}_{22}(x,t,w,v)\,\mathrm{d}w\ \mathrm{d}x \\ &+ \int_s^t \int_0^{u+s-x} \boldsymbol{x}_{11}(s,x,u,w)\boldsymbol{B}^1(x,w)\boldsymbol{x}_{22}(x,t,0,v)\,\mathrm{d}w\ \mathrm{d}x. \end{aligned}$$

Proof. Let $\boldsymbol{x}(s,t,u,v) = \prod_{s,u}^{t,v}(\boldsymbol{I} + \boldsymbol{Y}(x,w)\,\mathrm{d}(x,w))$. Then $\boldsymbol{x}(s,t,u,v)$ satisfies

$$\begin{aligned} &-\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right)\boldsymbol{x}(s,t,u,v) \\ &\quad = \begin{pmatrix} \boldsymbol{A}^0(s,u) & \boldsymbol{B}^0(s,u) \\ \boldsymbol{0} & \boldsymbol{C}^0(s,u) \end{pmatrix} \boldsymbol{x}(s,t,u,v) + \begin{pmatrix} \boldsymbol{A}^1(s,u) & \boldsymbol{B}^1(s,u) \\ \boldsymbol{0} & \boldsymbol{C}^1(s,u) \end{pmatrix} \boldsymbol{x}(s,t,0,v), \end{aligned} \tag{73}$$

with boundary condition $\boldsymbol{x}(s,s,u,v) = \delta_0(u-v)\boldsymbol{I}$. Let

$$\boldsymbol{x}(s,t,u,v) = \begin{pmatrix} \boldsymbol{x}_{11}(s,t,u,v) & \boldsymbol{x}_{12}(s,t,u,v) \\ \boldsymbol{x}_{21}(s,t,u,v) & \boldsymbol{x}_{22}(s,t,u,v) \end{pmatrix}.$$

First we show that $\boldsymbol{x}_{21}(s,t,u,v)=\mathbf{0}$. From (73),

$$\begin{aligned}-\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)&\boldsymbol{x}_{21}(s,t,u,v)\\&=\boldsymbol{C}^0(s,u)\boldsymbol{x}_{21}(s,t,u,v)+\boldsymbol{C}^1(s,u)\boldsymbol{x}_{21}(s,t,0,v)\end{aligned}$$

with boundary condition $\boldsymbol{x}_{21}(s,t,u,v)=\mathbf{0}$ for $s=t$. The zero–matrix is indeed a solution, so by uniqueness we conclude that it is the only solution.

Then with $\boldsymbol{x}_{21}(s,t,u,v)=\mathbf{0}$, the rest of the equations satisfy

$$-\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{x}_{11}(s,t,u,v)=\boldsymbol{A}^0(s,u)\boldsymbol{x}_{11}(s,t,u,v)+\boldsymbol{A}^1(s,u)\boldsymbol{x}_{11}(s,t,0,v), \tag{74}$$

$$-\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{x}_{22}(s,t,u,v)=\boldsymbol{C}^0(s,u)\boldsymbol{x}_{22}(s,t,u,v)+\boldsymbol{C}^1(s,u)\boldsymbol{x}_{22}(s,t,0,v), \tag{75}$$

$$\begin{aligned}-\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{x}_{12}(s,t,u,v)&=\boldsymbol{A}^0(s,u)\boldsymbol{x}_{12}(s,t,u,v)+\boldsymbol{B}^0(s,u)\boldsymbol{x}_{22}(s,t,u,v)\\&\quad+\boldsymbol{A}^1(s,u)\boldsymbol{x}_{12}(s,t,0,v)+\boldsymbol{B}^1(s,u)\boldsymbol{x}_{22}(s,t,0,v).\end{aligned} \tag{76}$$

From (74) and (75) it follows immediately that

$$\begin{aligned}\boldsymbol{x}_{11}(s,t,u,v)&=\prod_{s,u}^{t,v}(\boldsymbol{I}+\boldsymbol{A}(x,w)\,\mathrm{d}(x,w)),\\ \boldsymbol{x}_{22}(s,t,u,v)&=\prod_{s,u}^{t,v}(\boldsymbol{I}+\boldsymbol{C}(x,w)\,\mathrm{d}(x,w)).\end{aligned}$$

We then propose that

$$\begin{aligned}\boldsymbol{x}_{12}(s,t,u,v)=&\int_s^t\int_0^{u+x-s}\boldsymbol{x}_{11}(s,x,u,w)\boldsymbol{B}^0(x,w)\boldsymbol{x}_{22}(x,t,w,v)\,\mathrm{d}w\ \mathrm{d}x\\&+\int_s^t\int_0^{u+x-s}\boldsymbol{x}_{11}(s,x,u,w)\boldsymbol{B}^1(x,w)\boldsymbol{x}_{22}(x,t,0,v)\,\mathrm{d}w\ \mathrm{d}x\end{aligned} \tag{77}$$

and show that it satisfies (76).

Using Lemma 5.10 with $\boldsymbol{g}(s,x,u,w)=\boldsymbol{x}_{11}(s,x,u,w)\boldsymbol{B}^0(x,w)\boldsymbol{x}_{22}(x,t,w,v)$ and $\boldsymbol{g}(s,x,u,w)=\boldsymbol{x}_{11}(s,x,u,w)\boldsymbol{B}^1(x,w)\boldsymbol{x}_{22}(x,t,w,v)$, respectively, we get that

$$\begin{aligned}&\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{x}_{12}(s,t,u,v)\\&\quad=\int_s^t\int_0^{u+s-x}\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{x}_{11}(s,x,u,w)\,\boldsymbol{B}^0(x,w)\,\boldsymbol{x}_{22}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x\\&\qquad+\int_s^t\int_0^{u+s-x}\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{x}_{11}(s,x,u,w)\,\boldsymbol{B}^1(x,w)\,\boldsymbol{x}_{22}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x\\&\qquad-\boldsymbol{B}^0(s,u)\boldsymbol{x}_{22}(s,t,u,v)-\boldsymbol{B}^1(s,u)\boldsymbol{x}_{22}(s,t,0,v),\end{aligned}$$

where the last two terms are obtained using the boundary condition $\boldsymbol{x}_{11}(s,s,u,v) = \boldsymbol{I}\delta_0(u-v)$ so that

$$\int_0^u \boldsymbol{x}_{11}(s,s,u,w)\boldsymbol{B}^0(s,w)\boldsymbol{x}_{22}(s,t,w,v)\,\mathrm{d}w = \boldsymbol{B}^0(s,u)\boldsymbol{x}_{22}(s,t,u,v),$$
$$\int_0^u \boldsymbol{x}_{11}(s,s,u,w)\boldsymbol{B}^1(s,w)\boldsymbol{x}_{22}(s,t,0,v)\,\mathrm{d}w = \boldsymbol{B}^1(s,u)\boldsymbol{x}_{22}(s,t,0,v).$$

Then using (74) and (75) we get

$$\begin{aligned}
&\left(\frac{\partial}{\partial s}+\frac{\partial}{\partial u}\right)\boldsymbol{x}_{12}(s,t,u,v)\\
&\quad= -\int_s^t\int_0^{u+x-s}\Big(\boldsymbol{A}^0(s,u)\boldsymbol{x}_{11}(s,x,u,w)+\boldsymbol{A}^1(s,u)\boldsymbol{x}_{11}(s,x,0,w)\Big)\boldsymbol{B}^0(x,w)\,\boldsymbol{x}_{22}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x\\
&\qquad -\int_s^t\int_0^{u+x-s}\Big(\boldsymbol{A}^0(s,u)\boldsymbol{x}_{11}(s,x,u,w)+\boldsymbol{A}^1(s,u)\boldsymbol{x}_{11}(s,x,0,w)\Big)\boldsymbol{B}^1(x,w)\,\boldsymbol{x}_{22}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x\\
&\qquad -\boldsymbol{B}^0(s,u)\boldsymbol{x}_{22}(s,t,u,v)-\boldsymbol{B}^1(s,u)\boldsymbol{x}_{22}(s,t,0,v)\\
&\quad= -\boldsymbol{A}^0(s,u)\boldsymbol{x}_{12}(s,t,u,v)-\boldsymbol{A}^1(s,u)\int_s^t\int_0^{u+x-s}\boldsymbol{x}_{11}(s,x,0,w)\boldsymbol{B}^0(x,w)\,\boldsymbol{x}_{22}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x\\
&\qquad -\boldsymbol{A}^1(s,u)\int_s^t\int_0^{u+x-s}\boldsymbol{x}_{11}(s,x,0,w)\boldsymbol{B}^1(x,w)\,\boldsymbol{x}_{22}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x\\
&\qquad -\boldsymbol{B}^0(s,u)\boldsymbol{x}_{22}(s,t,u,v)-\boldsymbol{B}^1(s,u)\boldsymbol{x}_{22}(s,t,0,v)\\
&\quad\overset{(*)}{=} -\boldsymbol{A}^0(s,u)\boldsymbol{x}_{12}(s,t,u,v)\\
&\qquad -\boldsymbol{A}^1(s,u)\underbrace{\int_s^t\int_0^{x-s}\boldsymbol{x}_{11}(s,x,0,w)\Big(\boldsymbol{B}^0(x,w)\,\boldsymbol{x}_{22}(x,t,w,v)+\boldsymbol{B}^1(x,w)\,\boldsymbol{x}_{22}(x,t,0,v)\Big)\,\mathrm{d}w\,\mathrm{d}x}_{=\boldsymbol{x}_{12}(s,t,0,v)\text{ from (77)}}\\
&\qquad -\boldsymbol{B}^0(s,u)\boldsymbol{x}_{22}(s,t,u,v)-\boldsymbol{B}^1(s,u)\boldsymbol{x}_{22}(s,t,0,v)\\
&\quad= -\boldsymbol{A}^0(s,u)\boldsymbol{x}_{12}(s,t,u,v)-\boldsymbol{A}^1(s,u)\boldsymbol{x}_{12}(s,t,0,v)\\
&\qquad -\boldsymbol{B}^0(s,u)\boldsymbol{x}_{22}(s,t,u,v)-\boldsymbol{B}^1(s,u)\boldsymbol{x}_{22}(s,t,0,v),
\end{aligned}$$

which is exactly the right hand side of (76). The equality $(*)$, where $u+x-s$ is reduced to $x-s$, follows from the fact that $\boldsymbol{x}_{11}(s,x,0,w)$ will be zero for values of w larger than $x-s$, since a reset has taken place at time u. □

Lemma 5.24. *Let $\boldsymbol{A}(x) = (\boldsymbol{A}^0(x), \boldsymbol{A}^1(x))$ and $t \to r(t)$ an integrable function. Then*

$$\mathrm{e}^{-\int_s^t r(x)\,\mathrm{d}x}\prod_{s,u}^{t,v}(\boldsymbol{I}+\boldsymbol{A}(x,w)\,\mathrm{d}(x,w)) = \prod_{s,u}^{t,v}(\boldsymbol{I}+\tilde{\boldsymbol{A}}(x,w)\,\mathrm{d}(x,w)),$$

where $\tilde{\boldsymbol{A}}(x,w) = (\tilde{\boldsymbol{A}}^0(x,w), \tilde{\boldsymbol{A}}^1(x,w))$ with

$$\tilde{\boldsymbol{A}}^0(x,w) = \boldsymbol{A}^0(x,w) - r(x)\boldsymbol{I} \quad \textit{and} \quad \tilde{\boldsymbol{A}}^1(x,w) = \boldsymbol{A}^1(x,w).$$

Proof. Let

$$\boldsymbol{Z}(s,u,t,v) = \mathrm{e}^{-\int_s^t r(x)\,\mathrm{d}x} \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x).$$

Then

$$\begin{aligned}
&-\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \boldsymbol{Z}(s,u,t,v) \\
&\quad = -r(s)\mathrm{e}^{-\int_s^t r(x)\,\mathrm{d}x} \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) - \mathrm{e}^{-\int_s^t r(x)\,\mathrm{d}x} \left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right) \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) \\
&\quad = -r(s)\mathrm{e}^{-\int_s^t r(x)\,\mathrm{d}x} \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) \\
&\qquad + \mathrm{e}^{-\int_s^t r(x)\,\mathrm{d}x} \left(\boldsymbol{A}^0(s,u) \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) + \boldsymbol{A}^1(s,u) \prod_{s,0}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) \right) \\
&\quad = \mathrm{e}^{-\int_s^t r(x)\,\mathrm{d}x} \left(\left[\boldsymbol{A}^0(s,u) - r(s)\boldsymbol{I}\right] \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) + \boldsymbol{A}^1(s,u) \prod_{s,0}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(x)\,\mathrm{d}x) \right) \\
&\quad = \left[\boldsymbol{A}^0(s,u) - r(s)\boldsymbol{I}\right] \boldsymbol{Z}(s,t,u,v) + \boldsymbol{A}^1(s,u)\boldsymbol{Z}(s,t,0,v)
\end{aligned}$$

□

To compute the partial reserves $\boldsymbol{v}(s,t,u,v)$, (66), parametrise as follows.

Corollary 5.25. *Let*

$$\boldsymbol{Y}^0(s,u) = \begin{pmatrix} \Delta\boldsymbol{M}(s,u) - r(s)\boldsymbol{I} & \boldsymbol{R}^0(s,u) \\ \boldsymbol{0} & \Delta\boldsymbol{M}(s,u) \end{pmatrix}, \boldsymbol{Y}^1(s,u) = \begin{pmatrix} \nabla\boldsymbol{M}(s,u) & \boldsymbol{R}^1(s,u) \\ \boldsymbol{0} & \nabla\boldsymbol{M}(s,u) \end{pmatrix},$$

and $\boldsymbol{Y}(s,u) = (\boldsymbol{Y}^0(s,u), \boldsymbol{Y}^1(s,u))$. *Then the discounted reserves* $\boldsymbol{v}(s,t,u,v)$ *with deterministic interest rates* $r(t)$ *can be calculated as follows:*

$$\prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{Y}(x,w)\,\mathrm{d}(x,w)) = \begin{pmatrix} \mathrm{e}^{-\int_s^t r(x)\,\mathrm{d}x}\boldsymbol{p}(s,t,u,v) & \boldsymbol{v}(s,t,u,v) \\ \boldsymbol{0} & \boldsymbol{p}(s,t,u,v) \end{pmatrix}.$$

Proof. Follows directly from Van Loan, the definition (66) and Lemma 5.12 for the upper left block matrix. □

The Van Loan formula does not hold for $\prod_{s,u}^{t,v}$ directly, but only in combination with $\prod_{s,u}^{t,v}$, which we will refer to as a "hybrid" version of Van Loan's formula. This is entirely similar to (72), where the Chapman–Kolmogorov equation for the c.d.f. version of the transition probabilities also needs the presence of the density version.

Theorem 5.26 (Hybrid Van Loan formula)**.** *Let*

$$\boldsymbol{z}_{11}(s,u,t,v) = \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{A}(a,b)\,\mathrm{d}(a,b))$$

$$\boldsymbol{Z}_{22}(s,u,t,v) = \int_0^v \prod_{s,u}^{t,v} (\boldsymbol{I} + \boldsymbol{C}(a,b)\,\mathrm{d}(a,b))\,\mathrm{d}v$$

and

$$\begin{aligned}\boldsymbol{Z}_{12}(s,t,u,v) = &\int_s^t \int_0^{u+x-s} \boldsymbol{z}_{11}(s,u,x,w)\boldsymbol{B}^0(x,w)\boldsymbol{Z}_{22}(x,w,t,v)\,\mathrm{d}w\,\mathrm{d}x \\ &+ \int_s^t \int_0^{u+x-s} \boldsymbol{z}_{11}(s,u,x,w)\boldsymbol{B}^1(x,w)\boldsymbol{Z}_{22}(x,0,t,v)\,\mathrm{d}w\,\mathrm{d}x.\end{aligned}$$

Then the latter can be computed as the solution to the PDE

$$-\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right)\boldsymbol{Z}(s,t,u,v) = \boldsymbol{D}^0(s,u)\boldsymbol{Z}(s,t,u,v) + \boldsymbol{D}^1(s,u)\boldsymbol{Z}(s,t,0,v)$$

with initial condition

$$\boldsymbol{Z}(s,s,u,v) = \begin{pmatrix} \boldsymbol{I}\delta_0(v-u) & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I}1\{v \geq u\} \end{pmatrix},$$

where

$$\boldsymbol{D}^i(s,u) = \begin{pmatrix} \boldsymbol{A}^i(s,u) & \boldsymbol{B}^i(s,u) \\ \boldsymbol{0} & \boldsymbol{C}^i(s,u) \end{pmatrix}, \quad \boldsymbol{Z}(s,t,u,v) = \begin{pmatrix} \boldsymbol{z}_{11}(s,t,u,v) & \boldsymbol{Z}_{12}(s,t,u,v) \\ \boldsymbol{0} & \boldsymbol{Z}_{22}(s,t,u,v) \end{pmatrix}$$

for $i = 0,1$.

Remark 5.27: The integral equation form for the hybrid Van Loan is as follows.

$$\begin{aligned}\boldsymbol{Z}(s,t,u,v) = &\int_s^t \prod_s^w (\boldsymbol{I} + \boldsymbol{D}^0(y,u+y-s)\,\mathrm{d}y)\boldsymbol{D}^1(w,u+w-s)\boldsymbol{Z}(w,t,0,v)\,\mathrm{d}w \\ &+ \prod_s^t (\boldsymbol{I} + \boldsymbol{D}^0(y,u+y-s)\,\mathrm{d}y)\begin{pmatrix} \boldsymbol{I}\delta_0(v-u-(t-s)) & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I}1\{u+t-s \leq v\} \end{pmatrix}\end{aligned} \tag{78}$$

If we define the correponding integrated partial reserves,

$$V_{ij}(s,t,u,v) = \mathbb{E}\left(1\{Z(t) = j, U(t) \leq v\}\int_s^t \mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\,\mathrm{d}B(x)\middle| Z(s) = i, U(s) = u\right),$$

then by integrating (66),

$$\begin{aligned}\boldsymbol{V}(s,t,u,v) = &\int_s^t \mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u} \int_0^{u+x-s} \boldsymbol{p}(s,x,u,w)\boldsymbol{R}^0(x,w)\boldsymbol{P}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x + \\ &+ \int_s^t \mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u} \int_0^{u+x-s} \boldsymbol{p}(s,x,u,w)\boldsymbol{R}^1(x,w)\boldsymbol{P}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x.\end{aligned} \tag{79}$$

The we obtain from the hybrid Van Loan the following:

Corollary 5.28. *Let*

$$\boldsymbol{Y}^0(s,u) = \begin{pmatrix} \Delta \boldsymbol{M}(s,u) - r(s)\boldsymbol{I} & \boldsymbol{R}^0(s,u) \\ \boldsymbol{0} & \Delta \boldsymbol{M}(s,u) \end{pmatrix}, \boldsymbol{Y}^1(s,u) = \begin{pmatrix} \nabla \boldsymbol{M}(s,u) & \boldsymbol{R}^1(s,u) \\ \boldsymbol{0} & \nabla \boldsymbol{M}(s,u) \end{pmatrix},$$

and $\boldsymbol{Y}(s,u) = (\boldsymbol{Y}^0(s,u), \boldsymbol{Y}^1(s,u))$. *Then the solution to*

$$-\left(\frac{\partial}{\partial s} + \frac{\partial}{\partial u}\right)\boldsymbol{Z}(s,t,u,v) = \boldsymbol{Y}^0(s,u)\boldsymbol{Z}(s,t,u,v) + \boldsymbol{Y}^1(s,u)\boldsymbol{Z}(s,t,0,v)$$

with initial condition

$$\boldsymbol{Z}(s,s,u,v) = \begin{pmatrix} \boldsymbol{I}\delta_0(v-u) & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I}1\{v \geq u\} \end{pmatrix},$$

is given by

$$\boldsymbol{Z}(s,t,u,v) = \begin{pmatrix} \mathrm{e}^{-\int_s^t r(x)\,\mathrm{d}x}\boldsymbol{p}(s,t,u,v) & \boldsymbol{V}(s,t,u,v) \\ \boldsymbol{0} & \boldsymbol{P}(s,t,u,v) \end{pmatrix}.$$

Proof. Rewrite

$$\begin{aligned} \boldsymbol{V}(s,t,u,v) = &\int_s^t \int_0^{u+x-s} \left(\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w)\right)\boldsymbol{R}^0(x,w)\boldsymbol{P}(x,t,w,v)\,\mathrm{d}w\,\mathrm{d}x \\ &+ \int_s^t \int_0^{u+x-s} \left(\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w)\right)\boldsymbol{R}^1(x,w)\boldsymbol{P}(x,t,0,v)\,\mathrm{d}w\,\mathrm{d}x. \end{aligned}$$

By Lemma 5.24,

$$\mathrm{e}^{-\int_s^x r(u)\,\mathrm{d}u}\boldsymbol{p}(s,x,u,w) = \prod_{s,u}^{x,w}(\boldsymbol{I} + \tilde{\boldsymbol{M}}(s,y)\,\mathrm{d}y),$$

where $\tilde{\boldsymbol{M}}(s,y) = (\tilde{\boldsymbol{M}}^0(s,y), \tilde{\boldsymbol{M}}^1(s,y))$ with

$$\tilde{\boldsymbol{M}}^0(s,y) = \Delta \boldsymbol{M}(s,y) - r(s)\boldsymbol{I}, \quad \tilde{\boldsymbol{M}}^1(s,y) = \nabla \boldsymbol{M}(s,y),$$

from which the result follows from Theorem 5.26. □

CHAPTER 6

Random benefits and higher order moments

1. Random setup

The setup is identitcal to Section 1, except for the lump sum payments, $b_{ij}(t)$, which are now, in general, random. Therefore they will be denoted with capital letters $B_{ij}(t)$, which is both customary in probability theory and to destinguish random variables from deterministic functions. The rest of the notation follows that of Section 1.

Define the total accumulated and undiscounted benefits,

$$R_0(s,t) = \int_s^t b_{X(u)}(u)\,\mathrm{d}u + \sum_{i,j} \int_s^t B_{ij}(u)\,\mathrm{d}N_{ij}(u), \tag{80}$$

where $b_i(u)$ are the expected rates (therefore, deterministic) earned in state i, and $N_{ij}(u)$ denotes the counting process which increases one if a transition from i to j during time $[u, u+\mathrm{d}u)$ triggers a payment, which happens with probability $d_{ij}(u)\,\mathrm{d}u$. As in Section 1, the probability that a transition does not trigger a payment is $c_{ij}(u)\,\mathrm{d}u$, and the probability of a transition (of any kind) during $[u, u+\mathrm{d}u)$ is $\lambda_{ij}(u)\,\mathrm{d}u = c_{ij}(u)\,\mathrm{d}u + d_{ij}(u)\,\mathrm{d}u$.

The total discounted benefits, based on a deterministic interest rate $r(t)$, is then given by

$$\begin{aligned} R(s,t) &= \int_s^t \mathrm{e}^{-\int_s^u r(z)\,\mathrm{d}z} R_0(s,\mathrm{d}u) \\ &= \int_s^t \mathrm{e}^{-\int_s^u r(z)\,\mathrm{d}z} b_{X(u)}(u)\,\mathrm{d}u + \sum_{i,j} \int_s^t \mathrm{e}^{-\int_s^u r(z)\,\mathrm{d}z} B_{ij}(u)\,\mathrm{d}N_{ij}(u). \end{aligned} \tag{81}$$

We assume that the random benefits, $B_{ij}(u)$, are independent of $\{N_{ij}(x)\}_{x\geq 0}$.

2. The distribution of undiscounted benefits

Now define the benefit size distribution functions and moment generating distributions as

$$\begin{aligned} &F_{ij}^B(x;t) = \mathbb{P}(B_{ij}(t) \leq x) \\ &\hat{F}_{ij}^B(\theta;t) = \mathbb{E}\left[\mathrm{e}^{\theta B_{ij}(t)}\right], \quad \hat{\boldsymbol{F}}^B(\theta;t) = \{\hat{F}_{ij}^B(\theta;t)\}_{i,j\in E}, \end{aligned}$$

and let

$$\hat{G}_{ij}(\theta;s,t) = \mathbb{E}\left[1_{\{X(t)=j\}}\mathrm{e}^{\theta R_0(s,t)}\middle| X(s)=i\right], \quad \hat{\boldsymbol{G}}(\theta;s,t) = \{\hat{G}_{ij}(\theta;s,t)\}_{i,j=1,\ldots,p}.$$

Then the distribution of $R_0(s,t)$ is fully characterised by following result.

Theorem 6.1. *Let*

$$\boldsymbol{A}(\theta;u) = \boldsymbol{\Lambda}^1(u) \bullet \hat{\boldsymbol{F}}^B(\theta;u) + \boldsymbol{\Lambda}^{\boldsymbol{0}}(u) + \theta \boldsymbol{\Delta}(\boldsymbol{b}(t)). \tag{82}$$

where $\{\alpha_{ij}\} \bullet \{\beta_{ij}\} = \{\alpha_{ij}\beta_{ij}\}$ *denotes the Schur (entry–wise or Hadamard) matrix product. Then*

$$\hat{\boldsymbol{G}}(\theta;s,t) = \prod_s^t (\boldsymbol{I} + \boldsymbol{A}(\theta;u)\,\mathrm{d}u) \tag{83}$$

Proof. From the Markov property and conditional independence, for $s < u < t$ we have that

$$\begin{aligned}
&\mathbb{E}\left[1\{X(t)=j\}\mathrm{e}^{\theta R_0(s,t)}\Big| X(s)=i\right] \\
&= \sum_k \mathbb{E}\left[\mathrm{e}^{\theta R_0(s,u)}\mathrm{e}^{\theta R_0(u,t)}1\{X(u)=k\}1\{X(t)=j\}\Big| X(s)=i\right] \\
&= \sum_k \mathbb{E}\left[\mathrm{e}^{\theta R_0(s,u)}1\{X(u)=k\}\mathbb{E}\left(\mathrm{e}^{\theta R_0(u,t)}1\{X(t)=j\}\Big|\mathcal{F}_u\right)\Big| X(s)=i\right] \\
&= \sum_k \mathbb{E}\left[\mathrm{e}^{\theta R_0(s,u)}1\{X(u)=k\}\mathbb{E}\left(\mathrm{e}^{\theta R_0(u,t)}1\{X(t)=j\}\Big| X(u)=k\right)\Big| X(s)=i\right] \\
&= \sum_k \mathbb{E}\left(\mathrm{e}^{\theta R_0(s,u)}1\{X(u)=k\}\Big| X(s)=i\right)\mathbb{E}\left(\mathrm{e}^{\theta R_0(u,t)}1\{X(t)=j\}\Big| X(u)=k\right),
\end{aligned}$$

i.e.

$$\hat{\boldsymbol{G}}(\theta;s,t) = \hat{\boldsymbol{G}}(\theta;s,u)\hat{\boldsymbol{G}}(\theta;u,t).$$

Then write

$$\hat{\boldsymbol{G}}(\theta;s,t+\mathrm{d}t) - \hat{\boldsymbol{G}}(\theta;s,t) = \hat{\boldsymbol{G}}(\theta;s,t)\left(\hat{\boldsymbol{G}}(\theta;t,t+\mathrm{d}t) - \boldsymbol{I}\right) \tag{84}$$

and consider $\hat{\boldsymbol{G}}(\theta;t,t+\mathrm{d}t)$. By conditioning on whether there is an arrival or not during $[t,t+\mathrm{d}t)$, the expectation is then given by

$$\begin{aligned}
\hat{G}_{ij}(\theta;t,t+\mathrm{d}t) &= \left(\delta_{ij} + \lambda^0_{ij}(t)\,\mathrm{d}t\right)\mathrm{e}^{\theta b_i(t)\,\mathrm{d}t} + \lambda^1_{ij}(t)\,\mathrm{d}t\,\mathbb{E}\left(\mathrm{e}^{\theta b_i(t)\,\mathrm{d}t + \theta B_{ij}(t)}\right) \\
&= \left(\delta_{ij} + \lambda^0_{ij}(t)\,\mathrm{d}t\right)(1+\theta b_i(t)\,\mathrm{d}t) + \lambda^1_{ij}(t)\,\mathrm{d}t\hat{F}^B_{ij}(\theta;t)\,(1+\theta b_i(t)\,\mathrm{d}t) \\
&= \delta_{ij} + \delta_{ij}\theta b_i(t)\,\mathrm{d}t + \lambda^0_{ij}(t)\,\mathrm{d}t + \lambda^1_{ij}(t)\,\mathrm{d}t\,\hat{F}^B_{ij}(\theta;t),
\end{aligned}$$

which on matrix form is

$$\hat{\boldsymbol{G}}(\theta;t,t+\mathrm{d}t) = \boldsymbol{I} + \boldsymbol{\Lambda}^0(t)\,\mathrm{d}t + \boldsymbol{\Lambda}^1(t)\,\mathrm{d}t \bullet \hat{\boldsymbol{F}}(\theta;t) + \theta\boldsymbol{\Delta}(\boldsymbol{b}(t))\,\mathrm{d}t = \boldsymbol{I} + \boldsymbol{A}(\theta;t)\,\mathrm{d}t.$$

Then from (84),

$$\hat{\boldsymbol{G}}(\theta;s,t+\mathrm{d}t) - \hat{\boldsymbol{G}}(\theta;s,t) = \hat{\boldsymbol{G}}(\theta;s,t)\boldsymbol{A}(\theta;t)\,\mathrm{d}t,$$

proving the result. □

This is a strong result that suggests that we may simply replace all random variables by their corresponding moment generating functions for the sake of computing the moments generating function of $R_0(s,t)$.

3. Undiscounted moments

For the sake of computing moments we then proceed as follows. We now define the partial moments for $R_0(s,t)$ as

$$m_{ij}^{(k)}(s,t) = \mathbb{E}\left[1\{X(t)=j\}R_0(s,t)^k \middle| X(s)=i\right], \quad \boldsymbol{M}^{(k)}(s,t) = \{m_{ij}^{(k)}(s,t)\}_{i,j\in E},$$

and moments for the random rewards, $B_{ij}(t)$, as

$$\mu_{ij}^{(k)}(t) = \mathbb{E}\left(B_{ij}(t)^k\right), \quad \boldsymbol{\mu}^{(k)}(t) = \{\mu_{ij}^{(k)}(t)\}_{i,j\in E}.$$

The moments $\mu_{ij}^{(k)}(t)$ may not exist if the claim size distributions are heavy–tailed. In the following we *assume* the existence of moments up to the orders treated, and derive an expression for how to compute them.

The moments $\boldsymbol{\mu}^{(k)}(t)$ are then obtained by evaluating the k'th order derivative of the $\hat{\boldsymbol{F}}^B(\theta;t)$ at $\theta=0$.

For $k=0$, we are simply evaluating the expressions at $\theta=0$, giving

$$\begin{aligned}\boldsymbol{M}^{(0)}(s,t) &= \boldsymbol{P}(s,t)\\ \boldsymbol{C}^{(0)}(t) &\stackrel{\text{def}}{=} \boldsymbol{A}(\theta;u)\mid_{\theta=0} = \boldsymbol{\Lambda}(t).\end{aligned}$$

For $k\geq 1$, we get

$$\begin{aligned}\boldsymbol{C}^{(1)}(t) &\stackrel{\text{def}}{=} \frac{\partial}{\partial\theta}\boldsymbol{A}(\theta;u)\bigg|_{\theta=0} = \boldsymbol{\Lambda}^1(t)\bullet\frac{\partial}{\partial\theta}\hat{\boldsymbol{F}}^B(\theta;u)\bigg|_{\theta=0} + \boldsymbol{\Delta}(\boldsymbol{b}(u))\\ &= \boldsymbol{\Lambda}^1(t)\bullet\boldsymbol{\mu}^{(1)}(u) + \boldsymbol{\Delta}(\boldsymbol{b}(u))\\ \boldsymbol{C}^{(2)}(t) &\stackrel{\text{def}}{=} \frac{\partial^2}{\partial\theta^2}\boldsymbol{A}(\theta;u)\bigg|_{\theta=0} = \boldsymbol{\Lambda}^1(t)\bullet\frac{\partial^2}{\partial\theta^2}\hat{\boldsymbol{F}}^B(\theta;u)\bigg|_{\theta=0}\\ &= \boldsymbol{\Lambda}^1(t)\bullet\boldsymbol{\mu}^{(2)}(u).\end{aligned}$$

For $k\geq 3$, we get

$$\frac{\partial^k}{\partial\theta^k}\boldsymbol{A}(\theta;u) = \boldsymbol{\Lambda}^1(u)\bullet\frac{\partial^k}{\partial\theta^k}\hat{\boldsymbol{F}}^B(\theta;u),$$

so that

$$\boldsymbol{C}^{(k)}(u) \stackrel{\text{def}}{=} \frac{\partial^k}{\partial\theta^k}\boldsymbol{A}(\theta;u)\bigg|_{\theta=0} = \boldsymbol{\Lambda}^1(u)\bullet\boldsymbol{\mu}^{(k)}(u). \tag{85}$$

From (83) we get that

$$\begin{aligned}\frac{\partial}{\partial s}\frac{\partial^k}{\partial\theta^k}\hat{\boldsymbol{G}}(\theta;s,t) &= \frac{\partial^k}{\partial\theta^k}\frac{\partial}{\partial s}\hat{\boldsymbol{G}}(\theta;s,t)\\ &= -\frac{\partial^k}{\partial\theta^k}\left(\boldsymbol{A}(\theta;s)\hat{\boldsymbol{G}}(\theta;s,t)\right)\\ &= -\sum_{i=0}^{k}\binom{k}{i}\frac{\partial^i}{\partial\theta^i}\boldsymbol{A}(\theta;s)\frac{\partial^{k-i}}{\partial\theta^{k-i}}\hat{\boldsymbol{G}}(\theta;s,t),\end{aligned}$$

Introducing reduced moments,

$$\boldsymbol{M}_r^{(k)}(s,t) = \frac{\boldsymbol{M}^{(k)}(s,t)}{k!}, \quad \boldsymbol{C}_r^{(k)}(t) = \frac{\boldsymbol{C}^{(k)}(t)}{k!},$$

and evaluating r.h.s. and l.h.s. of the above differential equation at $\theta = 0$, we then get that

$$\frac{\partial}{\partial s}\boldsymbol{M}_r^{(k)}(s,t) = -\sum_{i=0}^{k}\boldsymbol{C}_r^{(i)}(s)\boldsymbol{M}_r^{(k-i)}(s,t) \tag{86}$$

For $k = 1$ this amounts to

$$\frac{\partial}{\partial s}\boldsymbol{M}_r^{(1)}(s,t) = -\boldsymbol{\Lambda}(s)\boldsymbol{M}_r^{(1)}(s,t) - \boldsymbol{C}_r^{(1)}(s)\boldsymbol{P}(s,t) \tag{87}$$

with initial condition $\boldsymbol{M}_r^{(1)}(s,t) = \boldsymbol{0}$ for $s = t$. The solution to this equation is given by

$$\boldsymbol{M}_r^{(1)}(s,t) = \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{\Lambda}(x)\,\mathrm{d}x)\boldsymbol{C}_r^{(1)}(u)\prod_u^t(\boldsymbol{I} + \boldsymbol{\Lambda}(x)\,\mathrm{d}x)\,\mathrm{d}u.$$

This follows directly by verification by differentiating the r.h.s. using Leibniz' rule. Applying Van Loan, we have hence proven the following result.

Theorem 6.2.

$$\prod_s^t\left(\boldsymbol{I} + \begin{pmatrix}\boldsymbol{\Lambda}(u) & \boldsymbol{C}_r^{(1)}(u)\\ \boldsymbol{0} & \boldsymbol{\Lambda}(u)\end{pmatrix}\mathrm{d}u\right) = \begin{pmatrix}\boldsymbol{P}(s,t) & \boldsymbol{M}_r^{(1)}(s,t)\\ \boldsymbol{0} & \boldsymbol{P}(s,t)\end{pmatrix}.$$

In general, the first k order reduced moments can be computed as follows.

Theorem 6.3 (Moments)**.**

$$\prod_s^t\left(\boldsymbol{I} + \begin{pmatrix}\boldsymbol{\Lambda}(u) & \boldsymbol{C}_r^{(1)}(u) & \cdots & \boldsymbol{C}_r^{(k)}(u)\\ \boldsymbol{0} & \boldsymbol{\Lambda}(u) & \cdots & \boldsymbol{C}_r^{(k-1)}(u)\\ & & \ddots & \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{\Lambda}(u)\end{pmatrix}\mathrm{d}u\right) = \begin{pmatrix}\boldsymbol{P}(s,t) & \boldsymbol{M}_r^{(1)}(s,t) & \cdots & \boldsymbol{M}_r^{(k)}(s,t)\\ \boldsymbol{0} & \boldsymbol{P}(s,t) & \cdots & \boldsymbol{M}_r^{(k-1)}(s,t)\\ & & \ddots & \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{P}(s,t)\end{pmatrix}.$$

Proof. Follows by simple verification that the equation system is equivalent to (86). □

4. Discounted moments

It is now easy to obtain the corresponding moments for $R(s,t)$ in the following way. Discounting can obviously been obtained by pre–discounting all rates and lump sum payment, i.e. for a fixed s, use rates

$$\tilde{b}_i(u) = \mathrm{e}^{-\int_s^u r(x)\,\mathrm{d}x} b_i(u), \quad i = 1,...,p,$$

and lump sum payments

$$\tilde{B}_{ij}(u) = \mathrm{e}^{-\int_s^u r(x)\,\mathrm{d}x} B_{ij}(u)$$

instead. But then

$$\tilde{\mu}_{ij}^{(k)}(u) = \mathbb{E}(\tilde{B}_{ij}(u)^k) = \mathrm{e}^{-k\int_s^u r(x)\,\mathrm{d}x}\mathbb{E}(B_{ij}(u)^k), \quad k \geq 1,$$

so

$$\tilde{\boldsymbol{\mu}}^{(k)}(u) = \{\tilde{\mu}_{ij}^{(k)}(u)\}_{i,j\in E} = \mathrm{e}^{-k\int_s^u r(x)\,\mathrm{d}x}\boldsymbol{\mu}^{(k)}(u).$$

Thus the corresponding matrices $\boldsymbol{C}^{(k)}(u)$ becomes

$$\begin{aligned}\tilde{\boldsymbol{C}}^{(0)}(u) &= \boldsymbol{\Lambda}(u)\\ \tilde{\boldsymbol{C}}^{(k)}(u) &= \mathrm{e}^{-k\int_s^u r(x)\,\mathrm{d}x}\boldsymbol{C}^{(k)}(u)\end{aligned}$$

and also

$$\tilde{\boldsymbol{C}}_r^{(k)}(u) = \mathrm{e}^{-k\int_s^u r(x)\,\mathrm{d}x}\boldsymbol{C}_r^{(k)}(u).$$

In principle the discounted moment can then be computed using Theorem 6.3 with these parameters. However,

$$\begin{aligned}\boldsymbol{V}(s,t) = \boldsymbol{V}^{(1)}(s,t) &= \tilde{\boldsymbol{M}}^{(1)}(s,t)\\ &= \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{\Lambda}(x)\,\mathrm{d}x)\tilde{\boldsymbol{C}}_r^{(1)}(u)\prod_u^t(\boldsymbol{I} + \boldsymbol{\Lambda}(x)\,\mathrm{d}x)\,\mathrm{d}u\\ &= \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{\Lambda}(x)\,\mathrm{d}x)\mathrm{e}^{-\int_s^u r(x)\,\mathrm{d}x}\boldsymbol{C}_r^{(1)}(u)\prod_u^t(\boldsymbol{I} + \boldsymbol{\Lambda}(x)\,\mathrm{d}x)\,\mathrm{d}u\\ &= \int_s^t \prod_s^u (\boldsymbol{I} + \boldsymbol{\Lambda}(x) - r(x)\boldsymbol{I}\,\mathrm{d}x)\boldsymbol{C}_r^{(1)}(u)\prod_u^t(\boldsymbol{I} + \boldsymbol{\Lambda}(x)\,\mathrm{d}x)\,\mathrm{d}u.\end{aligned}$$

Thus we get from Van Loan,

Theorem 6.4.

$$\prod_s^t\left(\boldsymbol{I} + \begin{pmatrix}\boldsymbol{\Lambda}(u) - r(u)\boldsymbol{I} & \boldsymbol{C}_r^{(1)}(u)\\ \boldsymbol{0} & \boldsymbol{\Lambda}(u)\end{pmatrix}\mathrm{d}u\right) = \begin{pmatrix}\exp(-\int_s^t r(x)\,\mathrm{d}x)\boldsymbol{P}(s,t) & \boldsymbol{V}(s,t)\\ \boldsymbol{0} & \boldsymbol{P}(s,t)\end{pmatrix}.$$

In many situation, it will be easier to compute the reserves using Theorem 6.4 rather than Theorem 6.3, since the element above the diagonal becomes more involved, even for the case of constant interest rate, where it must be discounted by the exponential funcion of the interest rate.

For higher order moments we proceed as follows.

From (86) we get

$$\frac{\partial}{\partial s}\boldsymbol{V}_r^{(k)}(s,t) = -\sum_{i=0}^{k}\tilde{\boldsymbol{C}}_r^{(i)}(s)\boldsymbol{V}_r^{(k-i)}(s,t). \tag{88}$$

In particular, $\boldsymbol{V}_r^{(0)}(s,t) = \boldsymbol{M}_r^{(0)}(s,t) = \boldsymbol{P}(s,t)$ and $\tilde{\boldsymbol{C}}_r^{(0)}(s) = \boldsymbol{\Lambda}(s)$.

Rewrite as

$$\frac{\partial}{\partial s}\boldsymbol{V}_r^{(k)}(s,t) + \boldsymbol{\Lambda}(s)\boldsymbol{V}_r^{(k)}(s,t) = -\sum_{i=1}^{k}\tilde{\boldsymbol{C}}_r^{(i)}(s)\boldsymbol{V}_r^{(k-i)}(s,t).$$

Multiply both side with $\prod_t^s(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)$ to get

$$\begin{aligned}\prod_t^s(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\frac{\partial}{\partial s}\boldsymbol{V}_r^{(k)}(s,t) + \prod_t^s(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\boldsymbol{\Lambda}(s)\boldsymbol{V}_r^{(k)}(s,t)\\ = -\sum_{i=1}^{k}\prod_t^s(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\tilde{\boldsymbol{C}}_r^{(i)}(s)\boldsymbol{V}_r^{(k-i)}(s,t).\end{aligned}$$

This is the same as

$$\frac{\partial}{\partial s}\left(\prod_t^s(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\boldsymbol{V}_r^{(k)}(s,t)\right) = -\sum_{i=1}^{k}\prod_t^s(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\tilde{\boldsymbol{C}}_r^{(i)}(s)\boldsymbol{V}_r^{(k-i)}(s,t).$$

Now integrate

$$\int_s^t\frac{\partial}{\partial y}\left(\prod_t^y(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\boldsymbol{V}_r^{(k)}(y,t)\right)\mathrm{d}y = -\sum_{i=1}^{k}\int_s^t\prod_t^y(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\tilde{\boldsymbol{C}}_r^{(i)}(y)\boldsymbol{V}_r^{(k-i)}(y,t)\,\mathrm{d}y.$$

The l.h.s. equals

$$\prod_t^t(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\boldsymbol{V}_r^{(k)}(t,t) - \prod_t^s(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\boldsymbol{V}_r^{(k)}(s,t) = -\prod_t^s(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\boldsymbol{V}_r^{(k)}(s,t).$$

Hence have that

$$\prod_t^s(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\boldsymbol{V}_r^{(k)}(s,t) = \sum_{i=1}^{k}\int_s^t\prod_t^y(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\tilde{\boldsymbol{C}}_r^{(i)}(y)\boldsymbol{V}_r^{(k-i)}(y,t)\,\mathrm{d}y.$$

Multiply both sides with $\prod_s^t(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\boldsymbol{V}_r^{(k)}(s,t)$ to get

$$\boldsymbol{V}_r^{(k)}(s,t) = \sum_{i=1}^{k}\int_s^t\prod_s^y(\boldsymbol{I}+\boldsymbol{\Lambda}(x)\,\mathrm{d}x)\tilde{\boldsymbol{C}}_r^{(i)}(y)\boldsymbol{V}_r^{(k-i)}(y,t)\,\mathrm{d}y.$$

Since $\tilde{\boldsymbol{C}}^{(k)}(u) = \mathrm{e}^{-k\int_s^u r(x)\,\mathrm{d}x}\boldsymbol{C}^{(k)}(u)$, this is readily rewritten as

Theorem 6.5.

$$\boldsymbol{V}_r^{(k)}(s,t) = \sum_{i=1}^{k}\int_s^t\prod_s^y(\boldsymbol{I}+[\boldsymbol{\Lambda}(x)-ir(x)\boldsymbol{I}]\,\mathrm{d}x)\boldsymbol{C}_r^{(i)}(y)\boldsymbol{V}_r^{(k-i)}(y,t)\,\mathrm{d}y.$$

For discounted moments, a version of Theorem 6.3 and a generalisation of Theorem 6.4 is as follows.

Theorem 6.6.

$$\prod_s^t \left(\boldsymbol{I} + \begin{pmatrix} \boldsymbol{\Lambda}(u) - kr(u)\boldsymbol{I} & \boldsymbol{C}_r^{(1)}(u) & \cdots & \boldsymbol{C}_r^{(k)}(u) \\ \boldsymbol{0} & \boldsymbol{\Lambda}(u) - (k-1)r(u)\boldsymbol{I} & \cdots & \boldsymbol{C}_r^{(k-1)}(u) \\ & & \ddots & \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{\Lambda}(u) \end{pmatrix} \mathrm{d}u \right)$$

$$= \begin{pmatrix} \boldsymbol{P}(s,t) & \boldsymbol{V}_r^{(1)}(s,t) & \cdots & \boldsymbol{V}_r^{(k)}(s,t) \\ \boldsymbol{0} & \boldsymbol{P}(s,t) & \cdots & \boldsymbol{V}_r^{(k-1)}(s,t) \\ & & \ddots & \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{P}(s,t) \end{pmatrix} \begin{pmatrix} \mathrm{e}^{-k\int_s^t r(u)\,\mathrm{d}u}\boldsymbol{I} & \boldsymbol{0} & \cdots & \boldsymbol{0} \\ \boldsymbol{0} & \mathrm{e}^{-(k-1)\int_s^t r(u)\,\mathrm{d}u}\boldsymbol{I} & \cdots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \ddots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{I} \end{pmatrix}.$$

Proof. Consider the derivative of a generic term on the r.h.s. (entrance of ith block row, $i = 0, ..., k$, and column row $i + j$ with $j = 0, ..., k - i$:

$$\frac{\partial}{\partial s}\left(\mathrm{e}^{-(k-i-j)\int_s^t r(u)\,\mathrm{d}u}\boldsymbol{V}_r^{(j)}(s,t)\right)$$
$$= r(s)(k-i-j)\mathrm{e}^{-(k-i-j)\int_s^t r(u)\,\mathrm{d}u}\boldsymbol{V}_r^{(j)}(s,t) + \mathrm{e}^{-(k-i-j)\int_s^t r(u)\,\mathrm{d}u}\frac{\partial}{\partial s}\boldsymbol{V}_r^{(j)}(s,t).$$

Inserting (88), we get that

$$\begin{aligned}
\frac{\partial}{\partial s}\left(\mathrm{e}^{-(k-i-j)\int_s^t r(u)\,\mathrm{d}u}\boldsymbol{V}_r^{(j)}(s,t)\right) &= (k-i-j)r(u)\left[\mathrm{e}^{-(k-i-j)\int_s^t r(u)\,\mathrm{d}u}\boldsymbol{V}_r^{(j)}(s,t)\right] \\
&\quad - \mathrm{e}^{-(k-i-j)\int_s^t r(u)\,\mathrm{d}u}\left[\sum_{m=0}^{j}\tilde{\boldsymbol{C}}_r^{(m)}(s)\boldsymbol{V}_r^{(j-m)}(s,t)\right] \\
&= (k-i-j)r(u)\left[\mathrm{e}^{-(k-i-j)\int_s^t r(u)\,\mathrm{d}u}\boldsymbol{V}_r^{(j)}(s,t)\right] \\
&\quad - \mathrm{e}^{-(k-i-j)\int_s^t r(u)\,\mathrm{d}u}\left[\boldsymbol{\Lambda}(s)\boldsymbol{V}_r^{(j)}(s,t) + \sum_{m=1}^{j}\tilde{\boldsymbol{C}}_r^{(m)}(s)\boldsymbol{V}_r^{(j-m)}(s,t)\right] \\
&= -\left(\boldsymbol{\Lambda}(s) - (k-i-j)r(s)\boldsymbol{I}\right)\left[\mathrm{e}^{-(k-i-j)\int_s^t r(u)\,\mathrm{d}u}\boldsymbol{V}_r^{(j)}(s,t)\right] \\
&\quad - \sum_{m=1}^{j}\boldsymbol{C}_r^{(m)}\left[\mathrm{e}^{-(k-i-j+m)\int_s^t r(u)\,\mathrm{d}u}\boldsymbol{V}_r^{(j-m)}(s,t)\right].
\end{aligned}$$

Hence the generic terms exactly solve the differential equation induced by the proposed product integral. □

As a consequence, $\boldsymbol{V}_r^{(k)}(s,t)$ equals the upper right block–matrix of the matrix exponential. In fact, the last block column contains the matrices $\boldsymbol{V}_r^{(i)}(s,t)$, $i = 0, ..., k$.

5. Derivative securities

We here provide a very short and actuarial approach to option pricing. Assume that a stockprice is given by a stochastic process $\{S(t)\}_{t\geq 0}$, in a market with constant interest rate $r > 0$. A time T the value of the stock is $S(T)$.

Assume that we want to make an agreement that $g(S(T))$ is paid out at time T instead of T. That could e.g. be with $g(x) = \max(x-c, 0)$, which would then amount to an European call option with strike c: if $S(T) < c$, then we would like to buy the stock at price $S(T)$, in which case the option is worthless, and if $S(T) > c$, then we would prefer to buy at price c, in which case the difference $S(T) - c$ must be priced and paid for. If instead we are interested in selling at time T, the corresponding arrangement would be $g(x) = \max(c-x, 0)$, which is referred to as an European put option.

The price for such arrangements can be evaluated by

$$p_g = \mathbb{E}\left(\mathrm{e}^{-rT} g\left(S(0)\mathrm{e}^{rT}\frac{S(T)}{\mathbb{E}(S(T))}\right)\right)$$

Example 6.7 (General Black–Scholes)**:** In the Black–Scholes model,

$$S(T) = S(0)\mathrm{e}^{(r-\frac{1}{2}\sigma^2)T+\sigma W_T}, \quad W_T \sim \mathrm{N}(0,T),$$

i.e. $S(T)$ has a log–normal distribution. Using knowledge regarding the moment generating function of the normal distribution,

$$\mathbb{E}(S(T)) = S(0)\mathrm{e}^{rT}$$

so

$$\begin{aligned} p_g &= \mathbb{E}\left(\mathrm{e}^{-rT} g\left(S(0)\mathrm{e}^{(r-\frac{1}{2}\sigma^2)T+\sigma W_T}\right)\right) \\ &= \frac{\mathrm{e}^{-rT}}{\sqrt{2\pi T}}\int_{-\infty}^{\infty} g\left(S(0)\mathrm{e}^{(r-\frac{1}{2}\sigma^2)T+\sigma y}\right) e^{-y^2/(2T)}\,\mathrm{d}y. \end{aligned}$$

Example 6.8 (Black–Scholes gauranteed minimum)**:** Here $g(x) = \max(x, G)$ for the guaranteed amount G. Then

$$\begin{aligned} p_g &= \frac{\mathrm{e}^{-rT}}{\sqrt{2\pi T}}\int_{-\infty}^{\infty} g\left(S(0)\mathrm{e}^{(r-\frac{1}{2}\sigma^2)T+\sigma y}\right) e^{-y^2/(2T)}\,\mathrm{d}y \\ &= \frac{\mathrm{e}^{-rT}}{\sqrt{2\pi T}}\int_{-\infty}^{\infty} \max\left(S(0)\mathrm{e}^{(r-\frac{1}{2}\sigma^2)T+\sigma y}, G\right) e^{-y^2/(2T)}\,\mathrm{d}y. \end{aligned}$$

Now

$$S(0)\mathrm{e}^{(r-\frac{1}{2}\sigma^2)T+\sigma y} \geq G$$

iff

$$y \geq \frac{\log(G/S(0)) - (r-\frac{1}{2}\sigma^2)T)}{\sigma} =: L$$

so

$$
\begin{aligned}
p_g &= \mathrm{e}^{-rT}\left(\int_{-\infty}^{L} G\frac{1}{\sqrt{2\pi T}}\mathrm{e}^{-y^2/(2T)}\,\mathrm{d}y + \int_{L}^{\infty}\mathrm{e}^{(r-\frac{1}{2}\sigma^2)T+\sigma y}\frac{1}{\sqrt{2\pi T}}\mathrm{e}^{-y^2/(2T)}\,\mathrm{d}y\right)\\
&= \mathrm{e}^{-rT}\left(\int_{-\infty}^{L/\sqrt{T}} G\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-y^2/2}\,\mathrm{d}y + \int_{L}^{\infty}\mathrm{e}^{(r-\frac{1}{2}\sigma^2)T+\sigma y}\frac{1}{\sqrt{2\pi T}}\mathrm{e}^{-y^2/(2T)}\,\mathrm{d}y\right)\\
&= \mathrm{e}^{-rT}G\left(\Phi(L/\sqrt{T}) + \int_{L}^{\infty}\mathrm{e}^{rT}\frac{1}{\sqrt{2\pi T}}\mathrm{e}^{-y^2/(2T)+\frac{2T\sigma y}{2T}-\frac{T^2\sigma^2}{2T}}\,\mathrm{d}y\right)\\
&= \mathrm{e}^{-rT}G\Phi(L/\sqrt{T}) + \int_{L}^{\infty}\frac{1}{\sqrt{2\pi T}}\mathrm{e}^{-(y-T\sigma)^2/(2T)}\,\mathrm{d}y\\
&= \mathrm{e}^{-rT}G\Phi(L/\sqrt{T}) + \int_{(L-T\sigma)/\sqrt{T}}^{\infty}\frac{1}{\sqrt{2\pi}}\mathrm{e}^{-y^2/2}\,\mathrm{d}y\\
&= \mathrm{e}^{-rT}G\Phi\left(\frac{\log(G/S(0)) - (r-\frac{1}{2}\sigma^2)T)}{\sqrt{T}\sigma}\right) + 1 - \Phi\left(\frac{\log(G/S(0)) - (r+\frac{1}{2}\sigma^2)T)}{\sqrt{T}\sigma}\right)\\
&= \mathrm{e}^{-rT}G\Phi\left(\frac{\log(G/S(0)) - (r-\frac{1}{2}\sigma^2)T)}{\sqrt{T}\sigma}\right) + \Phi\left(\frac{-\log(G/S(0)) + (r+\frac{1}{2}\sigma^2)T)}{\sqrt{T}\sigma}\right)
\end{aligned}
$$

In continuous time finance, models are usually written in terms of so–called stochastic differential equations (SDE). This subject goes beyond the scope of these lecture notes, nevertheless, we may provide the following intuitive introduction to the subject, which contains most of the ideas behind the rigorous theory.

If S_t denotes the price of a financial asset or portfolio, then an SDE is on the form

$$\mathrm{d}S_t = \mu(t,S_t)\,\mathrm{d}t + \sigma(t,S_t)\,\mathrm{d}W_t.$$

Here $\mu(x,y)$ and $\sigma(x,y)$ are determinisitic (known) functions. The former is called the drift and the latter the volatility. The process $\{W_t\}_{t\geq 0}$ is called a Browninan motion or Wiener process. The only thing we need to know about this process is that it has independent increment, i.e. if $s \leq t \leq u$ then $W_u - W_t$ is independent of $W_t - W_s$, and the $W_t - W_s \sim \mathrm{N}(0, t-s)$.

The symbol

$$\mathrm{d}W_t = W_{t+\mathrm{d}t} - W_t \sim \mathrm{N}(0,\mathrm{d}t).$$

We use the usual convention as earlier that $(\mathrm{d}t)^2 = 0$.

By definition,

$$
\begin{aligned}
\mathbb{E}(\mathrm{d}W_t) &= 0\\
\mathbb{E}((\mathrm{d}W_t)^2) &= \mathbb{V}\mathrm{ar}(\mathrm{d}W_t^2) = \mathrm{d}t.
\end{aligned}
$$

Furthermore,

$$\mathbb{V}\mathrm{ar}(\mathrm{d}W_t^2) = ... = 0.$$

Hence we have a random variable, $\mathrm{d}W_t^2$, with zero variance and mean $\mathrm{d}t$. Hence it must be equal to its mean (almost surely), i.e.

$$\mathrm{d}W_t^2 = \mathrm{d}t.$$

This constitutes the second multiplication rule. Finally,

$$\mathbb{E}(\mathrm{d}t \cdot \mathrm{d}W_t) = \mathrm{d}t \cdot 0 = 0.$$

We summarize these rules in the multiplication table:

$\times$	$\mathrm{d}t$	$\mathrm{d}W_t$
$\mathrm{d}t$	0	0
$\mathrm{d}W_t$	0	$\mathrm{d}t$

With this we can then derive a chain rule for computing $\mathrm{d}f(t,W_t)$. We look at the Taylor expansion of $f(t,x)$,

$$\begin{aligned} f(t+\Delta t, x+\Delta x) &= f(t,x) + f_t(t,x)\Delta t + f_x(t,x)\Delta x \\ &\quad + \frac{1}{2}\Big(f_{tt}(t,x)(\Delta t)^2 + 2f_{tx}(t,x)\Delta t\Delta x + f_{xx}(t,x)(\Delta x)^2\Big) + o((\Delta t)^2 + (\Delta x)^2). \end{aligned}$$

Now let replace Δt by $\mathrm{d}t$, x by W_t and Δx by $\mathrm{d}W_t$. Then we get

$$f(t+\mathrm{d}t, W_{t+\mathrm{d}t}) = f(t,W_t) + f_t(t,W_t)\,\mathrm{d}t + f_x(t,W_t)\,\mathrm{d}W_t + \frac{1}{2}f_{xx}(t,W_t)\,\mathrm{d}t.$$

Denoting

$$\mathrm{d}f(t,W_t) = f(t+\mathrm{d}t, W_{t+\mathrm{d}t}) - f(t,W_t),$$

we then obtain what is known as Itô's formula

$$\mathrm{d}f(t,W_t) = f_t(t,W_t)\,\mathrm{d}t + f_x(t,W_t)\,\mathrm{d}W_t + \frac{1}{2}f_{xx}(t,W_t)\,\mathrm{d}t. \tag{89}$$

With exactly the same argument, for the more general process S_t we would obtain

$$\mathrm{d}f(t,S_t) = f_t(t,S_t)\,\mathrm{d}t + f_x(t,S_t)\,\mathrm{d}W_t + \frac{1}{2}f_{xx}(t,W_t)(\mathrm{d}S_t)^2. \tag{90}$$

Expanding on

$$(\mathrm{d}S_t)^2 = (\mu(t,S_t)\,\mathrm{d}t + \sigma(t,S_t)\,\mathrm{d}W_t)^2 = \sigma^2(t,S_t)\,\mathrm{d}t,$$

where we have used the multiplication table, we get

$$\mathrm{d}f(t,S_t) = \left[f_t(t,S_t) + f_x(t,S_t)\mu(t,S_t) + \frac{1}{2}\sigma^2(t,S_t)\right]\mathrm{d}t + \sigma(t,S_t)\,\mathrm{d}W_t. \tag{91}$$

Example 6.9 (Example (6.7) continued)**:** Let us have a look at the Black–Scholes equation, but replacing then interest rate r by another constant drift μ, i.e.

$$S_t = S_0 \mathrm{e}^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}.$$

Then

$$S_t = f(t,W_t)$$

where

$$f(t,x) = S_0 \mathrm{e}^{(\mu - \frac{1}{2}\sigma^2)t + \sigma x}.$$

Now

$$\begin{aligned} f_t(t,x) &= \left(\mu - \frac{1}{2}\sigma^2\right) f(t,x) \\ f_x(t,x) &= \sigma f(t,x) \\ f_{xx}(t,x) &= \sigma^2 f(t,x). \end{aligned}$$

Then by (89), we get that

$$\begin{aligned} \mathrm{d}S_t &= \mathrm{d}f(t,W_t) \\ &= \left(\mu - \frac{1}{2}\sigma^2\right) f(t,W_t)\,\mathrm{d}t + \sigma f(t,W_t)\,\mathrm{d}W_t + \frac{1}{2}\sigma^2 f(t,W_t)\,\mathrm{d}t \\ &= \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t. \end{aligned}$$

This is the usual expression for the Black–Scholes model.

When computing prices, we replace μ by the risk–free interest rate r. The argumentation is based on a so–called no–arbitrage argument, but may be understood as follows. From an actuarial point of view, it seems reasonable that expectation of the present value of S_t should exactly be its price S_0. This is how fair premiums are computed. But

$$\begin{aligned} \mathbb{E}\left(\mathrm{e}^{-rt} S_t\right) &= \mathbb{E}\left(S_0 e^{(\mu - r - \frac{1}{2}\sigma^2)t + \sigma W_t}\right) \\ &= S_0 e^{(\mu - r - \frac{1}{2}\sigma^2)t} \mathbb{E}\left(e^{\sigma W_t}\right) \\ &= S_0 e^{(\mu - r - \frac{1}{2}\sigma^2)t} \mathrm{e}^{\frac{1}{2}\sigma^2 t} \\ &= S_0 \mathrm{e}^{(\mu - r)t}. \end{aligned}$$

The second to the last step is obtained through the moment generating function for the $\mathrm{N}(0,t)$ distribution.

So unless $\mu = r$, the expected present value of S_t will either be worth more than S_0 or less than S_0. If it is worth more, then taking a huge loan at interest rate r, could provide us with an *arbitrarily* large gain. Similarly, if it is less, then a selling argument yields the same. To avoid the possibility arbitrary large gains, present value of any future prices relating to S_t should be prices as if the drift of the process equals the interest rate.

The argument by replacing the drift by the interest is not valid in general, but a special feature of the Black–Scholes model. In general, one looks for a change of parameters such that the present value is exactly that of S_0. □

6. The unit-link reserve

In unit linked insurance policies, parts of the benefit will depend on exogenous funds like stock prices or investments. For example, an insurance policy may stipulate that the value of an investment portfolio being paid out at the time of death, or funding a pension scheme.

Let $S(x)$ denote the value of the exogenous funds, and let $\boldsymbol{R}(x,S(x))$ be the matrix that contains the expected rewards to be paid out during $[x,x+\mathrm{d}x)$ for the different transitions conditional on the value of $S(x)$. In general, $S(u)$ is random, and $\{S(u)\}_{u\geq 0}$ is assumed to be a Markov process. The time–t reserve is then defined as

$$\begin{aligned}\boldsymbol{V}(t,S(t)) &= \mathbb{E}\left(\int_t^T \prod_t^u (\boldsymbol{I}+(\boldsymbol{M}(v)-r(v)\boldsymbol{I})\,\mathrm{d}v)\boldsymbol{R}(u,S(u))\prod_u^T(\boldsymbol{I}+\boldsymbol{M}(v)\,\mathrm{d}v)\,\mathrm{d}u \,\middle|\, S(t)\right)\\ &= \int_t^T \prod_t^u (\boldsymbol{I}+(\boldsymbol{M}(v)-r(v)\boldsymbol{I})\,\mathrm{d}v)\mathbb{E}\Big(\boldsymbol{R}(u,S(u))\,|\,S(t)\Big)\prod_u^T(\boldsymbol{I}+\boldsymbol{M}(v)\,\mathrm{d}v)\,\mathrm{d}u.\end{aligned}$$

or

$$\boldsymbol{V}(t,y) = \int_t^T \prod_t^u (\boldsymbol{I}+(\boldsymbol{M}(v)-r(v)\boldsymbol{I})\,\mathrm{d}v)\mathbb{E}\Big(\boldsymbol{R}(u,S(u))\,|\,S(t)=y\Big)\prod_u^T(\boldsymbol{I}+\boldsymbol{M}(v)\,\mathrm{d}v)\,\mathrm{d}u.$$

Then $\boldsymbol{V}(t,y)$ can be computed by

$$\prod_t^T\left(\boldsymbol{I}+\begin{pmatrix}\boldsymbol{\Lambda}(u)-r\boldsymbol{I} & \mathbb{E}\Big(\boldsymbol{R}(u,S(u))\,|\,S(t)=y\Big)\\ \boldsymbol{0} & \boldsymbol{\Lambda}(u)\end{pmatrix}\mathrm{d}u\right)$$

Denote the transition densities of S by

$$p^{(S)}(t,u,x,y) = \mathbb{P}\Big(S(u)\in \mathrm{d}y\,|\,S(t)=x\Big).$$

Then

$$\mathbb{E}\Big(\boldsymbol{R}(u,S(u))\,|\,S(t)=x\Big) = \int_0^\infty \boldsymbol{R}(u,y)p^{(S)}(t,u,x,y)\,\mathrm{d}y.$$

Check with special cases, elaborate on the expectation measure $\mathbb{Q}$.

Example 6.10: Consider a unit–linked term life insurance with a guarantee for (x). We define a process with two states, active and dead, and the transition from active to dead will trigger a payment of $\max(S(t),G(t))$ if it happens at time $t<T$. Then

$$\boldsymbol{R}(t,S(t)) = \begin{pmatrix}-\bar{p}(t) & \mu_{x+t}\max(S(t),G(t))\\ 0 & 0\end{pmatrix},$$

where μ_{x+t} denotes the mortality rate for (x) at age $x+t$, $S(t)$ the stockprice at time t and $G(t)$ a guarantee, which is deterministic and may be thought of as a usual lump sum payment. The term $\bar{p}(t)$ denotes the premium rate. The premium pays for the purchase of the stock, $S(0)$, and for the guarantee $G(t)$. Hence a fair premium function $\bar{p}(t)$ is one which makes $V(0,S(0))=0$.

We must compute,

$$\mathbb{E}\Big(\boldsymbol{R}(u,S(u))\,|\,S(t)=y\Big)=\begin{pmatrix}-\bar{p}(u) & \mu_{x+u}\mathbb{E}\Big(\max(S(u),G(u))\,|\,S(t)=y\Big)\\ 0 & 0\end{pmatrix},$$

which amount to computing $\mathbb{E}\Big(\max(S(u),G(u))\,|\,S(t)=y\Big)$. We get (should be undiscounted!)

$$\begin{aligned}\mathbb{E}\Big(\max(S(u),G(u))\,|\,S(t)=y\Big)&=G(u)\Phi\left(\frac{\log(G(u)/y)-(r-\frac{1}{2}\sigma^2)(u-t))}{\sqrt{u-t}\sigma}\right)\\&\quad+\mathrm{e}^{r(u-t)}\Phi\left(\frac{-\log(G(u)/y)+(r+\frac{1}{2}\sigma^2)(u-t))}{\sqrt{u-t}\sigma}\right)\end{aligned}$$

Example 6.11 (Black–Scholes)**:** Here

$$S(u)=S(0)\mathrm{e}^{(r-\frac{1}{2}\sigma^2)u+\sigma W(u)},\quad W(u)\sim\mathrm{N}(0,u).$$

Example 6.12: Fund protection with more than one state,

$$S(u)=(X(u),\max_{0\le s\le u}X(s)).$$

where $X(u)$ is geometric Brownian motion.

Bibliographic notes

Product integration can be consulted in [**9**, **7**] and [**12**]. The particular approach taken in these notes is from [**3**]. The classical theory of SDE-driven interest rate models can be consulted in many textbooks. I may recommend [**4**]. The Markovian interest rate model goes at least back to [**10**]. In [**11**], a financial theory is developed for this kind of interest rate. The content of the current notes is taken from [**2**]. The development of a matrix–approach to reserves, higher order moments, and distributions of future payouts is from [**5**]. Time–inhomogeneous semi–Markov processes was first introduced in [**8**]. Of more recent papers we mention [**6**], which deals with cash–flows in semi–Markov models, and [**1**] which established PDEs for reserves and higher order moments (non–matrix version).

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APPENDIX A

R code for spouse model

```
prodint<-function(A,s,t,n){
  x0 <- s
  y0 <- diag(nrow(A(s)))
  h<-(t-s)/n
  for(i in 1:n) {
    s1 <- h * y0%*%A(x0)
    s2 <- h * (y0 +s1/2)%*%A(x0 + h/2)
    s3 <- h * (y0 +s2/2)%*%A(x0 + h/2)
    s4 <- h * (y0 + s3)%*%A(x0 + h)
    y0 <- y0 + s1/6 + s2/3 + s3/3+ s4/6
    x0 <- x0 + h
  }
  return(y0)
}

Lambda<-function(x){
  A<-matrix(0,4,4)
  A[1,2]<- 0.0005+10^(5.728+0.038*(x+6)-10)
  A[1,3]<- 0.0005+10^(5.88+0.038*x-10)
  A[2,4]<- 0.0005+10^(5.88+0.038*x-10)
  A[3,4]<- 0.0005+10^(5.728+0.038*(x+6)-10)
  evec<-rep(1,4)
  dsum<-as.vector(A%*%evec)
  return(A-diag(dsum))
}

R<-function(x,mu){
  if(x<=70){
    return(diag(c(-mu,-mu,0,0)))
  }
  else{
    return(diag(c(1.2,1.2,1.2,0)))
  }
}

dR<-function(x,mu){
  if(x<=70){
    return(diag(c(-1,-1,0,0)))
```

```
  }
  else{
    return(diag(c(0,0,0,0)))
  }
}

reserve<-function(t,TT,Lambda,R,mu,rint,N){
  dim<-nrow(Lambda(t))
  A11<-function(x){
    return(Lambda(x)-rint*diag(1,nrow=dim))
  }
  RM<-function(x){
    cbind(rbind(A11(x),matrix(0,dim,dim)),rbind(R(x,mu),Lambda(x)))
  }
  PRM<-prodint(RM,t,TT,N)
  RES<-PRM[1:dim,(dim+1):(2*dim)]
  return((RES%*%rep(1,dim))[1])
}

equiv_premium<-function(t,TT,Lambda,R,dR,mu,rint,numsteps){
  b=reserve(t,TT,Lambda,R,0.0,rint,numsteps)
  a=reserve(t,TT,Lambda,dR,0.0,rint,numsteps)
  return(-b/a)
}

equiv_premium(50,100,Lambda,R,dR,0.0,0.01,5000)

R1<-function(x,mu){
  if(x<=70){
    return(diag(c(-1.2,-1.2,0,0)))
  }
  else{
    return(diag(c(mu,mu,mu,0)))
  }
}

dR1<-function(x,mu){
  if(x<=70){
    return(diag(c(0,0,0,0)))
  }
  else{
    return(diag(c(1,1,1,0)))
  }
}
equiv_premium(50,100,Lambda,R1,dR1,0.0,0.01,5000)
```

APPENDIX B

A short guide to fitting Markovian interest rate models

PH fitting to weighted data works in the following way. We use the library matrixdist in R. Once installed, this is invoked by

```
library(matrixdist)
```

In the matrixdist package one may then create an object which is of phase–type by the command

```
ph(structure="general",dimension=dim)
```

This generates an random ph representation, which we may see how it looks like by

```
myph<-ph(structure="general",dimension=dim)
myph
```

Here "general" simply refers to a general phase–time structure, as opposed to sub–structures like e.g. Coxian, which would be then be set by "coxian".

Fitting to data is now a simple matter. Data $x_1, ..., x_n$, which may e.g. be read from a file, are stored in an array, say called data. Then the command "fit" can be used to perform a number of iterations in an EM algorithm, which converges towards to maximum likelihood estimate of its parameters. The following commands reads data, defines a five–dimensional phase–type object, and performs 1000 EM–iterations in a five dimensional phase–type setting with initial parameters given by ph5:

```
data<-scan(file="path-to-data/data.dat")
ph5<-ph(structure="general",dimension=5)
fit5 <- fit(ph5,y=data,stepsEM=1000)
```

The result, "fit5" is again a phase–type object, which may be refitted if another e.g. 2000 iterations are needed:

```
fit5 <- fit(fit5,y=data,stepsEM=2000)
```

One may also fit to weighted data. If a data point appears n of times, it may be defined as a weighted data point with weight n. The weight need to be an integer value,. This acutally allows us to fit to a histogram or theoretical distribution instead.

For a given theoretical distribution function F, choose a finite range $[a,b]$ and subdivide it into small intervals $[y_i, y_{i+1}]$, $i = 1, 2, ..., m-1$. Let x_i be the midpoint of the small intervals,

$x_i = (y_i + y_{i+1})/2$. Then we may consider the x_i's as weighted data with weights $F(y_{i+1}) - F(y_i)$. In particular, the latter method applied to bond prices (which behaves like a survival function).

Weighted data are fitted in the following way:

```
fit5<-fit(ph,y=data,weight=weights,stepsEM=1000)
```

Here "data" is an array containing the data points while "weights" is an array containing the corresponding weights. Hence they are of the same size.

We have assume a finite interval until now for the support of F. This may no be satisfactory if the remaining tail probability from the last x_n onwards is still large. This is often the case for bonds. In this case if would be better to treat b, the last interval limit, as a right censored data point. This can be handled in the following way:

```
fit(ph,y=uncens,weight=wuncens,rcen=cens,rcenweight=wcens,stepsEM=1000)
```

Here "ucens" is an array which contains the x_i's midpoints of the intervals, "wuncens" an array of the same size containing the wights, "rcens" an array (in this case of one point only) containing the cut–off of our support, and "wcens" an array of the same size as rcens which contains the remaning tail probability.

Once fitted, the fitted distribution can be visualized by accessing e.g the density or distribution functions by the commands

```
dens(fit5, seq(0,10, 0.01))
cdf(fit5, seq(0,10, 0.01))
```

where "seq(0,10, 0.01)" defines a range over 0 to 10 with steplengths 0.01 in which the functions are evaluated. One may e.g. plot the fitted density against a histogram using

```
hist(data,probability = T)
lines(seq(0,10, 0.01),dens(fit5, seq(0,10, 0.01)))
```

Regarding fitting to bond prices, we need to plot prices against the survival function of the fit. Assume that the bond prices are contained in the array "bond" with corresponding maturities in "mat". Then

```
plot(mat,bond,ylab="Bond price",xlab="Maturity")
lines(seq(0,300, 0.1), 1-cdf(ph10, seq(0,300, 0.1)))
```

plots the fitted ph distribution to bond prices in terms of its survival function. The exit rate vector contains the estimated interest rates used in order to achieve the fit.

If one prefers to use pre–specified interest rates, specified in the vector $\boldsymbol{r} = (r_1, ..., r_p)$, then one can replace the exit rate vector in each step of the EM algorithm by $\boldsymbol{r}$ instead. The is achieved by replacing the intensity matrix $\boldsymbol{T}$ obtained in an EM step by

$$\boldsymbol{T} + \boldsymbol{\Delta}(\boldsymbol{t}) - \boldsymbol{\Delta}(\boldsymbol{r}),$$

where $\boldsymbol{t}$ is the estimated exit rate vector. In R language:

```
for(i in 1:1000){
  ph <- fit(ph,y=uncens,weight=wuncens,rcen=cens,rcenweight=wcens,stepsEM=1)
  exit <- -rowSums(ph@pars$S)
  ph@pars$S <- ph@pars$S + diag(exit)-diag(r)
  }
}
```

The command “ph@pars$S” pulls the intensity matrix out of the phase–type object “ph”.